

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\A150NRL11EQ0011st Ammonia Chiller.pvdb

Date of Analysis : Sep 5, 2026 2:47pm

PV Elite 28, January 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 1600.000 x 2700.000 mm. ID
 Vessel Design Length, Tangent to Tangent 7008.97 mm.
 Specified Datum Line Distance 0.00 mm.
 Shell Side Design Temperature 85 °C
 Channel Side Design Temperature 85 °C
 Shell Side Design Pressure 25.000 bars
 Channel Side Design Pressure 191.000 bars
 Shop Shell Side Test Pressure 33.705 bars
 Shop Channel Side Test Pressure 252.080 bars
 Wind Design Code ASCE/SEI 7-22
 Earthquake Design Code ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	+N
Head	SA-516 70	...	...	K02700	+Q+IT
Head	SA-516 70	...	...	K02700	+N+IT
Cone	SA-516 70	...	...	K02700	+Q+IT
Flange	SA-350 LF2	1	...	K03011	+IT
Nozzle	SA-350 LF2	1	...	K03011	+IT
Nozzle Flg	SA-350 LF2	1	...	K03011	
Nozzle Flg	SA-105	...	...	K03504	+IT
Tubes	SA-334 6	...	...	K03006	+IT
Tubesheet	SA-350 LF2	1	...	K03011	+IT
Flg Bolting	SA-320 L7	...	<= 65	G41400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	+Q+IT

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
CH. COVER	191.098	0.00	198.000	1.5000	N/A	No
CH. COVER	191.098	0.00	216.000	1.5000	N/A	No
SH. FLG.	25.098	0.00	26.000	1.5000	N/A	No
SHELL-1	25.098	0.00	45.000	1.5000	N/A	No
SHELL-2	25.098	1.03	27.000	1.5000	N/A	No
K-SHELL	25.098	1.03	27.000	1.5000	N/A	No
SH. COVER	25.098	1.03	26.000	1.5000	No	No

Liquid Level: 1600.00 mm. Dens.: 0.001 kg./cm³ Sp. Gr.: 0.625

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Sphere	0.0	0.0	71.0	60.0	57.8	...	1.00	1.00
Cylinder	50.0	50.0	140.0	140.0	122.6	...	1.00	1.00



Body Flg	653.2	200.0	170.0	155.0	145.2	145.2	1.00	1.00
Cylinder	953.2	300.0	28.0	28.0	16.2	...	1.00	1.00
Conical	2858.2	1905.0	32.0	32.0	30.2	9.5	1.00	1.00
Cylinder	6958.2	4100.0	28.0	28.0	26.4	12.2	1.00	1.00
Ellipse	7009.0	50.8	32.0	27.0	26.1	8.6	1.00	1.00

Local Stress Analysis Results:

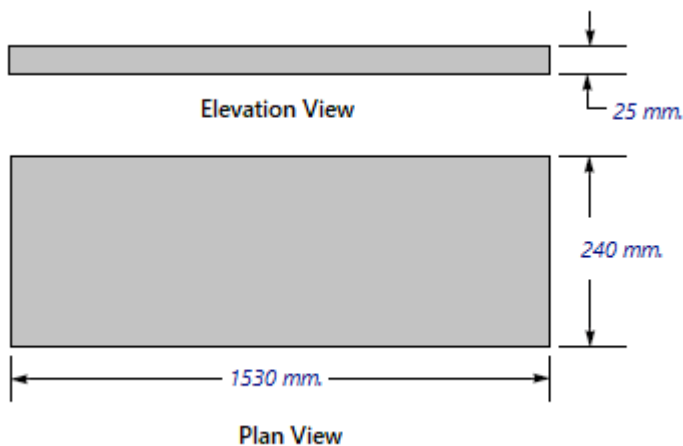
Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
CH. IN NOZ. A1	WRC-107/537	0.952	n/a	Passed
CH. OUT NOZ. B1	WRC-107/537	0.952	n/a	Passed
SS IN NOZ. A2	WRC-297	0.928	Shell	Passed
SS OUT NOZ. B2	WRC-297	0.928	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	200.000 mm.
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	1350.000 mm.
Wear Pad Width	240.000 mm.
Wear Pad Thickness	16.000 mm.
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	150.000 mm.
Baseplate Length	1530.000 mm.
Baseplate Thickness	25.000 mm.
Baseplate Width	240.000 mm.
Number of Ribs (including outside ribs)	4
Rib Thickness	16.000 mm.
Web Thickness	16.000 mm.
Height of Center Web	306.000 mm.
Number of Bolts in Baseplate	3

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	34168.93	Kgf
Maximum Transverse Saddle Shear Load	2513.95	Kgf
Maximum Longitudinal Saddle Shear Load	7541.84	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	30183.43	Kgf
Maximum Transverse Saddle Shear Load	295.23	Kgf
Maximum Longitudinal Saddle Shear Load	557.27	Kgf



Weights:

Fabricated	- Bare w/o removable internals	54610.2 kg.
Shop Test	- Fabricated + water (full)	87978.9 kg.
Shipping	- Fab. + rem. intls.+ shipping app.	54812.6 kg.
Erected	- Fab. + rem. intls.+ insul. (etc)	54812.6 kg.
Empty	- Fab. + intls. + details + wghts.	54812.6 kg.
Operating	- Empty + operating liquid (no ca)	80064.3 kg.

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Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
CH. IN NOZ A1	194	...	...	OK	47.74	OK	Passed
CH. OUT NOZ B1	194	...	...	OK	47.76	OK	Passed
MANHOLE M1	198	...	...	OK	40.08	OK	Passed
LIQ OUT NOZ B3	27	OK	...	OK	6.76	OK	Passed
SS IN NOZ. A2	27	OK	...	OK	9.83	OK	Passed
SS OUT NOZ. B2	26	OK	...	OK	9.83	OK	Passed
VENT NOZ V1	27	...	...	OK	4.92	OK	No Calc[*]
PR. IN P1	27	...	...	OK	4.92	OK	No Calc[*]
PR. IN NOZ. P2	27	...	...	OK	4.92	OK	No Calc[*]
SAFETY VALVE S1	27	...	...	OK	4.92	OK	No Calc[*]
LEVEL GUAGE L1	27	...	...	OK	4.92	OK	No Calc[*]
LEVEL GUAGE L2	27	...	...	OK	4.92	OK	No Calc[*]
LEVEL GUAGE L3	27	...	...	OK	4.92	OK	No Calc[*]
LEVEL GUAGE L4	27	...	...	OK	4.92	OK	No Calc[*]
DRAIN NOZ. D2	27	...	...	OK	4.92	OK	No Calc[*]
TEMP IN. T1	26	...	...	OK	4.92	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 26 Nozzle : TEMP IN. T1 [Shellside]
 Minimum MAWP Nozzles : 194 Nozzle : CH. OUT NOZ B1 [Tubeside]

[*] - This was a small opening and the areas were not computed.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm.	Layout Angle deg	Dia. Limit mm.
10	CH. IN NOZ A1	0.000	90.000	1225.200
10	CH. OUT NOZ B1	0.000	-45.000	1225.200
10	MANHOLE M1	0.000	-160.017	1022.000
50	LIQ OUT NOZ B3	1203.175	-90.000	198.325
60	SS IN NOZ. A2	6608.175	90.000	552.100
60	SS OUT NOZ. B2	4558.174	90.000	780.700
60	VENT NOZ V1	3258.175	90.000	113.325
60	PR. IN P1	3008.175	90.000	113.325
60	PR. IN NOZ. P2	6108.175	90.000	113.325
60	SAFETY VALVE S1	5858.175	90.000	113.325
60	LEVEL GUAGE L1	3008.175	30.481	113.325
60	LEVEL GUAGE L2	3008.175	-30.481	113.325
60	LEVEL GUAGE L3	3658.175	30.481	113.325
60	LEVEL GUAGE L4	3658.175	-30.481	113.325
60	DRAIN NOZ. D2	4508.175	-90.000	113.325
70	TEMP IN. T1	0.000	270.000	111.325

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

Diameter Limit clash between : CH. OUT NOZ B1 and CH. IN NOZ A1.
The Centerline Dist. is 973.81 while the sum of the limit Radii is 1225.20 mm..



*Diameter Limit clash between : MANHOLE M1 and CH. IN NOZ A1.
The Centerline Dist. is 901.13 while the sum of the limit Radii is 1123.60 mm..*

*Diameter Limit clash between : MANHOLE M1 and CH. OUT NOZ B1.
The Centerline Dist. is 931.22 while the sum of the limit Radii is 1123.60 mm..*

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
VENT NOZ V1	2.000 in	160	LWN	2.375	8.738	...	...	278.34	300
PR. IN P1	2.000 in	160	LWN	2.375	8.738	...	...	278.34	300
PR. IN NOZ. P2	2.000 in	160	LWN	2.375	8.738	...	...	278.34	300
SAFETY VALVE S1	2.000 in	160	LWN	2.375	8.738	...	...	278.34	300
LEVEL GUAGE L1	2.000 in	160	LWN	2.375	8.738	...	...	351.46	300
LEVEL GUAGE L2	2.000 in	160	LWN	2.375	8.738	...	...	351.46	300
LEVEL GUAGE L3	2.000 in	160	LWN	2.375	8.738	...	...	351.46	300
LEVEL GUAGE L4	2.000 in	160	LWN	2.375	8.738	...	...	351.46	300
DRAIN NOZ. D2	2.000 in	160	LWN	2.375	8.738	...	...	351.46	300
TEMP IN. T1	2.000 in	160	LWN	2.375	8.738	...	...	337.36	300
LIQ OUT NOZ B3	4.000 in	160	WNF	4.500	13.487	...	...	284.71	300
SS IN NOZ. A2	12.000 in	120	WNF	12.750	25.400	...	...	442.38	300
SS OUT NOZ. B2	18.000 in	120	WNF	18.000	34.925	...	...	456.03	300
MANHOLE M1	20.000 in Actual		RTJWn	23.937	50.000	...	...	478.50	1500
CH. IN NOZ A1	24.000 in Actual		None	27.937	50.000	...	...	512.48	...
CH. OUT NOZ B1	24.000 in Actual		None	27.937	50.000	...	...	512.48	...

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
VENT NOZ V1	SA-350 LF2	27.000	10.000	...	...	...
PR. IN P1	SA-350 LF2	27.000	10.000	...	...	...
PR. IN NOZ.	SA-350 LF2	27.000	10.000	...	...	...
SAFETY VALV	SA-350 LF2	27.000	10.000	...	...	...
LEVEL GUAGE	SA-350 LF2	30.000	10.000	...	...	...
LEVEL GUAGE	SA-350 LF2	27.000	10.000	...	...	...
LEVEL GUAGE	SA-350 LF2	30.000	10.000	...	...	...
LEVEL GUAGE	SA-350 LF2	27.000	10.000	...	...	...
DRAIN NOZ.	SA-350 LF2	30.000	10.000	...	...	...
TEMP IN. T1	SA-350 LF2	30.000	10.000	...	...	...
LIQ OUT NOZ	SA-350 LF2	32.000	10.000	...	...	...
SS IN NOZ.	SA-350 LF2	27.000	10.000	...	...	...
SS OUT NOZ.	SA-350 LF2	27.000	10.000	...	...	...
MANHOLE M1	SA-350 LF2	60.000	10.000	...	...	...
CH. IN NOZ	SA-350 LF2	60.000	10.000	...	...	...
CH. OUT NOZ	SA-350 LF2	60.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
VENT NOZ V1	3258.175	90.0	250.00	0.00	K-SHELL
PR. IN P1	3008.175	90.0	250.00	0.00	K-SHELL
PR. IN NOZ. P2	6108.175	90.0	250.00	0.00	K-SHELL
SAFETY VALVE S1	5858.175	90.0	250.00	0.00	K-SHELL
LEVEL GUAGE L1	3008.175	30.5	300.00	0.00	K-SHELL

LEVEL GUAGE L2	3008.175	-30.5	300.00	0.00	K-SHELL
LEVEL GUAGE L3	3658.175	30.5	300.00	0.00	K-SHELL
LEVEL GUAGE L4	3658.175	-30.5	300.00	0.00	K-SHELL
DRAIN NOZ. D2	4508.175	-90.0	300.00	0.00	K-SHELL
TEMP IN. T1	...	270.0	300.00	0.00	SH. COVER
LIQ OUT NOZ B3	1203.175	-90.0	250.00	0.00	SHELL-2
SS IN NOZ. A2	6608.174	90.0	400.00	0.00	K-SHELL
SS OUT NOZ. B2	4558.174	90.0	400.00	0.00	K-SHELL
MANHOLE M1	...	-160.0	300.00	0.00	CH. COVER
CH. IN NOZ A1	...	90.0	300.00	0.00	CH. COVER
CH. OUT NOZ B1	...	-45.0	300.00	0.00	CH. COVER

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	SPHERICAL HEAD: 71.0mm. THK X 1600.0mm. ID X 0.0mm.	SA-516 70
1	WELD NECK FLANGE: 170.0mm. THK X 1880.0mm. OD	SA-350 LF2
1	CYLINDER: 28.0mm. THK X 1600.0mm. ID X 300.0mm.	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 32.0mm. THK X 2700.0mm. ID X 50.8mm.	SA-516 70
1	INSULATION: 800mm X 70mm THK	COLD INSULATION
1	INSULATION: 50mm X 70mm THK	COLD INSULATION
5	BAFFLES: 1580.0mm DIA X 16.0mm THK	SA-516 70
14	TIE RODS: 16mm DIA X 5150mm	SA-193 B7M
14	SPACERS: 19.4mm DIA X 1.5mm THK X 5150.0mm	SA-334 6
1	TUBESHEET: 1880.0mm X 400.0mm THK	SA-350 LF2
1906	TUBES: 5200.0mm X 25.4mm DIA X 2.8mm THK	SA-334 6
1	INSULATION: 215mm X 70mm THK	COLD INSULATION
1	SADDLE: 200.0mm X 120 DEG	SA-516 70
1	INSULATION: 300mm X 70mm THK	COLD INSULATION
1	INSULATION: 1905mm X 70mm THK	COLD INSULATION
2	SADDLE: 300.0mm X 120 DEG	SA-516 70
1	INSULATION: 4100mm X 70mm THK	COLD INSULATION
1	INSULATION: 725mm X 70mm THK	COLD INSULATION
1	CLASS 1500 GR 1.1, 20.0" BLIND FLANGE(S)	SA-350 LF2
1	CLASS 300 GR 1.1, 4.0" BLIND FLANGE(S)	SA-350 LF2
8	CLASS 300 GR 1.1, 2.0" BLIND FLANGE(S)	SA-350 LF2
2	CLASS 300 GR 1.1, 2.0" BLIND FLANGE(S)	SA-105
1	GASKET: 1629.0mm. OD X 1603.0mm. ID	...
48	BODY FLANGE BOLTS: 36.0mm. DIA	SA-320 L7
48	NUTS FOR BODY FLANGE BOLTS: 36.0mm. DIA	...
1	CYLINDRICAL SEGMENT 140.0mm. THK X 3048.0mm. X 50.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 140.0mm. THK X 1978.5mm. X 50.0mm.	SA-516 70
1	ECC. CONICAL SEGMENT 32.0mm. THK X 5136.4mm. X 1905.0mm. X 3043.	SA-516 70
1	ECC. CONICAL SEGMENT 32.0mm. THK X 3345.7mm. X 1905.0mm. X 1982.	SA-516 70
2	CYLINDRICAL SEGMENT 28.0mm. THK X 3048.0mm. X 4100.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 28.0mm. THK X 2386.3mm. X 4100.0mm.	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
SH. FLG. Bolting	[11] [21]	!	-46	-46		0.580	28.000	1.00	No
SHELL-1	[8]	B	2	-22		0.575	28.000	1.00	No
SHELL-2	[8]	!	-35	-37		0.973	32.000	1.00	No
K-SHELL	[8]	!	-35	-37		0.969	28.000	1.00	No
SH. COVER	[10]	!	-35	-35		0.996	27.000	1.00	No
SH. COVER	[7]	!	-35	-44		0.842	32.000	1.00	No
FIXED SADDLE	[24]	B	-14	-39	-29	0.575	16.000	1.00	No
LIQ OUT NOZ B3	[1]	!	-46	-104		0.080	11.801	1.00	No
Nozzle Flg	[5]	!	-46	-79					
SS IN NOZ. A2	[1]	!	-46	-46		0.934	28.000	1.00	No
Nozzle Flg	[5]	!	-46	-79					
SS OUT NOZ. B2	[1]	!	-46	-46		0.934	28.000	1.00	No
Nozzle Flg	[5]	!	-46	-79					
VENT NOZ V1	[1]	!	-46	-104		0.068	7.645	1.00	No
Nozzle Flg	[5]	!	-46	-46					
PR. IN P1	[1]	!	-46	-104		0.068	7.645	1.00	No
Nozzle Flg	[5]	!	-46	-46					
PR. IN NOZ. P2	[1]	!	-46	-104		0.068	7.645	1.00	No
Nozzle Flg	[5]	!	-46	-46					
SAFETY VALVE S	[1]	!	-46	-104		0.068	7.645	1.00	No
Nozzle Flg	[5]	!	-46	-46					
LEVEL GUAGE L1	[1]	!	-46	-104		0.068	7.645	1.00	No
Nozzle Flg	[5]	!	-46	-46					
LEVEL GUAGE L2	[1]	!	-46	-104		0.068	7.645	1.00	No
Nozzle Flg	[5]	!	-46	-46					
LEVEL GUAGE L3	[1]	!	-46	-104		0.068	7.645	1.00	No
Nozzle Flg	[5]	!	-46	-46					
LEVEL GUAGE L4	[1]	!	-46	-104		0.068	7.645	1.00	No
Nozzle Flg	[5]	!	-46	-46					
DRAIN NOZ. D2	[1]	!	-46	-104		0.069	7.645	1.00	No
Nozzle Flg	[5]	!	-35	-35					
TEMP IN. T1	[1]	!	-46	-104		0.068	7.645	1.00	No
Nozzle Flg	[5]	A	-18	-48					
Tubesheet: SS	[13]	!	-46	-46		0.942	100.000	1.00	No
Warmest MDMT:			2	-22					
CH. COVER	[9]	!	-35	-36		0.976	60.000	1.00	Yes
CH. IN NOZ A1	[1]	!	-46	-46		0.962	60.000	1.00	No
CH. OUT NOZ B1	[1]	!	-46	-46		0.962	60.000	1.00	No
MANHOLE M1	[1]	!	-46	-46		0.796	50.000	1.00	Yes
Nozzle Flg	[5]	!	-46	-60					
Warmest MDMT:			-35	-36					

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	-22.0	-35.0	Fail

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3

- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (Kgfcmm2mm):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	25	bars
Channel Side Design Pressure	191	bars
Shell Side Design Temperature	85.0	°C
Channel Side Design Temperature	85.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Sour	
Service Type, Channel Side	Sour	
MDMT (CET), Shell Side	-35.0	°C
MDMT (CET), Tube Side	-35.0	°C
User defined MAWP, Shell Side	0	bars
User defined MAWP, Channel Side	0	bars
User defined MAPnc, Shell Side	0	bars
User defined MAPnc, Channel Side	0	bars
User defined Test Pres., Shell Side	0	bars
User defined Test Pres., Channel Side	0	bars

Projection of Nozzle from Vessel Top	0	mm.
Projection of Nozzle from Vessel Bottom	0	mm.
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS



Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 Km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	0
Base Elevation	300 mm.
Percent Wind for Hydrotect	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment	H or Hh 0 mm.
Distance Upwind of Crest	Lh 0 mm.
Distance from Crest to the Vessel	x 0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Seismic Design Code	ASCE/SEI 7-22
Importance Factor	1.500
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio	z/h 0.000
Amplification Factor	Ap 0.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Component resonance duct. fact.	[CAR] 0.000
Component strength factor	[Rpo] 0.000
Overstrength factor	[Omega0] 0.000
Response Acceleration Factor	[SMS] 0.326
Response Acceleration Factor	[SM1] 0.139
Building Importance Factor	[Ie] 1.000
Site Latitude	0.000 deg
Site Longitude	0.000 deg

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No



Use the MAWP to compute the MDMT Yes
 For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes
 For PWHT & P1 Materials the MDMT can be < -55F (-48C) No
 Using Metric Material Databases, ASME II D Yes
 Calculate B31.3 type stress for Nozzles with Loads Yes
 Reduce the MDMT due to lower membrane stress Yes
 Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/7
Element From Node	10
Element To Node	20
Element Type	Spherical
Description	CH. COVER
Distance "FROM" to "TO"	0 mm.
Inside Diameter	1600 mm.
Element Thickness	60 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	71 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	191 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +N+IT
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm.
Yield Stress, Operating	242 N./mm ²
UCS-66 Chart Curve Designation	Impact Tested
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	10
Detail Type	Liquid
Detail ID	Liquid 10
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	1600 mm.
Liquid Density	0.000624 kg./cm ³

Element From Node	10
Detail Type	Insulation
Detail ID	Ins: 20
Dist. from "FROM" Node / Offset dist	-800 mm.
Height/Length of Insulation	800 mm.
Thickness of Insulation	70 mm.
Density	0.40000E-04 kg./cm ³

Element From Node	10
Detail Type	Nozzle
Detail ID	CH. IN NOZ A1
Dist. from "FROM" Node / Offset dist	495.2 mm.
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	None
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	843.5 Kg
Grade of Attached Flange	None
Nozzle Matl	SA-350 LF2 +IT



Element From Node	10
Detail Type	Nozzle
Detail ID	CH. OUT NOZ B1
Dist. from "FROM" Node / Offset dist	495.2 mm.
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	None
Layout Angle	-45.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	843.5 Kg
Grade of Attached Flange	None
Nozzle Matl	SA-350 LF2 +IT

Element From Node	10
Detail Type	Nozzle
Detail ID	MANHOLE M1
Dist. from "FROM" Node / Offset dist	546 mm.
Nozzle Diameter	20 in.
Nozzle Schedule	None
Nozzle Class	1500
Layout Angle	-160.017
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	2649.7 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-350 LF2 +IT

Element#	2/7
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	CH. COVER
Distance "FROM" to "TO"	50 mm.
Inside Diameter	1600 mm.
Element Thickness	140 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	140 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	191 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +N
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid 20
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	1600 mm.
Liquid Density	0.000624 kg./cm³

Element From Node	20
Detail Type	Insulation
Detail ID	Ins: 20
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	50 mm.
Thickness of Insulation	70 mm.
Density	0.40000E-04 kg./cm³

Element#	3/7
Element From Node	30
Element To Node	40
Element Type	Flange



Description	SH. FLG.	
Distance "FROM" to "TO"	200	mm.
Flange Inside Diameter	1600	mm.
Element Thickness	155	mm.
Internal Corrosion Allowance	1.5	mm.
Nominal Thickness	170	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	25	bars
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	bars
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-350 LF2 +IT	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	225	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm.
Yield Stress, Operating	229.57	N./mm ²
UCS-66 Chart Curve Designation	Impact Tested	
External Pressure Chart Name	CS-2	
UNS Number	K03011	
Class / Thickness / Grade	1 ::	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid 30	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	1600	mm.
Liquid Density	0.000624	kg./cm ³
Element From Node	30	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	215	mm.
Thickness of Insulation	70	mm.
Density	0.40000E-04	kg./cm ³

Element#	4/7	
Element From Node	40	
Element To Node	50	
Element Type	Cylinder	
Description	SHELL-1	
Distance "FROM" to "TO"	300	mm.
Inside Diameter	1600	mm.
Element Thickness	28	mm.
Internal Corrosion Allowance	1.5	mm.
Nominal Thickness	28	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	25	bars
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	bars
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70 +Q	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	234	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm.
Yield Stress, Operating	262	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	



Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	40
Detail Type	Saddle
Detail ID	FIXED SADDLE
Dist. from "FROM" Node / Offset dist	150 mm.
Width of Saddle	200 mm.
Height of Saddle at Bottom	1350 mm.
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm.
Width of Wear Plate	240 mm.
Thickness of Wear Plate	16 mm.
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.6000002
Moment Factor	3.0
Dimension E at base (optional)	0 mm.
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm.
Element From Node	40
Detail Type	Liquid
Detail ID	Liquid: 40
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	1600 mm.
Liquid Density	0.000624 kg./cm ³
Element From Node	40
Detail Type	Insulation
Detail ID	Ins: 20
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	300 mm.
Thickness of Insulation	70 mm.
Density	0.40000E-04 kg./cm ³

Element#	5/7
Element From Node	50
Element To Node	60
Element Type	Conical
Description	SHELL-2
Distance "FROM" to "TO"	1905 mm.
Inside Diameter	1600 mm.
Element Thickness	32 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	32 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	25 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	1.0342 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +Q+IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Cone Diameter at "To" End	2700 mm.
Design Length of Cone	1905 mm.
Half Apex Angle of Cone	30.0 degrees
Toriconical (Y/N)	N
Weld is pre-Heated	No

Element From Node	50
Detail Type	Liquid
Detail ID	Liquid 50
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	1600 mm.



Liquid Density	0.000624	kg./cm ³
Element From Node	50	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	1905	mm.
Thickness of Insulation	70	mm.
Density	0.40000E-04	kg./cm ³
Element From Node	50	
Detail Type	Nozzle	
Detail ID	LIQ OUT NOZ B3	
Dist. from "FROM" Node / Offset dist	250	mm.
Nozzle Diameter	4	in.
Nozzle Schedule	160	
Nozzle Class	300	
Layout Angle	-90.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	38.321	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	

Element#	6/7	
Element From Node	60	
Element To Node	70	
Element Type	Cylinder	
Description	K-SHELL	
Distance "FROM" to "TO"	4100	mm.
Inside Diameter	2700	mm.
Element Thickness	28	mm.
Internal Corrosion Allowance	1.5	mm.
Nominal Thickness	28	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	25	bars
Design Temperature Internal Pressure	85	°C
Design External Pressure	1.0342	bars
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70 +Q+IT	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	60	
Detail Type	Saddle	
Detail ID	SLIDING SADDLE	
Dist. from "FROM" Node / Offset dist	500	mm.
Width of Saddle	300	mm.
Height of Saddle at Bottom	1900	mm.
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm.
Width of Wear Plate	400	mm.
Thickness of Wear Plate	16	mm.
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.1	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm.
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm.

Element From Node	60	
Detail Type	Saddle	
Detail ID	SLIDING SADDLE	
Dist. from "FROM" Node / Offset dist	3600	mm.
Width of Saddle	300	mm.
Height of Saddle at Bottom	1900	mm.



Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm.
Width of Wear Plate	400	mm.
Thickness of Wear Plate	16	mm.
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.1	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm.
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm.
Element From Node	60	
Detail Type	Liquid	
Detail ID	Liquid 60	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	1600	mm.
Liquid Density	0.000624	kg./cm ³
Element From Node	60	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	4100	mm.
Thickness of Insulation	70	mm.
Density	0.40000E-04	kg./cm ³
Element From Node	60	
Detail Type	Nozzle	
Detail ID	SS IN NOZ. A2	
Dist. from "FROM" Node / Offset dist	3750	mm.
Nozzle Diameter	12	in.
Nozzle Schedule	120	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	206.71	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	
Element From Node	60	
Detail Type	Nozzle	
Detail ID	SS OUT NOZ. B2	
Dist. from "FROM" Node / Offset dist	1700	mm.
Nozzle Diameter	18	in.
Nozzle Schedule	120	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	412.49	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	
Element From Node	60	
Detail Type	Nozzle	
Detail ID	VENT NOZ V1	
Dist. from "FROM" Node / Offset dist	400	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	160	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	10.683	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	
Element From Node	60	
Detail Type	Nozzle	
Detail ID	PR. IN P1	
Dist. from "FROM" Node / Offset dist	150	mm.



Nozzle Diameter 2 in.
 Nozzle Schedule 160
 Nozzle Class 300
 Layout Angle 90.0
 Blind Flange (Y/N) Y
 Weight of Nozzle (Used if > 0) 10.683 Kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 60
 Detail Type Nozzle
 Detail ID PR. IN NOZ. P2
 Dist. from "FROM" Node / Offset dist 3250 mm.
 Nozzle Diameter 2 in.
 Nozzle Schedule 160
 Nozzle Class 300
 Layout Angle 90.0
 Blind Flange (Y/N) Y
 Weight of Nozzle (Used if > 0) 10.683 Kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 60
 Detail Type Nozzle
 Detail ID SAFETY VALVE S1
 Dist. from "FROM" Node / Offset dist 3000 mm.
 Nozzle Diameter 2 in.
 Nozzle Schedule 160
 Nozzle Class 300
 Layout Angle 90.0
 Blind Flange (Y/N) Y
 Weight of Nozzle (Used if > 0) 10.683 Kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 60
 Detail Type Nozzle
 Detail ID LEVEL GUAGE L1
 Dist. from "FROM" Node / Offset dist 150 mm.
 Nozzle Diameter 2 in.
 Nozzle Schedule 160
 Nozzle Class 300
 Layout Angle 30.4806
 Blind Flange (Y/N) Y
 Weight of Nozzle (Used if > 0) 11.265 Kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 60
 Detail Type Nozzle
 Detail ID LEVEL GUAGE L2
 Dist. from "FROM" Node / Offset dist 150 mm.
 Nozzle Diameter 2 in.
 Nozzle Schedule 160
 Nozzle Class 300
 Layout Angle -30.4806
 Blind Flange (Y/N) Y
 Weight of Nozzle (Used if > 0) 11.265 Kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 60
 Detail Type Nozzle
 Detail ID LEVEL GUAGE L3
 Dist. from "FROM" Node / Offset dist 800 mm.
 Nozzle Diameter 2 in.
 Nozzle Schedule 160
 Nozzle Class 300
 Layout Angle 30.4806
 Blind Flange (Y/N) Y



Weight of Nozzle (Used if > 0) 11.265 Kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 60
 Detail Type Nozzle
 Detail ID LEVEL GUAGE L4
 Dist. from "FROM" Node / Offset dist 800 mm.
 Nozzle Diameter 2 in.
 Nozzle Schedule 160
 Nozzle Class 300
 Layout Angle -30.4806
 Blind Flange (Y/N) Y
 Weight of Nozzle (Used if > 0) 11.265 Kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 60
 Detail Type Nozzle
 Detail ID DRAIN NOZ. D2
 Dist. from "FROM" Node / Offset dist 1650 mm.
 Nozzle Diameter 2 in.
 Nozzle Schedule 160
 Nozzle Class 300
 Layout Angle -90.0
 Blind Flange (Y/N) Y
 Weight of Nozzle (Used if > 0) 11.265 Kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 60
 Detail Type Weight
 Detail ID DATASHEET WEIGH
 Dist. from "FROM" Node / Offset dist 0 mm.
 Miscellaneous Weight 6210 Kgf
 Offset from Element Centerline 0 mm.

Element# 7/7
 Element From Node 70
 Element To Node 80
 Element Type Elliptical
 Description SH. COVER
 Distance "FROM" to "TO" 50.8 mm.
 Inside Diameter 2700 mm.
 Element Thickness 27 mm.
 Internal Corrosion Allowance 1.5 mm.
 Nominal Thickness 32 mm.
 External Corrosion Allowance 0 mm.
 Design Internal Pressure 25 bars
 Design Temperature Internal Pressure 85 °C
 Design External Pressure 1.0342 bars
 Design Temperature External Pressure 85 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70 +Q+IT
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Elliptical Head Factor 2.0
 Weld is pre-Heated No

Element From Node 70
 Detail Type Liquid
 Detail ID Liquid 70
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Liquid 1600 mm.
 Liquid Density 0.000624 kg./cm³

Element From Node 70
 Detail Type Insulation



Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	725.8	mm.
Thickness of Insulation	70	mm.
Density	0.40000E-04	kg./cm ³
Element From Node	70	
Detail Type	Nozzle	
Detail ID	TEMP IN. T1	
Dist. from "FROM" Node / Offset dist	700	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	160	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	11.265	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	

XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
CH. COVER		...	...	...	...
CH. COVER		50	...	50	...
SH. FLG.		653.175	...	200	...
SHELL-1		953.175	...	300	...
SHELL-2		2858.18	...	1905	...
K-SHELL		6958.17	...	4100	...
SH. COVER		7008.97	...	50.8	...

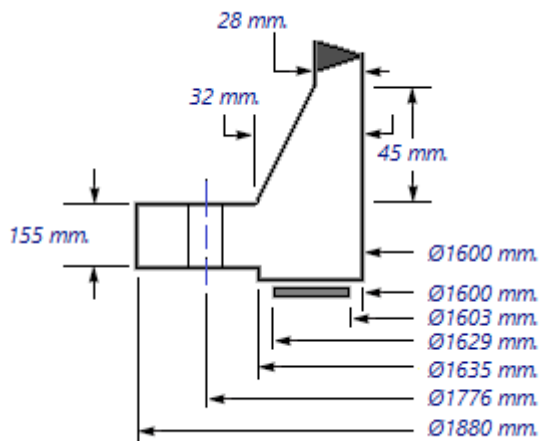
The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Flange Input Data Values **Description: SH. FLG** :

SH. FLG.

Flange Type	Integral Weld Neck		
Design Pressure	P	25.10	bars
Design Temperature		85	°C
Internal Corrosion Allowance	ci	1.5000	mm.
External Corrosion Allowance	ce	0.0000	mm.
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1600.000	mm.
Flange Outside Diameter	A	1880.000	mm.
Flange Thickness	t	155.0000	mm.
Thickness of Hub at Small End	go	28.0000	mm.
Thickness of Hub at Large End	gl	32.0000	mm.
Length of Hub	h	45.0000	mm.
Flange Material	SA-350 LF2 +IT		
Flange Material UNS number	K03011		
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material	SA-320 L7		
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	1776.000	mm.
Nominal Bolt Diameter	a	36.0000	mm.
Type of Threads	TEMA Metric		
Number of Bolts	48		
Flange Face Outside Diameter	Fod	1635.000	mm.
Flange Face Inside Diameter	Fid	1600.000	mm.
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	1629.000	mm.
Gasket Inside Diameter	Gi	1603.000	mm.
Gasket Factor	m	3.7500	
Gasket Design Seating Stress	y	62.06	N./mm ²
Gasket Thickness	tg	3.1750	mm.
Raised face depth	Fd	0.0000	mm.



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Hub Small End Required Thickness due to Internal Pressure:
 $= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= (25.1 \cdot (1600.0/2 + 1.5)) / (138.0 \cdot 1.0 - 0.6 \cdot 25.1) + Ca$



= 16.2388 mm.

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 26.5) / (801.5 + 0.6 \cdot 26.5)$$

$$= 44.736 \text{ bars}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 28.0) / (800.0 + 0.6 \cdot 28.0)$$

$$= 47.303 \text{ bars}$$

Corroded Flange Thickness, $t_c = T - c_i$	153.500 mm.
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	1603.000 mm.
Corroded Large Hub, $g1_{cor} = g1 - c_i$	30.500 mm.
Corroded Small Hub, $g0_{cor} = g0 - c_i$	26.500 mm.
Code R Dimension, $R = ((C - B_{cor}) / 2) - g1_{cor}$	56.000 mm.
Gasket Contact Width, $N = (G_o - G_i) / 2$	13.000 mm.
Basic Gasket Width, $b_o = N / 2$	6.500 mm.
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	6.425 mm.
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	1616.151 mm.

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi / 4 \cdot G^2 \cdot P$$

$$= \pi / 4 \cdot 1616.1509^2 \cdot 25.098$$

$$= 525036.688 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 1616.1509 \cdot 3.75 \cdot 25.1$$

$$= 62614.121 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi / 4 \cdot B_{cor}^2 \cdot P$$

$$= \pi / 4 \cdot 1603.0^2 \cdot 25.098$$

$$= 516526.906 \text{ Kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 525037 - 516527$$

$$= 8509.819 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_o,1 + W_o,2 + H'p, 0)$$

$$= \max(525037 + 62614 + 0, 0)$$

$$= 587650.812 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 62.06 \cdot 6.4246 \cdot \pi \cdot 1616.151 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 206406.406 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((587651 + 0 + 4 \cdot 0 / 1616.15) / 172, 206406 / 172)$$

$$= 335.064 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 36.0 + 6 \cdot 153.5 / (3.75 + 0.5)$$

$$= 288.706 \text{ mm.}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1776.0 \cdot \sin(\pi / 48)$$

$$= 116.156 \text{ mm.}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{\frac{Bs}{2a + t}}, 1)$$

$$= \max(\sqrt{\frac{116.156}{2 \cdot 36.0 + 153.5}}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	335.064	354.247	
Radial Distance between Hub and Bolts:	47.250	56.000	
Radial Distance between Bolts and Edge:	37.500	52.000	
Circ. Spacing between the Bolts:	82.600	116.156	288.706

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 354.247 \cdot 172.0 / (62.06 \cdot \pi(1629.0 + 1603.0))$$

$$= 9.670 \text{ mm.}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 354.2472$$

$$= 621295.69 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 587651 - 525037$$

$$= 62614.09 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1776.0 - 1616.1509) / 2$$

$$= 79.9246 \text{ mm.}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((1776.0 - 1603.0) / 2 + 79.9246)$$

$$= 83.2123 \text{ mm.}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1776.0 - 1603.0 - 30.5) / 2$$

$$= 71.2500 \text{ mm.}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	516527.	71.2500	1.0000	36803.
Face Pressure, Mt	8510.	83.2123	1.0000	708.
Gasket Load, Mg	62614.	79.9246	1.0000	5005.
Gasket Seating, Matm	621296.	79.9246	1.0000	49658.

Total Moment for Operation, Mo 42516. Kg-m.
 Total Moment for Gasket seating, Mg 49658. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 206.106 mm.
 Hub Ratio, h/h0 = HL / H0 0.218
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.151

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.910
 Factor V per 4.16.5 0.484



Factor f per 4.16.5

1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 1880.0 / 1603.0$$

$$= 1.173$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.173^2 \cdot \log_{10}(1.173))) / (1.173^2 - 1)) / (1.173 - 1)$$

$$= 12.258$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.173^2(1 + 8.55246 \cdot \log_{10}(1.173)) - 1) / ((1.0472 + 1.9448 \cdot 1.173^2) \cdot (1.173 - 1))$$

$$= 1.850$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.173^2(1 + 8.55246 \cdot \log_{10}(1.173)) - 1) / (1.36136(1.173^2 - 1) \cdot (1.173 - 1))$$

$$= 13.471$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.173^2 + 1) / (1.173^2 - 1)$$

$$= 6.327$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 13.471 \cdot 26.5^2 \cdot 206.106 / 0.484$$

$$= 4025616.250 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.91 / 206.106$$

$$= 0.004 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / (T + t^3/d)$$

$$= (153.5 \cdot 0.004 + 1) / (1.85 + 153.5^3 / 4025616)$$

$$= 1.805$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1.0 \cdot 42516 / 1603.0) / (1.8053 \cdot 30.5^2)$$

$$= 154.87 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1.0 \cdot 49658 / 1603.0) / (1.8053 \cdot 30.5^2)$$

$$= 180.89 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (1.9034 \cdot 42516 / 1603.0) / (1.8053 \cdot 153.5^2)$$

$$= 11.64 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (1.9034 \cdot 49658 / 1603.0) / (1.8053 \cdot 153.5^2)$$

$$= 13.59 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (12.2582 \cdot 42516 / (153.5^2 \cdot 1603.0)) - 6.3268 \cdot 12$$

$$= 61.68 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRA$$

$$= (12.2582 \cdot 49658 / (153.5^2 \cdot 1603.0)) - 6.3268 \cdot 14$$

$$= 72.04 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (155 + \max(12, 62)) / 2$$

$$= 108.28 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHA + \max(SRa, STa)) / 2$$

$$= (181 + \max(14, 72)) / 2$$

$$= 126.47 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 587651 / 354.2472$$

$$= 162.68 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (206406 / 354.2472)$$

$$= 57.14 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	154.87	207.00	180.89 207.00
Radial Flange	11.64	138.00	13.59 138.00
Tangential Flange	61.68	138.00	72.04 138.00
Maximum Average	108.28	138.00	126.47 138.00
Bolting	162.68	172.00	57.14 172.00

Minimum Required Flange Thickness 145.161 mm.

Estimated M.A.W.P. (Operating) 27 bars

Estimated Finished Weight of Flange at given Thk. 972.9 kg.

Estimated Unfinished Weight of Forging at given Thk 1186.3 kg.

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 49657.8 / 1.0 \cdot 9806.637 \cdot 0.484 / (1.62 \cdot 202713 \cdot 26.5^2 \cdot 206.106 \cdot 0.3)$$

$$= 0.862 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 42515.9 / 1.0 \cdot 9806.637 \cdot 0.484 / (1.62 \cdot 199054 \cdot 26.5^2 \cdot 206.106 \cdot 0.3)$$

$$= 0.752 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 49657.8 / 1.0 \cdot 9806.637 \cdot 0.484 / (1.805 \cdot 202713 \cdot 26.5^2 \cdot 206.106 \cdot 0.3)$$

$$= 0.773 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 42515.9 / 1.0 \cdot 9806.637 \cdot 0.484 / (1.805 \cdot 199054 \cdot 26.5^2 \cdot 206.106 \cdot 0.3)$$

$$= 0.674 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	149.03	207.00	172.38 207.00
Radial Flange	12.00	138.00	13.88 138.00
Tangential Flange	57.68	138.00	66.71 138.00
Maximum Average	103.35	138.00	119.54 138.00
Bolting	162.68	172.00	57.14 172.00

Minimum Req'd Thickness, ambient, no Corrosion: 137.846 mm.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
CH. COVER		191.1	71	1.5	1600	138	1
CH. COVER		191.1	140	1.5	1600	138	2
SH. FLG.		25.098	170	1.5	1600	138	3
SHELL-1		25.098	28	1.5	1600	138	4
SHELL-2		25.098	32	1.5	2700	138	5
K-SHELL		25.098	28	1.5	2700	138	6
SH. COVER		25.098	32	1.5	2700	138	7

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
CH. COVER		191	198.435	203.925	60	57.7784	1
CH. COVER		191	215.949	218.535	140	122.557	2
SH. FLG.		25	26.4369	26.535	155	145.161	3
SHELL-1		25	44.6379	47.303	28	16.2389	4
SHELL-2		25	26.5534	27.9817	32	30.2027	5
K-SHELL		25	26.6439	28.2682	28	26.353	6
SH. COVER		25	25.9269	27.5427	27	26.0902	7

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

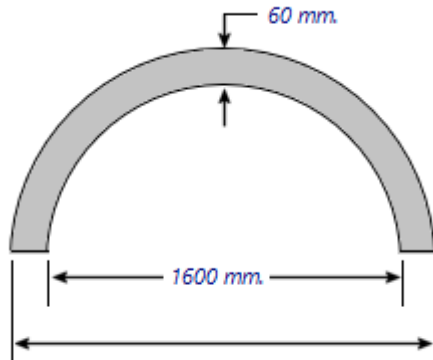
Shell Side MAWP = 26 bars
 Shell Side MAPnc = 27 bars
 Channel Side MAWP = 194 bars
 Channel Side MAPnc = 204 bars

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: CH. COVER

Spherical Head Design Information:					
Design Pressure	P	191.098 bars	Int. Design Temperature	85 °C	
Material		SA-516 70	Ext. Design Temperature	85 °C	
Length		0.000 mm.	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1600.000 mm.	Actual Stress	132.8 N./mm ²	
Spec. Min. Thk t _e , or s _a		60.0000 mm.	Spec. Nominal Thk.	71.000 mm.	
Surface Area		47666.8 cm ²	SG of Contents	0.625	
Element Volume		1072522.5 cm ³	Weight of Contents	669.9 Kg _f	
Empty Weight		2415.0 Kg _f	Operating Weight	3084.9 Kg _f	
			Comp. All. Ope. Stress	147.3 N./mm ²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 20.000 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 10.000 mm.

Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:
 $= (P \cdot R) / (2 \cdot S \cdot E - 0.2 \cdot P)$ per UG-27 (d)
 $= (191.098 \cdot 801.5) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 191.098)$
 $= 56.2784 + 1.5000 = 57.7784 \text{ mm.}$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.098 bars
 $= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t)$ per UG-27 (d)
 $= (2 \cdot 138.0 \cdot 1.0 \cdot 58.5) / (801.5 + 0.2 \cdot 58.5)$
 $= 199 - 0.1 = 198 \text{ bars}$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t)$ per UG-27 (d)
 $= (2 \cdot 138.0 \cdot 1.0 \cdot 60.0) / (800.0 + 0.2 \cdot 60.0)$
 $= 204 \text{ bars}$



Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned} &= (P \cdot (R + 0.2 \cdot t)) / (2 \cdot E \cdot t) \\ &= (191.098 \cdot (801.5 + 0.2 \cdot 58.5)) / (2 \cdot 1.0 \cdot 58.5) \\ &= 132.829 \text{ N./mm}^2 \end{aligned}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 6.373 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Spherical Portion:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 20.000 mm.

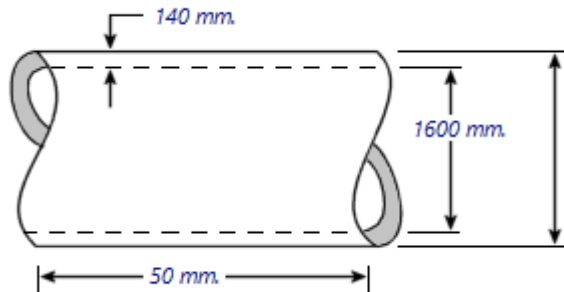
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 10.000 mm.

Material treatment options: +N+IT

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Internal Pressure Calculation Results:

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Internal Pressure Results for: CH. COVER

Cylindrical Shell Design Information:					
Design Pressure	P	191.098 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1600.000 mm.	Actual Stress		122.1 N./mm ²
Spec. Min. Thk t _e , or s _a		140.0000 mm.	Spec. Nominal Thk.		140.000 mm.
Surface Area		2953.1 cm ²	SG of Contents		0.625
Element Volume		100549.0 cm ³	Weight of Contents		62.8 Kg
Empty Weight		296.6 Kg	Operating Weight		359.4 Kg
			Comp. All. Ope. Stress		147.3 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (191.098 \cdot 801.5) / (138.0 \cdot 1.0 - 0.6 \cdot 191.098)$$

$$= 121.0566 + 1.5000 = 122.5566 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.098 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 138.5) / (801.5 + 0.6 \cdot 138.5)$$

$$= 216 - 0.1 = 216 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 140.0) / (800.0 + 0.6 \cdot 140.0)$$

$$= 219 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (191.098 \cdot (801.5 + 0.6 \cdot 138.5)) / (1.0 \cdot 138.5)$$

$$= 122.062 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 8.046 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

Minimum Design Metal Temperature Results:

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm.. This component must be impact tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

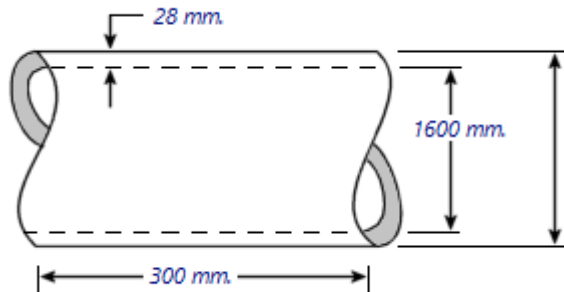
MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	193.907 bars
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-35 °C
Governing Thk.	140.000 mm.	Basic MDMT	-17 °C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-17 °C
		Joint Efficiency E*	100 %

Material treatment options: +N

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: SHELL-1

Cylindrical Shell Design Information:					
Design Pressure	P	25.098 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		300.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1600.000 mm.	Actual Stress		77.4 N./mm ²
Spec. Min. Thk t _e , or sa		28.0000 mm.	Spec. Nominal Thk.		28.000 mm.
Surface Area		15607.4 cm ²	SG of Contents		0.625
Element Volume		313506.7 cm ³	Weight of Contents		195.8 Kg
Empty Weight		333.0 Kg	Operating Weight		528.8 Kg
			Comp. All. Ope. Stress		139.8 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +Q

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (25.098 \cdot 801.5) / (138.0 \cdot 1.0 - 0.6 \cdot 25.098)$$

$$= 14.7389 + 1.5000 = 16.2389 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.098 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 26.5) / (801.5 + 0.6 \cdot 26.5)$$

$$= 45 - 0.1 = 45 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 28.0) / (800.0 + 0.6 \cdot 28.0)$$

$$= 47 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (25.098 \cdot (801.5 + 0.6 \cdot 26.5)) / (1.0 \cdot 26.5)$$

$$= 77.420 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.720 %

Minimum Design Metal Temperature Results:



Govrn. thk, tg = 28.0, tr = 15.231, c = 1.5 mm., $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (tg - c) = 0.575$, Temp. Reduction = 25 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 2 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -22 °C

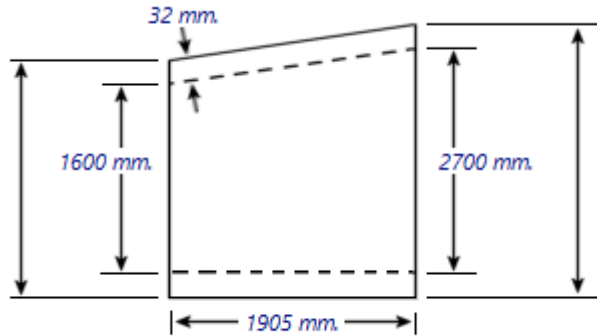
MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	25.927 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-35 °C
Governing Thk.	28.000 mm.	Basic MDMT	2 °C
Stress/Thk. Ratio	0.575	Temperature Difference	25 °C
UCS-66(c) red.	No	Computed Min. Temp.	-22 °C
		Joint Efficiency E^*	100 %

Material treatment options: +Q

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: SHELL-2

Conical Head/Section Design Information:					
Design Pressure	P	25.098 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		1905.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	2700.000 mm.	Actual Stress		130.0 N./mm ²
Spec. Min. Thk t _e , or sa		32.0000 mm.	Spec. Nominal Thk.		32.000 mm.
Surface Area		138530.6 cm ²	SG of Contents		0.625
Element Volume		5228091.0 cm ³	Weight of Contents		2101.5 Kgf
Empty Weight		3748.3 Kgf	Operating Weight		5849.8 Kgf
			Comp. All. Ope. Stress		144.0 N./mm ²
			To End Diameter		1600.000 mm.
Line of Support	Both Ends				

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +Q+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D) / (2 \cdot \cos(a) \cdot (S \cdot E - 0.6 \cdot P))$$
 per Appendix 1-4 (e)

$$= (25.098 \cdot 2703.4641) / (2 \cdot 0.866 \cdot (138.0 \cdot 1.0 - 0.6 \cdot 25.098))$$

$$= 28.7027 + 1.5000 = 30.2027 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.098 bars

$$= (2 \cdot S \cdot E \cdot \cos(a)) / (D + 1.2 \cdot t \cdot \cos(a))$$
 per App 1-4(e)

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 30.5 \cdot 0.866) / (2703.464 + 1.2 \cdot 30.5 \cdot 0.866)$$

$$= 27 - 0.1 = 27 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot \cos(a)) / (D + 1.2 \cdot t \cdot \cos(a))$$
 per App 1-4(e)

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 32.0 \cdot 0.866) / (2700.0 + 1.2 \cdot 32.0 \cdot 0.866)$$

$$= 28 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (D + 1.2 \cdot t \cdot \cos(a))) / (2 \cdot E \cdot t \cdot \cos(a))$$

$$= (25.098 \cdot (2703.4641 + 1.2 \cdot 30.5 \cdot 0.866)) / (2 \cdot 1.0 \cdot 30.5 \cdot 0.866)$$

$$= 129.954 \text{ N./mm}^2$$



% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.955 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +Q+IT

EXTERNAL PRESSURE RESULTS for: SHELL-2

Cone From 50 to 60 Ext. Chart: CS-2 at 85 °C

Elastic Modulus from Chart: CS-2 at 85 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
26.414	2773.90	1525.93	105.02	0.5501	0.0023856	107.54

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 107.5402) / (3 \cdot 105.0172) = 13.6529 \text{ bars}$$

Note:

The cone thickness used in the calculation has been modified per UG-33(f), $t_e = t \cdot \cos(\alpha)$.

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
6.963	2773.90	1525.93	398.38	0.5501	0.0003091	30.90

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 30.9025) / (3 \cdot 398.3816) = 1.0342 \text{ bars}$$

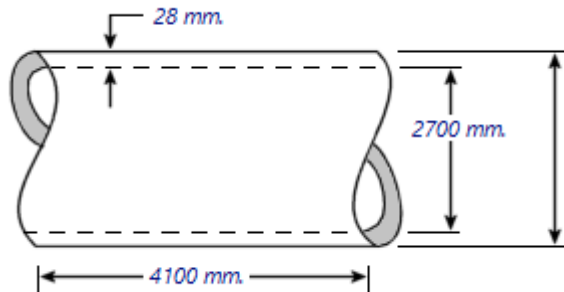
Note:

The cone thickness used in the calculation has been modified per UG-33(f), $t_e = t \cdot \cos(\alpha)$.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: K-SHELL

Cylindrical Shell Design Information:					
Design Pressure	P	25.098 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		4100.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	2700.000 mm.	Actual Stress		129.5 N./mm ²
Spec. Min. Thk t _e , or s _a		28.0000 mm.	Spec. Nominal Thk.		28.000 mm.
Surface Area		354987.5 cm ²	SG of Contents		0.625
Element Volume		19551760.0 cm ³	Weight of Contents		12212.4 Kg
Empty Weight		7625.5 Kg	Operating Weight		19837.9 Kg
			Comp. All. Ope. Stress		129.2 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +Q+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (25.098 \cdot 1351.5) / (138.0 \cdot 1.0 - 0.6 \cdot 25.098)$$

$$= 24.8530 + 1.5000 = 26.3530 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.098 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 26.5) / (1351.5 + 0.6 \cdot 26.5)$$

$$= 27 - 0.1 = 27 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 28.0) / (1350.0 + 0.6 \cdot 28.0)$$

$$= 28 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (25.098 \cdot (1351.5 + 0.6 \cdot 26.5)) / (1.0 \cdot 26.5)$$

$$= 129.514 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.026 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +Q+IT

EXTERNAL PRESSURE RESULTS for: K-SHELL

Cylindrical Shell From 60 to 70 Ext. Chart: CS-2 at 85 °C

Elastic Modulus from Chart: CS-2 at 85 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
26.500	2756.00	4375.80	104.00	1.5877	0.0007773	77.18

$$MAEP = (4*B)/(3*(Do/t)) = (4*77.1761)/(3*104.0) = 9.8938 \text{ bars}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.722	2756.00	4375.80	257.04	1.5877	0.0001994	19.94

$$MAEP = (4*B)/(3*(Do/t)) = (4*19.9399)/(3*257.0444) = 1.0343 \text{ bars}$$

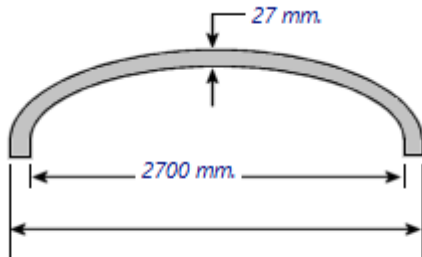
Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
26.500	2756.00	98548.04	104.00	35.7576	0.0001027	10.27

$$MAEP = (4*B)/(3*(Do/t)) = (4*10.2689)/(3*104.0) = 1.3165 \text{ bars}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: SH. COVER

Elliptical Head Design Information:					
Design Pressure	P	25.098 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		50.800 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	2700.000 mm.	Actual Stress		133.1 N./mm ²
Spec. Min. Thk t _e , or sa		27.0000 mm.	Spec. Nominal Thk.		32.000 mm.
Surface Area		87224.4 cm ²	SG of Contents		0.625
Element Volume		2867870.5 cm ³	Weight of Contents		1141.6 Kg _f
Empty Weight		2323.7 Kg _f	Operating Weight		3465.3 Kg _f
			Comp. All. Ope. Stress		129.2 N./mm ²
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 33.750 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 16.875 mm.

Material treatment options: +Q+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (25.098 \cdot 2703.0 \cdot 0.999) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 25.098)$$

$$= 24.5902 + 1.5000 = 26.0902 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.098 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 25.5) / (0.999 \cdot 2703.0 + 0.2 \cdot 25.5)$$

$$= 26 - 0.1 = 26 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 27.0) / (1.0 \cdot 2700.0 + 0.2 \cdot 27.0)$$

$$= 28 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (25.098 \cdot (0.999 \cdot 2703.0 + 0.2 \cdot 25.5)) / (2 \cdot 1.0 \cdot 25.5)$$

= 133.082 N./mm²

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (25.098 \cdot 1351.5) / (138.0 \cdot 1.0 - 0.6 \cdot 25.098) + 1.5$$

$$= 26.353 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.098 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 30.5) / (1351.5 + 0.6 \cdot 30.5)$$

$$= 31 - 0.1 = 31 \text{ bars}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (2703.0 / (2 \cdot 676.5)))^2 / 6$$

$$= 0.998523$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.053 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 33.750 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 16.875 mm.

Material treatment options: +Q+IT

EXTERNAL PRESSURE RESULTS for: SH. COVER

Elliptical Head From 70 to 80 Ext. Chart: CS-2 at 85 °C

Elastic Modulus from Chart: CS-2 at 85 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
25.500	2754.00	108.00	0.0012860	92.66

$$MAEP = B / (K_0 \cdot Do / t) = 92.6577 / (0.9 \cdot 108.0) = 9.5321 \text{ bars}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
7.131	2754.00	386.22	0.0003596	35.95

$$MAEP = B / (K_0 \cdot Do / t) = 35.9531 / (0.9 \cdot 386.2188) = 1.0343 \text{ bars}$$

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	33.705 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	32.500 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	34.231 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	28.520 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	35.750 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	35.750 bars
App 27-4	= M.A.W.P.	25.927 bars

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	252.080 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	248.300 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	264.837 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	213.298 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	273.130 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	273.130 bars
App 27-4	= M.A.W.P.	193.907 bars

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
CH. COVER	175.3	234.0	0.749	252.24
CH. COVER	161.1	234.0	0.689	252.24
SHELL-1	104.5	234.0	0.446	33.86
SHELL-2	175.9	234.0	0.752	33.97
K-SHELL	175.3	234.0	0.749	33.97
SH. COVER	180.1	234.0	0.770	33.97

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
CH. IN NOZ A1	Nozzle	138.00	138.00	1.000
CH. OUT NOZ B1	Nozzle	138.00	138.00	1.000
MANHOLE M1	Nozzle	138.00	138.00	1.000
LIQ OUT NOZ B3	Nozzle	138.00	138.00	1.000
SS IN NOZ. A2	Nozzle	138.00	138.00	1.000
SS OUT NOZ. B2	Nozzle	138.00	138.00	1.000
VENT NOZ V1	Nozzle	138.00	138.00	1.000
PR. IN P1	Nozzle	138.00	138.00	1.000
PR. IN NOZ. P2	Nozzle	138.00	138.00	1.000
SAFETY VALVE S1	Nozzle	138.00	138.00	1.000
LEVEL GUAGE L1	Nozzle	138.00	138.00	1.000
LEVEL GUAGE L2	Nozzle	138.00	138.00	1.000
LEVEL GUAGE L3	Nozzle	138.00	138.00	1.000
LEVEL GUAGE L4	Nozzle	138.00	138.00	1.000
DRAIN NOZ. D2	Nozzle	138.00	138.00	1.000
TEMP IN. T1	Nozzle	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
CH. COVER	138.00	138.00	1.000
CH. COVER	138.00	138.00	1.000



SH. FLG.	138.00	138.00	1.000
SHELL-1	138.00	138.00	1.000
SHELL-2	138.00	138.00	1.000
K-SHELL	138.00	138.00	1.000
SH. COVER	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	118.00	118.00	1.000
Tubesheet Material	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
CH. IN NOZ A1	174.44	179.40	0.972
CH. OUT NOZ B1	174.44	179.40	0.972
MANHOLE M1	148.02	179.40	0.825
LIQ OUT NOZ B3	14.84	179.40	0.083
SS IN NOZ. A2	21.66	179.40	0.121
SS OUT NOZ. B2	21.87	179.40	0.122
VENT NOZ V1	12.80	179.40	0.071
PR. IN P1	12.80	179.40	0.071
PR. IN NOZ. P2	12.80	179.40	0.071
SAFETY VALVE S1	12.80	179.40	0.071
LEVEL GUAGE L1	12.80	179.40	0.071
LEVEL GUAGE L2	12.80	179.40	0.071
LEVEL GUAGE L3	12.80	179.40	0.071
LEVEL GUAGE L4	12.80	179.40	0.071
DRAIN NOZ. D2	12.80	179.40	0.071
TEMP IN. T1	12.80	179.40	0.071

Elements Suitable for Test Pressure.

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Input Echo, Tubesheet Number 1, Description: U-TS

Shell Data:

Main Shell Description: SHELL-1

Shell Test Pressure		33.70 bars
Shell Thickness	ts	28.0000 mm.
Shell Internal Corrosion Allowance	cas	0.0000 mm.
Shell External Corrosion Allowance	casext	0.0000 mm.
Inside Diameter of Shell	Ds	1600.000 mm.
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: CH. COVER

Channel Type:		Cylinder
Channel Test Pressure		252.08 bars
Channel Thickness	tc	140.0000 mm.
Channel Internal Corrosion Allowance	cac	0.0000 mm.
Channel External Corrosion Allowance	cacext	0.0000 mm.
Inside Diameter of Channel	Dc	1600.000 mm.
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	1906
Tube Wall Thickness	et	2.7700 mm.
Tube Outside Diameter	D	25.4000 mm.
Straight Tube Length (bet. inner tubsht faces) L		5200.00 mm.
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-334 6
Tube Material UNS Number		K03006
Is this a Welded Tube		No
Tube Material Specification used	Smls. & wld. tube	
Tube Allowable Stress at Temperature		118.00 N./mm ²
Tube Allowable Stress At Ambient		118.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	240.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm.
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.7700 mm.
Groove Weld Leg	ag	2.7700 mm.
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	770.800 mm.
Largest Center-to-Center Tube Distance	U1	76.2000 mm.
Length of Expanded Portion of Tube	ltx	0.0000 mm.
Tube-side pass partition groove depth	hg	0.0000 mm.

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral with Channel, Conf. e

Tubesheet Design Metal Temperature	T	21.11 °C
Tubesheet Material		SA-350 LF2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03011
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	400.0000 mm.
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm.
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm.
Tubesheet Outside Diameter	A	1880.000 mm.

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		Yes
Is the Tubesheet Flat on Tubeside?		Yes
Is the Tubesheet Flat on Shellside?		No
Step 1 Diameter on the Shellside	ds1	1632.00 mm.
Step 1 Depth on the Shellside	hs1	6.00 mm.
Step 2 Diameter on the Shellside	ds2	1597.00 mm.
Step 2 Depth on the Shellside	hs2	6.00 mm.
Calculated Tubesheet Diameter per Div 2, 4.18.4		1880.00 mm.

Note: Tubesheet diameter is now: 1880.000 mm. Div 2, 4.18.4.

Area of the Untubed Lanes	AL	1181.6 cm ²
Junction Stress Reduction option		Perform Plastic Calculation
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Shell
Flange Outside Diameter	A	1880.000 mm.
Flange Inside Diameter	B	1600.000 mm.
Flange Face Outside Diameter	Fod	1635.000 mm.
Flange Face Inside Diameter	Fid	1600.000 mm.
Gasket Outside Diameter	Go	1629.000 mm.
Gasket Inside Diameter	Gi	1603.000 mm.
Small end Hub thk.	g0	28.0000 mm.
Large end Hub thk.	g1	32.0000 mm.
Gasket Factor,	m	3.75
Gasket Design Seating Stress	y	62.06 N./mm ²
Flange Facing Sketch		Code Sketch 1a
Gasket Thickness	tg	3.1750 mm.
Full face Gasket Flange Option		Program Selects

Bolting Information:

Diameter of Bolt Circle	C	1776.000 mm.
Nominal Bolt Diameter	dB	36.0000 mm.
Thread Series		TEMA Metric Thread
Number of Bolts	n	48
Bolt Material		SA-320 L7
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm.

Note: The full bolt load Sa*Ab, was used for all load cases.

Tubesheet Integral with		Channel
Tubesheet Extended as Flange		Yes
Thickness of Extended Portion of Tubesheet	Tf	388.0000 mm.
Is Bolt Load Transferred to the Tubesheet		Yes

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.7583	25.40	5970.80	33.50	50.0000	0.0009803	84.69

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 84.6873) / (3 \cdot 33.4971) = 33.7073$ bars

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	2.4997 mm.
Allowable Internal Pressure at Corroded thickness	281.95 bars
Required Internal Design Pressure	252.08 bars
Allowable External Pressure at Corroded thickness	187.80 bars
Required External Design Pressure	33.70 bars
Required Thickness due to Shell Side pressure	0.7583 mm.

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

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Gasket Contact Width,	$N = (G_{oc} - G_{ic}) / 2$	13.000 mm.
Basic Gasket Width,	$b_0 = N / 2$	6.500 mm.
Effective Gasket Width,	$b = \sqrt{b_0} \cdot 2.5$	6.422 mm.
Gasket Reaction Diameter,	$G = G_o - 2.0 \cdot b$	1616.156 mm.

Warning: Available bolt area, 354.247 cm² for the flange is < required 449.953 cm²

Note that for the pressure test case, ASME VIII-2 paragraph 4.18.13 (c)(9) does not require that the bolt area requirement be met.

Results for ASME U-tube Tubesheet Calculations for Configuration e:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$= 2 \cdot r_o + dt$
 $= 2 \cdot 770.8 + 25.4$
 $= 1567.000$ mm.

Basic ligament efficiency for shear [μ]:

$= (p - dt) / p$
 $= (31.75 - 25.4) / 31.75$
 $= 0.200$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ψ]:

$= G_s / D_o$ (Configurations d, e, f)
 $= 1616.1555 / 1567.0$
 $= 1.0314$

Ratio [ϕ]:

$= D_c / D_o$ (Configurations a, e, f)
 $= 1600.0 / 1567.0$
 $= 1.0211$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$= l_{tx} / h$
 $= 0 / 400.0$
 $= 0.000$ (must be $0 \leq \rho \leq 1$)

Effective tube hole Diameter [d^*]:

$= \max(dt - 2t_t (E_t/E) \cdot (S_t/S) \cdot (\rho); dt - 2t_t)$
 $= \max(25.4 - 2 \cdot 2.77 (0.20270E+09 / 2.0270E+09) \cdot$

$$\begin{aligned} & (117/138) \cdot (0); 25.4 - 2 \cdot 2.77 \\ & = 25.4000 \text{ mm.} \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned} & = p / \sqrt{(1 - 4 \cdot \min(A; 4 \cdot D_o \cdot p)) / (\pi \cdot D_o^2))} \\ & = 31.75 / \sqrt{(1 - 4 \cdot \min(1181.56; 4 \cdot 1567.0 \cdot 31.75)) / (\pi \cdot 1567.0^2))} \\ & = 32.7697 \text{ mm.} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} & = (p^* - d^*) / p^* \\ & = (32.7697 - 25.4) / 32.7697 \\ & = 0.2249 \end{aligned}$$

$h/p = 12.598425$; $\mu^* = 0.224894$; $E^* = 37562056 \text{ KPa}$.
 $E^*/E = 0.185307$; $\nu^* = 0.424937$

Note: As h/p (12.598) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:
 β_s , k_s , λ_s , δ_s and ω_s are all = 0

Channel coefficient [β_c]:

$$\begin{aligned} & = ((12(1 - \nu_c^2))^{1/4}) / \sqrt{(D_c + t_c) t_c)} \\ & = ((12(1 - 0.3^2))^{1/4}) / \sqrt{(1600.0 + 140.0) 140.0)} \\ & = 0.0037 \text{ 1/mm.} \end{aligned}$$

Channel coefficient [k_c]:

$$\begin{aligned} & = \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_s^2)) \\ & = 0.004 \cdot 2027012 \cdot 140.0^3 / (6(1 - 0.3^2)) \\ & = 3752017408.0000 \text{ bars} \cdot \text{mm.}^2 \end{aligned}$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:
 $CqDelta = CqTheta = CmTheta = CmDelta = 1$

Channel coefficient [λ_c]:

$$\begin{aligned} & = 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + \\ & \quad h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta) \\ & = 6(1600.0 + 140.0 / \cos(0)) \cdot 0.003752017408 / 400.0^3 \cdot (1.0 + \\ & \quad 400.0 \cdot 0.004 \cdot (1.0 + 1.0) / 2 + (400.0 \cdot 0.004)^2 / 2 \cdot 1.0) \\ & = 2177964.2 \text{ bars} \end{aligned}$$

Channel coefficient [δ_c]:

$$\begin{aligned} & = D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c))) \\ & = 1600.0^2 / (4 \cdot 0.2027012 \cdot 140.0) \cdot (1 - 1600.0 \cdot 0.3 / (2(1600.0 + 140.0))) \\ & = 0.19441 \text{E-01 mm./N./mm}^2 \end{aligned}$$

Channel coefficient [zetac]:
 $zetac = 0$

Parameter [ω_c]:

$$\begin{aligned} & = \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \\ & \quad \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)] \\ & = 1.021 \cdot 0.3752017408 [0.019441(1.0 + 0.004 \cdot 1.0 \cdot 400.0) - \\ & \quad 0(1.0/400.0 + 1.0 \cdot 0.004/2)] \\ & = 67848.2656 \text{ mm.}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$\begin{aligned} & = A / D_o \\ & = 1880.0 / 1567.0 \\ & = 1.1997 \end{aligned}$$

Coefficient [F]:

$$\begin{aligned} & = (1 - \nu^*) / E^* \cdot (\lambda_c + E \cdot \ln(K)) \\ & = (1 - 0.42) / 37562056 \cdot (2177964 + 0.2027012 \cdot \ln(1.2)) \\ & = 3.8995 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= Do^2/16 \cdot [(ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt]$$

$$= 1567.0^2/16 \cdot [(1.031 - 1) \cdot (1.031^2 + 1) \cdot 33.705 - (1.021 - 1) \cdot (1.021^2 + 1) \cdot 252.08]$$

$$= -1329228.0000 \text{ bars} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + (\omega c \cdot Pt) + W* \cdot (C - Gs) / (2 \cdot \pi \cdot Do)$$

$$= -1329228 + (17103168) + 621295 \cdot (1776.0 - 1616.156) / (2 \cdot \pi \cdot 1567.0)$$

$$= 16763064.0000 \text{ bars} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ((M*) - Do^2/32 \cdot F(Ps - Pt)) / (1 + F)$$

$$= ((16763064) - 1567.0^2/32 \cdot 3.9 (33.7 - 252.08)) / (1 + 3.9)$$

$$= 16758056.0000 \text{ bars} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + Do^2/64 \cdot (3 + v*) \cdot (Ps - Pt)$$

$$= 16758056 + 1567.0^2/64 \cdot (3 + 0.425) \cdot (33.7 - 252.08)$$

$$= -11937390.0000 \text{ bars} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(| Mp | ; | Mo |)$$

$$= \max(| 16758056 | ; | -0.1E+08 |)$$

$$= 16758056.0000 \text{ bars} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu* (h - hg')^2)$$

$$= 6 \cdot 16758056 / (0.2249 (400.0 - 0)^2)$$

$$= 279.4485 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot \beta T \cdot Sy$$

$$= 2 \cdot 0.95 \cdot 250.0$$

$$= 474.99 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu* (h - hg')^2)$$

$$= 6 \cdot 16428145 / (0.2249 (303.8198 - 0)^2)$$

$$= 474.8480 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + Cats + Catc$$

$$= 303.8198 + 0 + 0$$

$$= 303.8198 \text{ mm.}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(HreqB ; HreqS)$$

$$= \max(303.8198 ; 225.1442)$$

$$= 303.8198 \text{ mm.}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$= | Ps - Pt |$$

$$= | 33.7 - 252.08 |$$

$$= 218.375 \text{ bars}$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / Do \cdot 0.8 \cdot \beta T \cdot Sy$$

$$= 4 \cdot 0.2 \cdot 400.0 / 1567.0 \cdot 0.8 \cdot 0.95 \cdot 249.995$$

$$= 38.800 \text{ N./mm}^2$$

| Ps - Pt | is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness

for shear is for information only.

Minimum required thickness for Shear [HreqS]:
 $= 1/(4 \cdot \mu) \cdot (Do / (0.8 \cdot \beta T \cdot Sy)) \cdot |Ps - Pt| + Cats + Catc$
 $= 1/(4 \cdot 0.2) \cdot (1567.0/189.997) \cdot |33.7 - 252.08| + 0$
 $= 225.1442 \text{ mm.}$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Channel thickness of: 140.000 mm.
 Min. Channel len. adjacent to the tubesheet is: 851.915mm.

Tube Metal Area [At]:
 $= \pi (do - t)t$
 $= \pi (25.4 - 2.77)2.77$
 $= 1.969 \text{ cm}^2$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:
 $= \pi \cdot t (do - t)S$
 $= \pi \cdot 2.77 (25.4 - 2.77)227.996$
 $= 4578.387 \text{ Kgf}$

Groove Weld Strength [Fg]:
 $= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag)Sw \text{ (but not } > Ft)$
 $= 0.85 \cdot \pi \cdot 2.77 (25.4 + 0.67 \cdot 2.77)228.0$
 $= 4578.3872 \text{ Kgf}$

Weld Strength Factor [fw]:
 $= S / Sw \text{ where } Sw = \min(S \text{ and } St)$
 $= 227.996/227.996$
 $= 1.000$

Design Strength Ratio [fd]:
 $= 1 \text{ for full strength welds}$

Fillet Weld Strength [Ff]:
 $= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af)Sw \text{ (but not } > Ft)$
 $= 0.55 \cdot \pi \cdot 2.77 (25.4 + 0.67 \cdot 2.77)228.0$
 $= 3032.8518 \text{ Kgf}$

Minimum Length of Weld [ar]:
 $= 2 \cdot \sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t)fw \cdot fd) - 0.75 \cdot do}$
 $= 2 \cdot \sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.77 \cdot (25.4 - 2.77)1.0 \cdot 1.0) - 0.75 \cdot 25.4}$
 $= 3.372 \text{ mm.}$

Minimum Required Fillet Weld Leg [afr]: 1.6859 mm.
 Minimum Required Groove Weld Leg [agr]: 1.6859 mm.

Tube-Tubesheet Jt allowable, 4578.39 is $\geq$ tube strength 4578.39 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	279.4 $\leq$	475.0 N./mm ²	Ok
Tubesheet shear stress	106.9 $\leq$	190.0 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm.)	Required	Actual	P/F
Tubesheet Thickness :	303.820	400.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.686	2.770	Ok

Tube-Tubesheet Groove Weld Leg : 1.686 2.770 Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA		---- Tubesheet Stresses				Case Type	Pass/Fail
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd		
D1uc	322.764	55.333	309	475	123	190	Ps+Pt D1	Ok
D2uc	118.904	...	44	475	17	190	Ps+Pt D2	Ok
D3uc	303.820	...	279	475	107	190	Ps+Pt D3	Ok
Max:	322.7636	55.333	mm.	0.651		0.650	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Req'd. Thk.		Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
	Shell	Channel					
D1uc	...	...	...	...	...	370	Ok
D2uc	...	...	...	...	...	370	Ok
D3uc	...	...	...	...	...	370	Ok
Max	...	...		...		...	

Note:
 Because there was no net pressure differential across the tubesheet due to
 the specified pressures psd,min and ptd,min; load case D4 was not processed as
 the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm.)	Required	Actual	P/F
Tubesheet thickness :	322.764	400.000	Ok
Tubesheet thickness flanged extension :	55.333	388.000	Ok
Tube thickness :	2.500	2.770	Ok
Tube-tubesheet fillet weld leg :	1.686	2.770	Ok
Tube-tubesheet groove weld leg :	1.686	2.770	Ok

Given Shell Thickness: 28.0000 mm.
 Given Channel Thickness: 140.0000 mm.

Min Channel length of thk, (140.000) adj. to tubesheet: 851.915 mm.

Note: This is a full strength Tube to Tubesheet Joint.



External Pressure Calculation Results Summary:

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External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	No Calc	1720	58.5	0.0085029	122.731
20	30	316.667	1880	138.5	0.096694	122.731
30	40	No Calc	...	153.5	No Calc	No Calc
40	50	300	1656	26.5	0.021	122.731
50	60	1525.93	2773.9	30.5	0.0023856	107.54
60	70	4375.8	2756	26.5	0.00077727	77.1761
70	80	No Calc	2754	25.5	0.001286	92.6577

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	60	No Calc	...	83.4808
20	30	140	No Calc	...	120.548
30	40	155	145.161	...	No Calc
40	50	28	No Calc	...	26.185
50	60	32	9.54009	1.03419	13.6529
60	70	28	12.2219	1.03419	9.8938
70	80	27	8.63067	1.03419	9.53213

Minimum

10

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	316.667	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	300	No Calc	No Calc	No Calc
50	60	1525.93	1525.93	No Calc	No Calc
60	70	4375.8	98548	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	2415.03	1072522	2368.19	1078566	241.503
20	30	296.57	100549	293.645	100926	29.657
30	40	1061.92	433644	1049.34	433984	106.192
40	50	332.977	313507	315.429	315771	33.2977
50	60	3748.3	5228091	3575.42	5250400	374.83
60	70	7625.46	19551760	7220.93	19603964	762.547
70	80	2323.69	2867870	2214.77	2877117	232.369
Total		17803	29567942.00	17037	29660728.00	1780



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
10	Liqd	669.918	-266.667	...	...	Liquid 10
10	Insl	15.6983	-400	...	...	Ins: 20
10	Nozl	927.853	-628.313	495.2	...	CH. IN NOZ A1
10	Nozl	927.853	-628.313	-350.159	350.159	CH. OUT NOZ B1
10	Nozl	2914.64	-584.708	-186.591	-513.128	MANHOLE M1
20	Liqd	62.8048	25	...	...	Liquid 20
20	Insl	0.85866	25	...	...	Ins: 20
30	Liqd	270.862	100	...	...	Liquid 30
30	Insl	3.07522	107.5	...	...	Ins: 20
40	Sadl	381.861	150	1053	...	FIXED SADDLE
40	Liqd	195.822	150	...	...	Liquid: 40
40	Insl	4.56014	150	...	...	Ins: 20
50	Liqd	2101.52	704.526	275	...	Liquid 50
50	Insl	43.948	952.5	...	...	Ins: 20
50	Nozl	42.1531	250	-1393.66	...	LIQ OUT NOZ B3
60	Sadl	862.35	500	1603	...	SLIDING SADDLE
60	Sadl	862.35	3600	1603	...	SLIDING SADDLE
60	Liqd	12212.4	2050	550	...	Liquid 60
60	Insl	102.04	2050	...	...	Ins: 20
60	Nozl	227.387	3750	1486.52	...	SS IN NOZ. A2
60	Nozl	453.738	1700	1543.67	...	SS OUT NOZ. B2
60	Nozl	11.7516	400	1371.42	...	VENT NOZ V1
60	Nozl	11.7516	150	1371.42	...	PR. IN P1
60	Nozl	11.7516	3250	1371.42	...	PR. IN NOZ. P2
60	Nozl	11.7516	3000	1371.42	...	SAFETY VALVE S1
60	Nozl	12.3915	150	695.651	1181.9	LEVEL GUAGE L1
60	Nozl	12.3915	150	-695.651	1181.9	LEVEL GUAGE L2
60	Nozl	12.3915	800	695.651	1181.9	LEVEL GUAGE L3
60	Nozl	12.3915	800	-695.651	1181.9	LEVEL GUAGE L4
60	Nozl	12.3915	1650	-1371.42	...	DRAIN NOZ. D2
60	Wght	6210	...	...	...	DATASHEET WEIGH
70	Liqd	1141.61	275.8	550	...	Liquid 70
70	Insl	32.2352	362.9	...	...	Ins: 20
70	Nozl	12.3915	627.97	-700	...	TEMP IN. T1
20	FTsh	6172.62	250	...	...	U-TS
20	Tube	19953.4	2650	...	...	Tubes
30	Baf1	235.51	1030	...	...	Baffles
30	Baf1	235.51	2060	...	...	Baffles
30	Baf1	235.51	3090	...	...	Baffles
30	Baf1	235.51	4120	...	...	Baffles
30	Baf1	236.253	5150	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	2106.6
Liquid	16654.9
Insulation	202.4
Nozzles	5615.0
Weights	6210.0
Exchanger Components	27304.3
Liquid in Tubes	2386.8

Sum of the Detail Weights 60480.0 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	19584.3	19584.3	19584.3	19584.3	19584.3	19584.3
Saddles	2106.6	2106.6	2106.6	2106.6	2106.6	2106.6
Nozzles	5615.0	5615.0	5615.0	5615.0	5615.0	5615.0
Exchanger	27304.3	27304.3	27304.3	27304.3	27304.3	27304.3
Insulation	...	...	202.4	202.4	202.4	202.4
Ope Weights	...	...	...	...	...	6210.0
Ope. Liquid	...	...	...	...	...	16654.9
Tube Ope Lqd	...	...	...	...	...	2386.8
Test Liquid	...	29549.9	...	...	...	...
Tube Tst Lqd	...	3818.8	...	...	...	...
Totals	54610.2	87978.9	54812.6	54812.6	54812.6	80064.3

Field Installation Options:

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	54610.2 kg.
Shop Test Wt.	- Fabricated weight + water (full)	87978.9 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	54812.6 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	54812.6 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	54812.6 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	80064.3 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	79221.5 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	3821156.8 cm ³
Weight of Ope Liq in tubes :	2386.8 kg.
Weight of Water in tubes :	3818.8 kg.

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	47666.8	40212.4
20	30	2953.1	2513.27
30	40	20045.9	10807.1
40	50	15607.4	15079.6
50	60	138531	133927
60	70	354987	347774



70	80	87224.4	83331.5
Total		667015.7	633645.5 cm ²
Total		718.0	682.1 ft ²

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Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
MANHOLE M1	238.12	255.10	85	1500	GR 1.1	...	...	...	...
LIQ OUT NOZ B3	47.63	51.00	85	300	GR 1.1	...	...	...	...
SS IN NOZ. A2	47.63	51.00	85	300	GR 1.1	...	...	...	...
SS OUT NOZ. B2	47.63	51.00	85	300	GR 1.1	...	...	...	...
VENT NOZ V1	47.63	51.00	85	300	GR 1.1	...	...	...	...
PR. IN P1	47.63	51.00	85	300	GR 1.1	...	...	...	...
PR. IN NOZ. P2	47.63	51.00	85	300	GR 1.1	...	...	...	...
SAFETY VALVE S	47.63	51.00	85	300	GR 1.1	...	...	...	...
LEVEL GUAGE L1	47.63	51.00	85	300	GR 1.1	...	...	...	...
LEVEL GUAGE L2	47.63	51.00	85	300	GR 1.1	...	...	...	...
LEVEL GUAGE L3	47.63	51.00	85	300	GR 1.1	...	...	...	...
LEVEL GUAGE L4	47.63	51.00	85	300	GR 1.1	...	...	...	...
DRAIN NOZ. D2	47.63	51.00	85	300	GR 1.1	...	...	...	...
TEMP IN. T1	47.63	51.00	85	300	GR 1.1	...	...	...	...

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 47.631 bars
 Lowest Flange Pressure Rating was (Amb)[ShellSide]: 51.000 bars

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 238.123 bars
 Lowest Flange Pressure Rating was (Amb)[TubeSide]: 255.100 bars

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-1, UG-44(b) for more information.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 Km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	0
Base Elevation	300 mm.
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.194 \cdot 0.981) / (1 + 1.7 \cdot 3.4 \cdot 0.194)))$
 $= \min(0.85, 0.916)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.736
Structure Height to Diameter ratio	3.146

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:
 Exposure Category: D from Table 26.11-1
 $\alpha = 11.5$; $Z_g = 589788$ mm.

Effective Height [z]:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 1900.0 + 300.0 = 2200.0$ mm.
 $= 7.218$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z [Kz]:
 Because z (7.218 ft.) < 15 ft.
 $= 2.41(15 / Z_g)^{2/\alpha}$
 $= 2.41(15 / 1935.0)^{2/11.5}$
 $= 1.035$

Type of Hill: No Hill

Wind Directionality Factor [Kd]:
 $= 1$, per Table 26.6-1

As there is No Hill Present: [Kzt]:
 $K_1 = 0$, $K_2 = 0$, $K_3 = 0$

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.035 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 111.85^2)$$

$$= 33.1 \text{ psf } [161.849] \text{ Kgs/m}^2$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 33.149 \cdot 0.85 \cdot 0.736 \cdot 14.624$$

$$= 303.2 \text{ lbs. } [137.5] \text{ Kg}$$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
CH. COVER	2200.0	0.000	0.000	0.000	1.035	1.000	161.849
CH. COVER	2200.0	0.000	0.000	0.000	1.035	1.000	161.849
SH. FLG.	2200.0	0.000	0.000	0.000	1.035	1.000	161.849
SHELL-1	2200.0	0.000	0.000	0.000	1.035	1.000	161.849
SHELL-2	2200.0	0.000	0.000	0.000	1.035	1.000	161.849
K-SHELL	2200.0	0.000	0.000	0.000	1.035	1.000	161.849
SH. COVER	2200.0	0.000	0.000	0.000	1.035	1.000	161.849

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf
DATASHEET WEI	0.00	2200.00	161.85	0.00

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	2200	2232	13586.5	161.849	151.275
20	30	2200	2424	1212	161.849	13.4947
30	40	2200	2088	4176	161.849	46.4964
40	50	2200	2155.2	6465.6	161.849	71.9893
50	60	2200	2824.8	53812.4	161.849	599.159
60	70	2200	3475.2	142483	161.849	1586.44
70	80	2200	3472.8	19311.3	161.849	215.016

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-22	
Importance Factor		1.500	
Force Modification Factor R		3.000	
Site Class		D	
Component Elevation Ratio	z/h	0.000	
Amplification Factor	Ap	0.000	
Force Factor		0.000	
Consider Vertical Acceleration		Yes	
Minimum Acceleration Multiplier		0.000	
User Value of Sds (used if > 0)		0.000	
User Value of Sd1 (used if > 0)		0.000	
Component resonance duct. fact.	[CAR]	0.000	
Component strength factor	[Rpo]	0.000	
Overstrength factor	[Omega0]	0.000	
Response Acceleration Factor	[SMS]	0.326	
Response Acceleration Factor	[SM1]	0.139	
Building Importance Factor	[Ie]	1.000	
Site Latitude		0.000	deg
Site Longitude		0.000	deg

Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.326 = 0.217$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.139 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(10.66270.75)$
 = 0.118 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = $(1/\text{Natural Frequency}) = (1/33.0)$
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of $(1.7 \cdot 0.118, 0.03)$ per 12.8.2
 = 0.030

As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.217 \cdot 77115 \cdot 1.5$
 = 7541.839 Kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.217 \cdot 77115$
 = 3351.9 Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	800	7711.49	591.517
20	30	800	7711.49	591.517
30	40	800	7711.49	591.517
40	50	800	7711.49	591.517
40	50	800	7711.49	591.517
50	60	800	7711.49	591.517

60	0	1350	7711.49	998.185
60	70	1350	7711.49	998.185
60	70	1350	7711.49	998.185
70	80	1350	7711.49	998.185

The Earthquake Weight does not include the weight of the saddles.

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Shop/Field Installation Options :

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	4164.053 mm.
Center of Gravity of the Liquid	4311.796 mm.
Center of Gravity of the Insulation	4049.621 mm.
Center of Gravity of the Nozzles	228.088 mm.
Center of Gravity of the Added Weights (Operating)	2858.175 mm.
Center of Gravity of the Tubesheet(s)	250.000 mm.
Center of Gravity of the Tubes	2650.000 mm.
Center of Gravity of the Baffles	3544.474 mm.
Center of Gravity of Bare Shell New and Cold	3470.881 mm.
Center of Gravity of Bare Shell Corroded	3443.231 mm.
Vessel CG in the Operating Condition	2901.822 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	2507.512 mm.
Vessel CG in the Test Condition	3117.946 mm.

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ASME Horizontal Vessel Analysis: Bundle pullout, Left Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

ASME Horizontal Vessel Analysis: Bundle pullout, Right Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

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ASME Horizontal Vessel Analysis: Bundle pullout, Left Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

ASME Horizontal Vessel Analysis: Bundle pullout, Right Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

There are more than two saddles present in this model. The load on the saddle supports has been computed as the total weight divided by the number of saddles. Ensure that this is an appropriate assumption for this vessel model. It may be necessary to manually enter the bending moments at the saddle through the Saddle Input dialog. Also note that the dimension Ls is taken to be the distance between the adjoining saddles (first two or last two).

Note:

Wear Pad Width (240.00) is less than $1.56 \cdot \sqrt{R_m \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

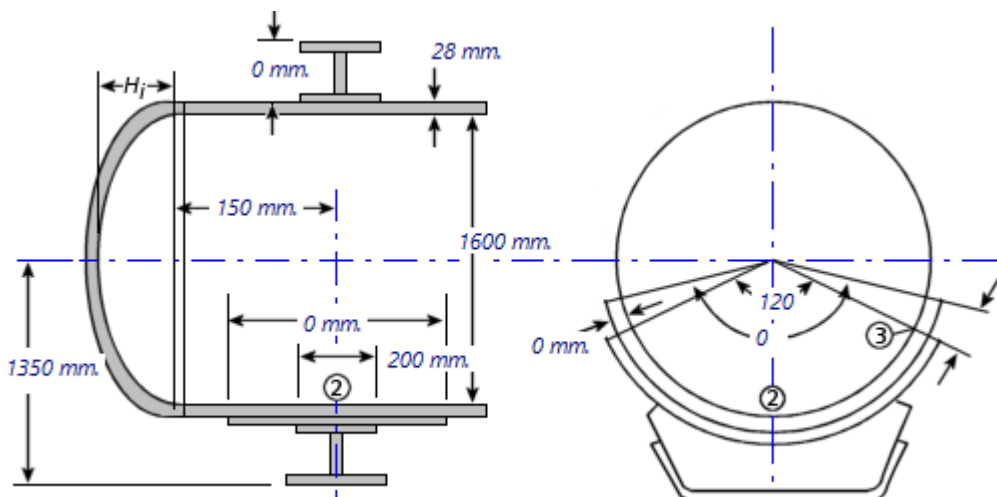
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(200.0 + 1.56 \cdot \sqrt{814.7501 \cdot 26.5}, 2 \cdot 150.0)$$

$$= 300.0000 \text{ mm.}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	814.75	mm.
Shell Thickness used in this Case	t	28.000	mm.
Stiffened Vessel Length per 4.15.6	L	6555.80	mm.
Distance from Saddle to Vessel tangent	a1 or a	150.00	mm.
Saddle Width	b1 or b	200.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1350.00	mm.
Coefficient of Friction	mu	0.60	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		33787.07	Kgf
Pressure used in Analysis	P	25.049	bars

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top	of Midspan	29.28	138.00
Long. Stress at Bottom	of Midspan	47.74	138.00
Long. Stress at Top	of Saddles	38.23	138.00

Long. Stress at Bottom of Saddles	38.79	138.00
Tangential Shear in Shell	13.50	110.40
Tangential Shear in Head	2.33	110.40
Circ. Stress at Horn of Saddle	16.63	172.50
Circ. Compressive Stress in Shell	2.21	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$
 $= 3.0(2683.9/3 + 0) \cdot 1350.0/1461.851$
 $= 2478.5 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot \text{B} / \text{Ls}$
 $= \max(649.49, 0) \cdot 1350.0/2555.0$
 $= 343.2 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot \text{B} / \text{Ls}$
 $= \max(7541.84, 0) \cdot 1350.0/2555.0$
 $= 3984.9 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$
 $= 3.0(7542/3 + 0) \cdot 1350.0/1461.851$
 $= 6964.8 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$
 $= 26822 + \max(343, 2479, 3985, 6965)$
 $= 33787.1 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1587.169(0.0 + \pi/4(1.796 \cdot 1.2)^2)$
 $= 6369.188 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	34168.93	Kgf
Transverse Shear Load	2513.95	Kgf
Longitudinal Shear Load	7541.84	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$
 $= \min(2(1600.0/2 + 28.0 + 16.0)\sin(120.0/2), 2 \cdot 814.75)$
 $= 1461.851 \text{ mm.}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0132	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is less than $\text{Rm}/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2)/(2a \cdot L))/(1 + (4\text{hm})/3L)]$
 $= -33787 \cdot 150.0 [1 - (1 - 150.0/6555.8 + (814.75^2 - 0.0^2)/(2 \cdot 150.0 \cdot 6555.8))/(1 + (4 \cdot 0.0)/(3 \cdot 6555.8))]$
 $= 1594.7 \text{ Kg-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4((1 + 2(\text{Rm}^2 - \text{hm}^2)/(L^2))/(1 + (4\text{hm})/3L) - 4a/L)$
 $= 33787 \cdot 6556/4((1 + 2(815^2 - 0^2)/(6556^2))/(1 + (4 \cdot 0)/(3 \cdot 6556)) - 4 \cdot 150/6556)$
 $= 52018.9 \text{ Kg-m.}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t)$
 $= 25.049 \cdot 814.75 / (2 \cdot 26.5) - 52018.9 / (\pi \cdot 814.8^2 \cdot 26.5)$
 $= 29.28 \text{ N./mm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$
 $= 25.049 \cdot 814.75 / (2 \cdot 26.5) + 52018.9 / (\pi \cdot 814.8^2 \cdot 26.5)$
 $= 47.74 \text{ N./mm}^2$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:
 $= P \cdot R_m / (2t) - M_1 / (\pi \cdot R_m^2 \cdot t)$
 $= 25.049 \cdot 814.75 / (2 \cdot 26.5) - 1594.7 / (\pi \cdot 814.8^2 \cdot 26.5)$
 $= 38.23 \text{ N./mm}^2$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:
 $= P \cdot R_m / (2t) + M_1 / (\pi \cdot R_m^2 \cdot t)$
 $= 25.049 \cdot 814.75 / (2 \cdot 26.5) + 1594.7 / (\pi \cdot 814.8^2 \cdot 26.5)$
 $= 38.79 \text{ N./mm}^2$

Maximum Shear Force in the Saddle (4.15.3) [T]:
 $= Q(L-2a) / (L + (4 \cdot h_m / 3))$
 $= 33787(6555.8 - 2 \cdot 150.0) / (6555.8 + (4 \cdot 0.0 / 3))$
 $= 32240.9 \text{ Kgf}$

Shear Stress in the shell no rings, stiffened (4.15.13) [τ_3]:
 $= K_3 \cdot Q / (R_m \cdot t)$
 $= 0.8799 \cdot 33787 / (814.7501 \cdot 26.5)$
 $= 13.50 \text{ N./mm}^2$

Shear Stress in the head, shell stiffened (4.15.14) [τ_{3*}]:
 $= K_3 \cdot Q / (R_m \cdot t_h)$
 $= 0.8799 \cdot 33787 / (814.7501 \cdot 153.5)$
 $= 2.33 \text{ N./mm}^2$

Decay Length (4.15.20) [x_1, x_2]:
 $= 0.78 \cdot \sqrt{(R_m \cdot t)}$
 $= 0.78 \cdot \sqrt{(814.75 \cdot 26.5)}$
 $= 114.612 \text{ mm.}$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:
 $= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$
 $= -0.7603 \cdot 33787 \cdot 0.1 / (26.5(200.0 + 114.61 + 114.61))$
 $= -2.21 \text{ N./mm}^2$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:
 $= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$
 $= -33787 / (4 \cdot 26.5(200.0 + 114.612 + 114.612)) -$
 $3 \cdot 0.0132 \cdot 33787 / (2 \cdot 26.5^2)$
 $= -16.63 \text{ N./mm}^2$

Effective reinforcing plate width (4.15.24) [B1]:
 $= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$
 $= \min(200.0 + 1.56 \cdot \sqrt{(814.75 \cdot 26.5)}, 2 \cdot 150.0)$
 $= 300.00 \text{ mm.}$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1530.0000	mm.
Baseplate Thickness	Bpthk	25.0000	mm.
Baseplate Width	Bpwid	240.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	16.0000	mm.
Web Thickness	Webtk	16.0000	mm.
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	242.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.600	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$
 $= ((1.56 \cdot \sqrt{(801.5 \cdot 26.5)} + 240.0)^3) / 12$
 $= 72.477 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$
 $= 72.477 + 123.848 (190.187 - 13.25)^2$
 $= 38845.180 \text{ cm}^4$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	467.4	26.5	13.3	123.8	164099.1	0.388E+05
Wearplate	240.0	16.0	34.5	38.4	132480.0	0.932E+04
Web	16.0	481.0	283.0	77.0	2177967.8	0.215E+05
BasePlate	240.0	25.0	536.0	60.0	3216000.0	0.718E+05
Totals	...	...	...	299.2	5690547.0	0.141E+06

Distance to Centroid [C1]:
 $= AY / A$
 $= 5690547 / 299.208$
 $= 190.187 \text{ mm.}$

Angle [β]:
 $= 180 - \text{Saddle Angle} / 2$
 $= 180 - 120.0 / 2$
 $= 120.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 $= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$
 $= 0.2035$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.204 \cdot 33787.07$
 $= 6876.4019 \text{ Kgf}$

Tension Stress, St = (Fh/As) = 3.8456 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 145.1982 N./mm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$
 $= 1350.0 - 801.5 \cdot \sin(120.0/2) / 1.0472$
 $= 687.165 \text{ mm.}$

Bending Moment, M = Fh * d = 4725.3149 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 6.2323 N./mm²
 Allowed Stress, Sa = 2/3 * Yield Str = 161.3313 N./mm²

Minimum Thickness of Baseplate per Moss:

$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$
 $= \sqrt{3(33787 + 382)240.0 / (2 \cdot 1530.0 \cdot 161.331)}$
 $= 22.107 \text{ mm.}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120.0/2) (800.0 + 28.0 + 16.0)$
 $= 1461.851 \text{ mm.}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nribs} - 1)$
 $= 1461.8508 / (4 - 1)$
 $= 487.284 \text{ mm.}$

Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 487.2836 \cdot 240.0 / 2$$

$$= 584.740 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 34168.934 / (1530.0 \cdot 240.0)$$

$$= 9.305 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 584.7 \cdot 9.31$$

$$= 5441.164 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 29.44 + 38.983$$

$$= 68.423 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 5441.2 / 68.4227$$

$$= 7.799 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	16.0	184.0	108.0	29.4	317951.9	0.179E+04
Web	243.6	16.0	8.0	39.0	31186.1	730.
Totals	...	...	...	68.4	349138.1	0.252E+04

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 200.0 - 16.0$$

$$= 184.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 349138.094 / 68.423$$

$$= 51.027 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(ytot, Saddle \text{ Width} - ytot)$$

$$= \max(51.027, 200.0 - 51.027)$$

$$= 148.973 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (2516.2 / 68.423) }$$

$$= 60.642 \text{ mm.}$$

Length of Outer Rib [L]:

$$= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk$$

$$= 1350.0 - \cos(120.0 / 2) (800.0 + 28.0 + 16.0) - 25.0$$

$$= 903.000 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 0.19904E+09 / 242.0) }$$

$$= 127.422$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 903.0 / 60.642$$

$$= 14.891$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 7541.8 / (2 \cdot 1530.0) \cdot 487.284 \cdot 903.0 / 2$$

$$= 542.255 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ (14.8907 < 127.4221) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (14.89)^2 / (2 \cdot 127.42^2)) 242 /$$

$$(5/3 + 3 \cdot (14.89) / (8 \cdot 127.42) - (14.89^3) / (8 \cdot 127.42^3))$$

$$= 140.5 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 7.8 / 140.53 + (542.26 \cdot 148.973 / 25162118) / 161.33$$

$$= 0.251$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	184.0	108.0	29.4	317951.9	0.238E+04
Web	487.3	16.0	8.0	78.0	62372.3	602.
Totals	...	...	...	107.4	380324.2	0.298E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 380324.25 / 107.405$$

$$= 35.410 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(35.41, \text{abs}(200.0 - 35.41))$$

$$= 164.590 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + W_{pdthk})^2 - (Pitch/2)^2 } - B_{pthk}$$

$$= 1350.0 - \sqrt{ (844.0 + 16.0)^2 - (487.284/2)^2 } - 25.0$$

$$= 516.932 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 2984.3 / 107.405 }$$

$$= 52.712 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 516.932 / 52.712$$

$$= 9.807$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 7541.839 / (2 \cdot 1530.0)$$

$$= 2.465 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 2.465 \cdot 487.284 \cdot 516.932$$

$$= 620.839 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 620.839 / 181316.016$$

$$= 33.579 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (9.8068 < 127.4221) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.81)^2 / (2 \cdot 127.42^2)) 242 /$$

$$(5/3 + 3 \cdot (9.81) / (8 \cdot 127.42) - (9.81^3) / (8 \cdot 127.42^3))$$

= 142.3 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $Sc/Sca + (M_{base,c} \cdot C1/I) / Sba$
 = $9.72/142.31 + (620.84 \cdot 164.59/2984.275) / 161.33$
 = 0.276

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	3
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	150.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	85.0 mm.
Nominal Bolt Diameter	Bnd	30.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.9088 cm ²
Saddle Load QO (Weight)	QO	27204.1 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	3984.9 Kg
Maximum Transverse Force	Ft	2513.9 Kg
Maximum Longitudinal Force	Fl	7541.8 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_{b,l}]:

= $F_l / (Bolt\ Area \cdot Number\ of\ Bolts)$
 = $7541.8 / (3.909 \cdot 3)$
 = 63.1 N./mm². Must be less than 100.0 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:

= $F_t / (Bolt\ Area \cdot Number\ of\ Bolts)$
 = $2513.9 / (3.909 \cdot 3)$
 = 21.0 N./mm². Must be less than 100.0 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $F_l / (BoltShearAllowable \cdot Nbolts)$
 = $7541.84 / (100.0 \cdot 3.0)$
 = 2.4654 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot F_t + \text{Sum of X Moments}$
 = $1350.0 \cdot 2513.95 + 0.0$
 = 3393.89 Kg-m.

Eccentricity (e):

= $Rmom / QO$
 = $3393.89 / 27204.14$
 = 124.75 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= $\max[Bltarearl, Bltarears, Bltareart]$
 = $\max[0.0, 2.4654, 0.0]$
 = 2.4654 cm²

MDMT Calculations for: FIXED SADDLE, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 16.0, c = 1.5 mm., E* = 1.0

Thickness Ratio = $tr \cdot E / (tg - c) = 0.575$, Temp. Reduction = 25 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -39 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (400.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$
 $= \min(300.0 + 1.56 \cdot \sqrt{1364.75 \cdot 26.5}, 2 \cdot 500.0)$
 $= 596.6703 \text{ mm.}$

Input and Calculated Values:

Vessel Mean Radius	Rm	1364.75	mm.
Shell Thickness used in this Case	t	28.000	mm.
Stiffened Vessel Length per 4.15.6	L	6555.80	mm.
Distance from Saddle to Vessel tangent	a1 or a	500.00	mm.
Saddle Width	b1 or b	300.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	689.25	mm.
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1900.00	mm.
Coefficient of Friction	mu	0.10	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		32757.10	Kgf
Pressure used in Analysis	P	25.015	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	62.28	138.00
Long. Stress at Bottom of Midspan	66.55	138.00
Long. Stress at Top of Saddles	64.42	138.00
Long. Stress at Bottom of Saddles	64.41	138.00
Tangential Shear in Shell	7.82	110.40
Tangential Shear in Head	8.12	110.40
Circ. Stress at Horn of Saddle	20.18	172.50
Addl. Stress in Head as Stiffener	133.69	172.50
Circ. Compressive Stress in Shell	1.54	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(2683.9/3 + 0) \cdot 1900.0/2414.4792$
 $= 2112.0 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl, wind; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(1688.71, 0) \cdot 1900.0/3100.0$
 $= 1035.0 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq; Sum of X Forces}) \cdot B / L_s$$

$$= \max(7541.84, 0) \cdot 1900.0/3100.0$$

$$= 4622.4 \text{ Kg}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(7542/3 + 0) \cdot 1900.0/2414.4792$$

$$= 5934.8 \text{ Kg}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl, Fwt, Fsl, Fst})$$

$$= 26822 + \max(1035, 2112, 4622, 5935)$$

$$= 32757.1 \text{ Kg}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587.169(0.0 + \pi/4(2.896 \cdot 1.2)^2)$$

$$= 16560.307 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	33619.45	Kgf
Transverse Shear Load	2513.95	Kgf
Longitudinal Shear Load	7541.84	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(2700.0/2 + 28.0 + 16.0)\sin(120.0/2), 2 \cdot 1364.75)$$

$$= 2414.479 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0132	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is less than Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -32757 \cdot 500.0 [1 - (1 - 500.0/6555.8 + (1364.75^2 - 689.25^2)/(2 \cdot 500.0 \cdot 6555.8))/(1 + (4 \cdot 689.25)/(3 \cdot 6555.8))]$$

$$= -69.1 \text{ Kg-m}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 32757 \cdot 6556/4((1 + 2(1365^2 - 689^2)/(6556^2))/(1 + (4 \cdot 689)/(3 \cdot 6556)) - 4 \cdot 500/6556)$$

$$= 33748.9 \text{ Kg-m}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 25.015 \cdot 1364.75/(2 \cdot 26.5) - 33748.9/(\pi \cdot 1364.8^2 \cdot 26.5)$$

$$= 62.28 \text{ N/mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 25.015 \cdot 1364.75/(2 \cdot 26.5) + 33748.9/(\pi \cdot 1364.8^2 \cdot 26.5)$$

$$= 66.55 \text{ N/mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ3]:

$$= P \cdot R_m/(2t) - M1/(\pi \cdot R_m^2 \cdot t)$$

$$= 25.015 \cdot 1364.75/(2 \cdot 26.5) - 69.1/(\pi \cdot 1364.8^2 \cdot 26.5)$$

$$= 64.42 \text{ N/mm}^2$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ4]:

$$= P \cdot R_m/(2t) + M1/(\pi \cdot R_m^2 \cdot t)$$

$$= 25.015 \cdot 1364.75/(2 \cdot 26.5) + 69.1/(\pi \cdot 1364.8^2 \cdot 26.5)$$

$$= 64.41 \text{ N/mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:
 $= Q(L-2a)/(L+(4 \cdot hm/3))$
 $= 32757(6555.8 - 2 \cdot 500.0)/(6555.8 + (4 \cdot 689.25/3))$
 $= 24347.4 \text{ Kg}$

Shear Stress in the shell no rings, stiffened (4.15.13) [τ_3]:
 $= K_3 \cdot Q / (R_m \cdot t)$
 $= 0.8799 \cdot 32757 / (1364.75 \cdot 26.5)$
 $= 7.82 \text{ N./mm}^2$

Shear Stress in the head, shell stiffened (4.15.14) [τ_3^*]:
 $= K_3 \cdot Q / (R_m \cdot t_h)$
 $= 0.8799 \cdot 32757 / (1364.75 \cdot 25.5)$
 $= 8.12 \text{ N./mm}^2$

Membrane stress in the head as a stiffener (4.15.16) [σ_5]:
 $= K_4 \cdot Q / (R_m \cdot t_h) + ((P \cdot R_i) / (2 \cdot t_h)) \cdot (R_i / h_m)$
 $= 0.4011 \cdot 32757 / (1364.75 \cdot 25.5) +$
 $((25.02 \cdot 1351.5) / (2 \cdot 25.5)) \cdot (1351.5 / 689.25)$
 $= 133.69 \text{ N./mm}^2$

Decay Length (4.15.20) [x_1, x_2]:
 $= 0.78 \cdot \sqrt{(R_m \cdot t)}$
 $= 0.78 \cdot \sqrt{(1364.75 \cdot 26.5)}$
 $= 148.335 \text{ mm.}$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:
 $= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$
 $= -0.7603 \cdot 32757 \cdot 0.1 / (26.5(300.0 + 148.34 + 148.34))$
 $= -1.54 \text{ N./mm}^2$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_7^*]:
 $= -Q / (4 \cdot t(b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$
 $= -32757 / (4 \cdot 26.5(300.0 + 148.335 + 148.335)) -$
 $12 \cdot 0.013 \cdot 32757 \cdot 1364.75 / (6555.799 \cdot 26.5^2)$
 $= -20.18 \text{ N./mm}^2$

Effective reinforcing plate width (4.15.24) [B1]:
 $= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$
 $= \min(300.0 + 1.56 \cdot \sqrt{(1364.75 \cdot 26.5)}, 2 \cdot 500.0)$
 $= 596.67 \text{ mm.}$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2490.0000	mm.
Baseplate Thickness	Bpthk	25.0000	mm.
Baseplate Width	Bpwid	340.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	16.0000	mm.
Web Thickness	Webtk	16.0000	mm.
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	242.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.100	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$
 $= ((1.56 \cdot \sqrt{(1351.5 \cdot 26.5)} + 400.0)^3) / 12$
 $= 107.816 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$
 $= 107.816 + 184.235(175.499 - 13.25)^2$
 $= 48607.152 \text{ cm}^4$

	mm.	mm.	mm.	cm ²	mm. ³	cm ⁴
Shell	695.2	26.5	13.3	184.2	244111.5	0.486E+05
Wearplate	400.0	16.0	34.5	64.0	220800.0	0.127E+05
Web	16.0	481.0	283.0	77.0	2177968.5	0.237E+05
BasePlate	340.0	25.0	536.0	85.0	4556001.0	0.111E+06
Totals	...	...	...	410.2	7198881.0	0.196E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 7198881/410.195$$

$$= 175.499 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 32757.102$$

$$= 6666.7808 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 2.8934 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 145.1984 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1900.0 - 1351.5 \cdot \sin(120.0/2) / 1.0472$$

$$= 782.319 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 5215.6484 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 4.5895 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 161.3315 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(32757 + 862)340.0 / (2 \cdot 2490.0 \cdot 161.332)}$$

$$= 20.459 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1350.0 + 28.0 + 16.0)$$

$$= 2414.479 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nrings} - 1)$$

$$= 2414.479 / (5 - 1)$$

$$= 603.620 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 603.6198 \cdot 340.0/2$$

$$= 1026.154 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 33619.453 / (2490.0 \cdot 340.0)$$

$$= 3.971 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1026.2 \cdot 3.97$$

$$= 4074.973 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 45.44 + 48.29$$

$$= 93.730 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

$$= 4075.0/93.7296$$

$$= 4.264 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	16.0	284.0	158.0	45.4	717951.9	0.577E+04
Web	301.8	16.0	8.0	48.3	38631.7	0.256E+04
Totals	...	...	...	93.7	756583.6	0.833E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 300.0 - 16.0$$

$$= 284.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 756584/93.73$$

$$= 80.720 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{Saddle Width} - ytot)$$

$$= \max(80.72, 300.0 - 80.72)$$

$$= 219.280 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(8331.9/93.73)}$$

$$= 94.283 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1900.0 - \cos(120.0/2) (1350.0 + 28.0 + 16.0) - 25.0$$

$$= 1178.000 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19904\text{E}+09/242.0)}$$

$$= 127.422$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1178.0/94.283$$

$$= 12.494$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 7541.8 / (2 \cdot 2490.0) \cdot 603.62 \cdot 1178.0/2$$

$$= 538.437 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($12.4943 < 127.4221$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (12.49)^2 / (2 \cdot 127.42^2)) 242 / (5/3 + 3 \cdot (12.49) / (8 \cdot 127.42) - (12.49^3) / (8 \cdot 127.42^3))$$

$$= 141.4 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 4.26/141.39 + (538.44 \cdot 219.28/83318912) / 161.33$$

$$= 0.116$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	284.0	158.0	45.4	717951.9	0.778E+04
Web	603.6	16.0	8.0	96.6	77263.3	0.225E+04
Totals	...	...	...	142.0	795215.2	0.100E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 795215/142.019$$

$$= 55.994 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(55.994, \text{abs}(300.0 - 55.994))$$

$$= 244.006 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 1900.0 - \cos(1394.0/2) (25.0 + 0.0 + 0.0) - 0.0$$

$$= 481.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (10027.5/142.019) }$$

$$= 84.028 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 481.0/84.028$$

$$= 5.724$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 7541.839/(2 \cdot 2490.0)$$

$$= 1.514 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.514 \cdot 603.62 \cdot 481.0$$

$$= 439.709 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 439.709/410953.75$$

$$= 10.493 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < Cc$ (5.7243 < 127.4221) per AISC E2-1 [Sca]:

$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$$

$$= (1 - (5.72)^2 / (2 \cdot 127.42^2)) 242 / (5/3 + 3 \cdot (5.72) / (8 \cdot 127.42) - (5.72^3) / (8 \cdot 127.42^3))$$

$$= 143.6 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 5.54/143.6 + (439.71 \cdot 244.006/10027.528) / 161.33$$

$$= 0.104$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7



Bolt Allowable Stress	Stba	150.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	330.0 mm.
Nominal Bolt Diameter	Bnd	30.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.9088 cm ²
Saddle Load QO (Weight)	QO	27684.6 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	4622.4 Kgf
Maximum Transverse Force	Ft	2513.9 Kgf
Maximum Longitudinal Force	Fl	7541.8 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
 = Fl / (Bolt Area • Number of Bolts)
 = 7541.8/(3.909 • 4)
 = 47.3 N./mm². Must be less than 100.0 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
 = Ft / (Bolt Area • Number of Bolts)
 = 2513.9/(3.909 • 4)
 = 15.8 N./mm². Must be less than 100.0 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = Fl / (BoltShearAllowable • Nbolts)
 = 7541.84/(100.0 • 4.0)
 = 1.8490 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = B • Ft + Sum of X Moments
 = 1900.0 • 2513.95 + 0.0
 = 4776.59 Kg-m.

Eccentricity (e):
 = Rmom / QO
 = 4776.59/27684.63
 = 172.53 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0.0, 1.849, 0.0]
 = 1.8490 cm²

Note: Saddle and wear plate material are both specified as impact tested.

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

There are more than two saddles present in this model. The load on the saddle supports has been computed as the total weight divided by the number of saddles. Ensure that this is an appropriate assumption for this vessel model. It may be necessary to manually enter the bending moments at the saddle through the Saddle Input dialog. Also note that the dimension Ls is taken to be the distance between the adjoining saddles (first two or last two).

Note:

Wear Pad Width (240.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

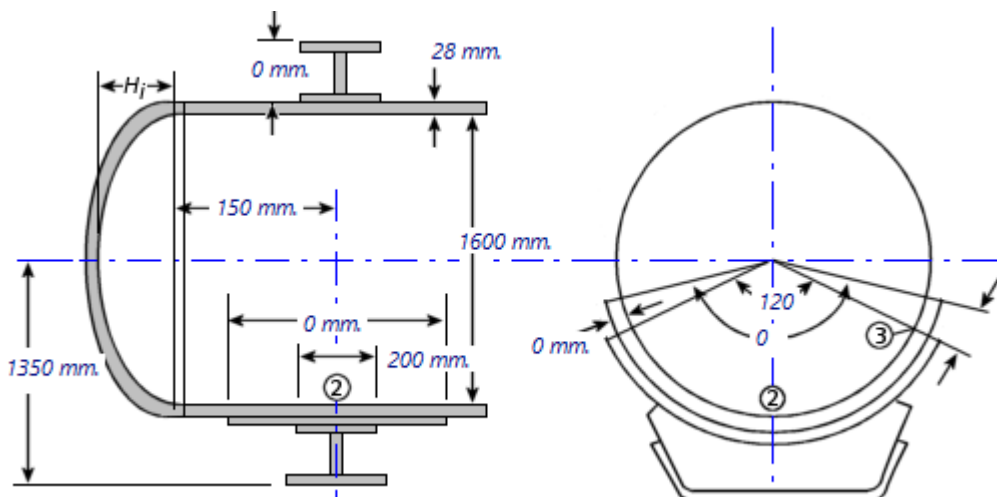
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(200.0 + 1.56 \cdot \sqrt{814.7501 \cdot 26.5}, 2 \cdot 150.0)$$

$$= 300.0000 \text{ mm.}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	814.75	mm.
Shell Thickness used in this Case	t	28.000	mm.
Stiffened Vessel Length per 4.15.6	L	6555.80	mm.
Distance from Saddle to Vessel tangent	a1 or a	150.00	mm.
Saddle Width	b1 or b	200.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1350.00	mm.
Coefficient of Friction	mu	0.60	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		29442.04	Kgf
Pressure used in Analysis	P	33.785	bars

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top	of Midspan	43.90	234.00
Long. Stress at Bottom	of Midspan	59.98	234.00
Long. Stress at Top	of Saddles	51.69	234.00

Long. Stress at Bottom of Saddles	52.19	234.00
Tangential Shear in Shell	11.77	138.58
Tangential Shear in Head	2.03	122.36
Circ. Stress at Horn of Saddle	14.49	350.99
Circ. Compressive Stress in Shell	1.93	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$
 $= 3.0(885.7/3 + 0) \cdot 1350.0/1461.851$
 $= 817.9 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot \text{B} / \text{Ls}$
 $= \max(214.33, 0) \cdot 1350.0/2555.0$
 $= 113.2 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$
 $= 28624 + \max(113, 818, 0, 0)$
 $= 29442.0 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1587.169(0.0 + \pi/4(1.796 \cdot 1.2)^2)$
 $= 6369.188 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	29823.89	Kgf
Transverse Shear Load	295.23	Kgf
Longitudinal Shear Load	214.33	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$
 $= \min(2(1600.0/2 + 28.0 + 16.0)\sin(120.0/2), 2 \cdot 814.75)$
 $= 1461.851 \text{ mm.}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0132	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is less than $\text{Rm}/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2)/(2a \cdot L))/(1 + (4\text{hm})/3L)]$
 $= -29442 \cdot 150.0 [1 - (1 - 150.0/6555.8 + (814.75^2 - 0.0^2)/(2 \cdot 150.0 \cdot 6555.8))/(1 + (4 \cdot 0.0)/(3 \cdot 6555.8))]$
 $= 1389.6 \text{ Kg-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4((1 + 2(\text{Rm}^2 - \text{hm}^2)/(L^2))/(1 + (4\text{hm}^2)/(3L)) - 4a/L)$
 $= 29442 \cdot 6556/4((1 + 2(815^2 - 0^2)/(6556^2))/(1 + (4 \cdot 0)/(3 \cdot 6556)) - 4 \cdot 150/6556)$
 $= 45329.2 \text{ Kg-m.}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot \text{Rm}/(2t) - \text{M2}/(\pi \cdot \text{Rm}^2 t)$
 $= 33.785 \cdot 814.75/(2 \cdot 26.5) - 45329.2/(\pi \cdot 814.8^2 \cdot 26.5)$
 $= 43.90 \text{ N./mm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot \text{Rm}/(2t) + \text{M2}/(\pi \cdot \text{Rm}^2 \cdot t)$
 $= 33.785 \cdot 814.75/(2 \cdot 26.5) + 45329.2/(\pi \cdot 814.8^2 \cdot 26.5)$
 $= 59.98 \text{ N./mm}^2$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$= P \cdot R_m / (2t) - M_1 / (\pi \cdot R_m^2 \cdot t)$$

$$= 33.785 \cdot 814.75 / (2 \cdot 26.5) - 1389.6 / (\pi \cdot 814.8^2 \cdot 26.5)$$

$$= 51.69 \text{ N./mm}^2$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot R_m / (2t) + M_1 / (\pi \cdot R_m^2 \cdot t)$$

$$= 33.785 \cdot 814.75 / (2 \cdot 26.5) + 1389.6 / (\pi \cdot 814.8^2 \cdot 26.5)$$

$$= 52.19 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 29442(6555.8 - 2 \cdot 150.0) / (6555.8 + (4 \cdot 0.0 / 3))$$

$$= 28094.7 \text{ Kg}$$

Shear Stress in the shell no rings, stiffened (4.15.13) [τ_3]:

$$= K_3 \cdot Q / (R_m \cdot t)$$

$$= 0.8799 \cdot 29442 / (814.7501 \cdot 26.5)$$

$$= 11.77 \text{ N./mm}^2$$

Shear Stress in the head, shell stiffened (4.15.14) [τ_{3*}]:

$$= K_3 \cdot Q / (R_m \cdot t_h)$$

$$= 0.8799 \cdot 29442 / (814.7501 \cdot 153.5)$$

$$= 2.03 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(814.75 \cdot 26.5)}$$

$$= 114.612 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 29442 \cdot 0.1 / (26.5(200.0 + 114.61 + 114.61))$$

$$= -1.93 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -29442 / (4 \cdot 26.5(200.0 + 114.612 + 114.612)) - 3 \cdot 0.0132 \cdot 29442 / (2 \cdot 26.5^2)$$

$$= -14.49 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(200.0 + 1.56 \cdot \sqrt{(814.75 \cdot 26.5)}, 2 \cdot 150.0)$$

$$= 300.00 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1530.0000	mm.
Baseplate Thickness	Bpthk	25.0000	mm.
Baseplate Width	Bpwid	240.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	16.0000	mm.
Web Thickness	Webtk	16.0000	mm.
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.600	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(801.5 \cdot 26.5)} + 240.0)^3) / 12$$

$$= 72.477 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 72.477 + 123.848(190.187 - 13.25)^2$$

$$= 38845.180 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	467.4	26.5	13.3	123.8	164099.1	0.388E+05
Wearplate	240.0	16.0	34.5	38.4	132480.0	0.932E+04
Web	16.0	481.0	283.0	77.0	2177967.8	0.215E+05
BasePlate	240.0	25.0	536.0	60.0	3216000.0	0.718E+05
Totals	...	...	...	299.2	5690547.0	0.141E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 5690547/299.208$$

$$= 190.187 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 29442.035$$

$$= 5992.0923 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 3.3510 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1350.0 - 801.5 \cdot \sin(120.0/2) / 1.0472$$

$$= 687.165 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 4117.6367 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 5.4308 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{ 3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress}) }$$

$$= \sqrt{ 3(29442 + 382)240.0 / (2 \cdot 1530.0 \cdot 173.33) }$$

$$= 19.926 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (800.0 + 28.0 + 16.0)$$

$$= 1461.851 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1461.8508 / (4 - 1)$$

$$= 487.284 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 487.2836 \cdot 240.0/2$$

$$= 584.740 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 29823.895 / (1530.0 \cdot 240.0)$$

$$= 8.122 \text{ Kgf/cm}^2$$

Axial Load [P]:
 = $A_p \cdot B_p$
 = $584.7 \cdot 8.12$
 = 4749.247 Kgf

Area of the Rib and Web [Ar]:
 = Rib Area + Web Area
 = $29.44 + 38.983$
 = 68.423 cm²

Compressive Stress [Sc]:
 = P/Ar
 = $4749.2/68.4227$
 = 6.807 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	16.0	184.0	108.0	29.4	317951.9	0.179E+04
Web	243.6	16.0	8.0	39.0	31186.1	730.
Totals	...	...	...	68.4	349138.1	0.252E+04

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = $200.0 - 16.0$
 = 184.000 mm.

Distance to Centroid from Datum [ytot]:
 = AY / A
 = $349138.094/68.423$
 = 51.027 mm.

Distance to Centroid [C1]:
 = max(ytot, Saddle Width - ytot)
 = max(51.027, 200.0 - 51.027)
 = 148.973 mm.

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{2516.2/68.423}$
 = 60.642 mm.

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = $1350.0 - \cos(120.0/2)(800.0 + 28.0 + 16.0) - 25.0$
 = 903.000 mm.

Intermediate Term [Cc]:
 = $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
 = $\sqrt{(2 \cdot \pi^2 \cdot 0.19904\text{E}+09/260.0)}$
 = 122.933

Slenderness ratio [KL/r]:
 = KL/r
 = $1 \cdot 903.0/60.642$
 = 14.891

Bending Moment [Rm]:
 = $Fl / (2 \cdot B_{plen}) \cdot e \cdot L / 2$
 = $214.3 / (2 \cdot 1530.0) \cdot 487.284 \cdot 903.0/2$
 = 15.410 Kg-m.

Compressive Allowable, $KL/r < C_c$ (14.8907 < 122.9326) per AISC E2-1 [Sca]:
 = $(1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
 = $(1 - (14.89)^2 / (2 \cdot 122.93^2)) 260 / (5/3 + 3 \cdot (14.89) / (8 \cdot 122.93) - (14.89^3) / (8 \cdot 122.93^3))$
 = 150.8 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 6.81/150.76 + (15.41 \cdot 148.973/25162118) / 173.33$$

$$= 0.050$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	184.0	108.0	29.4	317951.9	0.238E+04
Web	487.3	16.0	8.0	78.0	62372.3	602.
Totals	...	...	...	107.4	380324.2	0.298E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 380324.25/107.405$$

$$= 35.410 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(35.41, \text{abs}(200.0 - 35.41))$$

$$= 164.590 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + Wpdthk)^2 - (Pitch/2)^2 } - Bpthk$$

$$= 1350.0 - \sqrt{ (844.0 + 16.0)^2 - (487.284/2)^2 } - 25.0$$

$$= 516.932 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 2984.3/107.405 }$$

$$= 52.712 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 516.932/52.712$$

$$= 9.807$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 214.33/(2 \cdot 1530.0)$$

$$= 0.070 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.07 \cdot 487.284 \cdot 516.932$$

$$= 17.644 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 17.644/181316.016$$

$$= 0.954 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < Cc$ (9.8068 < 122.9326) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (9.81)^2 / (2 \cdot 122.93^2)) 260 /$$

$$(5/3 + 3 \cdot (9.81) / (8 \cdot 122.93) - (9.81^3) / (8 \cdot 122.93^3))$$

$$= 152.8 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 8.49/152.76 + (17.64 \cdot 164.59/2984.275) / 173.33$$

$$= 0.061$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	3
Total Number of Bolts in Tension/Baseplate	Nbt	1

Bolt Material Specification	SA-193 B7
Bolt Allowable Stress	Stba 172.00 N./mm ²
Bolt Corrosion Allowance	Bca 1.5 mm.
Distance from Bolts to Edge	Edgedis 85.0 mm.
Nominal Bolt Diameter	Bnd 30.0000 mm.
Thread Series	Series TEMA Metric
BasePlate Allowable Stress	S 138.00 N./mm ²
Area Available in a Single Bolt	BltArea 3.9088 cm ²
Saddle Load QO (Weight)	QO 29006.0 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL 113.2 Kgf
Maximum Transverse Force	Ft 295.2 Kgf
Maximum Longitudinal Force	Fl 214.3 Kgf
Saddle Bolted to Steel Foundation	No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:
 = Fl / (Bolt Area • Number of Bolts)
 = 214.3 / (3.909 • 3)
 = 1.8 N./mm². Must be less than 103.2 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:
 = Ft / (Bolt Area • Number of Bolts)
 = 295.2 / (3.909 • 3)
 = 2.5 N./mm². Must be less than 103.2 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = Fl / (BoltShearAllowable • Nbolts)
 = 214.33 / (103.2 • 3.0)
 = 0.0679 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = B • Ft + Sum of X Moments
 = 1350.0 • 295.23 + 0.0
 = 398.56 Kg-m.

Eccentricity (e):
 = Rmom / QO
 = 398.56 / 29005.98
 = 13.74 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0.0, 0.0679, 0.0]
 = 0.0679 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
 Wear Pad Width (400.00) is less than 1.56*sqrt(rm*t)
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 = min(b + 1.56*sqrt(Rm • t), 2a)
 = min(300.0 + 1.56*sqrt(1364.75 • 26.5), 2 • 500.0)
 = 596.6703 mm.

Input and Calculated Values:

Vessel Mean Radius	Rm	1364.75 mm.
Shell Thickness used in this Case	t	28.000 mm.

Stiffened Vessel Length per 4.15.6	L	6555.80	mm.
Distance from Saddle to Vessel tangent	a1 or a	500.00	mm.
Saddle Width	b1 or b	300.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	689.25	mm.
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1900.00	mm.
Coefficient of Friction	mu	0.10	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		29321.08	Kgf
Pressure used in Analysis	P	33.839	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	85.23	234.00
Long. Stress at Bottom of Midspan	89.05	234.00
Long. Stress at Top of Saddles	87.14	234.00
Long. Stress at Bottom of Saddles	87.14	234.00
Tangential Shear in Shell	7.00	138.58
Tangential Shear in Head	7.27	128.98
Circ. Stress at Horn of Saddle	18.06	350.99
Addl. Stress in Head as Stiffener	179.16	234.00
Circ. Compressive Stress in Shell	1.38	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (885.7/3 + 0) \cdot 1900.0/2414.4792$
 $= 697.0 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(557.27, 0) \cdot 1900.0/3100.0$
 $= 341.6 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 28624 + \max(342, 697, 0, 0)$
 $= 29321.1 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1587.169 (0.0 + \pi/4 (2.896 \cdot 1.2)^2)$
 $= 16560.307 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	30183.43	Kgf
Transverse Shear Load	295.23	Kgf
Longitudinal Shear Load	557.27	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(2700.0/2 + 28.0 + 16.0) \sin(120.0/2), 2 \cdot 1364.75)$
 $= 2414.479 \text{ mm.}$

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011
 K5 = 0.7603 K6 = 0.0529 K7 = 0.0132 K8 = 0.3405
 K9 = 0.2711 K10 = 0.0581 K1* = 0.1923

Note: Dimension a is less than Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -29321 \cdot 500.0 [1 - (1 - 500.0/6555.8 + (1364.75^2 - 689.25^2)/(2 \cdot 500.0 \cdot 6555.8))/(1 + (4 \cdot 689.25)/(3 \cdot 6555.8))]$$

$$= -61.8 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 29321 \cdot 6556/4 ((1 + 2(1365^2 - 689^2)/(6556^2))/(1 + (4 \cdot 689)/(3 \cdot 6556)) - 4 \cdot 500/6556)$$

$$= 30208.8 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$$

$$= 33.839 \cdot 1364.75/(2 \cdot 26.5) - 30208.8/(\pi \cdot 1364.8^2 \cdot 26.5)$$

$$= 85.23 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 33.839 \cdot 1364.75/(2 \cdot 26.5) + 30208.8/(\pi \cdot 1364.8^2 \cdot 26.5)$$

$$= 89.05 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$= P \cdot Rm/(2t) - M1/(\pi \cdot Rm^2 \cdot t)$$

$$= 33.839 \cdot 1364.75/(2 \cdot 26.5) - 61.8/(\pi \cdot 1364.8^2 \cdot 26.5)$$

$$= 87.14 \text{ N./mm}^2$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot Rm/(2t) + M1/(\pi \cdot Rm^2 t)$$

$$= 33.839 \cdot 1364.75/(2 \cdot 26.5) + 61.8/(\pi \cdot 1364.8^2 \cdot 26.5)$$

$$= 87.14 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot hm/3))$$

$$= 29321(6555.8 - 2 \cdot 500.0)/(6555.8 + (4 \cdot 689.25/3))$$

$$= 21793.5 \text{ Kg}$$

Shear Stress in the shell no rings, stiffened (4.15.13) [τ_3]:

$$= K3 \cdot Q / (Rm \cdot t)$$

$$= 0.8799 \cdot 29321/(1364.75 \cdot 26.5)$$

$$= 7.00 \text{ N./mm}^2$$

Shear Stress in the head, shell stiffened (4.15.14) [τ_{3*}]:

$$= K3 \cdot Q / (Rm \cdot th)$$

$$= 0.8799 \cdot 29321/(1364.75 \cdot 25.5)$$

$$= 7.27 \text{ N./mm}^2$$

Membrane stress in the head as a stiffener (4.15.16) [σ_5]:

$$= K4 \cdot Q/(Rm \cdot th) + ((P \cdot Ri)/(2 \cdot th)) \cdot (Ri/hm)$$

$$= 0.4011 \cdot 29321/(1364.75 \cdot 25.5) + ((33.84 \cdot 1351.5)/(2 \cdot 25.5)) \cdot (1351.5/689.25)$$

$$= 179.16 \text{ N./mm}^2$$

Decay Length (4.15.20) [x1, x2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1364.75 \cdot 26.5)}$$

$$= 148.335 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 29321 \cdot 0.1/(26.5(300.0 + 148.34 + 148.34))$$

$$= -1.38 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_7^*]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -29321 / (4 \cdot 26.5(300.0 + 148.335 + 148.335)) -$$

$$12 \cdot 0.013 \cdot 29321 \cdot 1364.75 / (6555.799 \cdot 26.5^2)$$

$$= -18.06 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(300.0 + 1.56 \cdot \sqrt{1364.75 \cdot 26.5}, 2 \cdot 500.0)$$

$$= 596.67 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2490.0000	mm.
Baseplate Thickness	Bpthk	25.0000	mm.
Baseplate Width	Bpwid	340.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	16.0000	mm.
Web Thickness	Webtk	16.0000	mm.
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.100	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{R \cdot t} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{1351.5 \cdot 26.5} + 400.0) 26.5^3) / 12$$

$$= 107.816 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C_1 - Y)^2$$

$$= 107.816 + 184.235 (175.499 - 13.25)^2$$

$$= 48607.152 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	695.2	26.5	13.3	184.2	244111.5	0.486E+05
Wearplate	400.0	16.0	34.5	64.0	220800.0	0.127E+05
Web	16.0	481.0	283.0	77.0	2177968.5	0.237E+05
BasePlate	340.0	25.0	536.0	85.0	4556001.0	0.111E+06
Totals	...	...	...	410.2	7198881.0	0.196E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 7198881 / 410.195$$

$$= 175.499 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K_1 \cdot Q$$

$$= 0.204 \cdot 29321.078$$

$$= 5967.4756 \text{ Kg}$$

$$\text{Tension Stress, } S_t = (F_h / A_s) = 2.5899 \text{ N./mm}^2$$

$$\text{Allowed Stress, } S_a = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1900.0 - 1351.5 \cdot \sin(120.0/2)/1.0472$$

$$= 782.319 \text{ mm.}$$

$$\text{Bending Moment, } M = F_h \cdot d = 4668.5586 \text{ Kg-m.}$$

$$\text{Bending Stress, } S_b = (M \cdot C_1 / I) = 4.1081 \text{ N./mm}^2$$

$$\text{Allowed Stress, } S_a = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(29321 + 862)340.0 / (2 \cdot 2490.0 \cdot 173.33)}$$

$$= 18.702 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1350.0 + 28.0 + 16.0)$$

$$= 2414.479 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2414.479 / (5 - 1)$$

$$= 603.620 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 603.6198 \cdot 340.0/2$$

$$= 1026.154 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 30183.426 / (2490.0 \cdot 340.0)$$

$$= 3.565 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 1026.2 \cdot 3.57$$

$$= 3658.497 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 45.44 + 48.29$$

$$= 93.730 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 3658.5 / 93.7296$$

$$= 3.828 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	16.0	284.0	158.0	45.4	717951.9	0.577E+04
Web	301.8	16.0	8.0	48.3	38631.7	0.256E+04
Totals	...	...	...	93.7	756583.6	0.833E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 300.0 - 16.0$$

$$= 284.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 756584 / 93.73$$

$$= 80.720 \text{ mm.}$$

Distance to Centroid [C1]:
 = max(ytot, Saddle Width - ytot)
 = max(80.72, 300.0 - 80.72)
 = 219.280 mm.

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{8331.9/93.73}$
 = 94.283 mm.

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1900.0 - cos(120.0/2)(1350.0 + 28.0 + 16.0) - 25.0
 = 1178.000 mm.

Intermediate Term [Cc]:
 = $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
 = $\sqrt{(2 \cdot \pi^2 \cdot 0.19904\text{E}+09/260.0)}$
 = 122.933

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 1178.0/94.283
 = 12.494

Bending Moment [Rm]:
 = FI / (2 • Bplen) • e • L / 2
 = 557.3 / (2 • 2490.0) • 603.62 • 1178.0/2
 = 39.786 Kg-m.

Compressive Allowable, KL/r < Cc (12.4943 < 122.9326) per AISC E2-1 [Sca]:
 = $(1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$
 = $(1 - (12.49)^2 / (2 \cdot 122.93^2)) 260 /$
 = $(5/3 + 3 \cdot (12.49) / (8 \cdot 122.93) - (12.49^3) / (8 \cdot 122.93^3))$
 = 151.7 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
 = Sc/Sca + (Rm • C1 / I) / Sba
 = 3.83/151.73 + (39.79 • 219.28/83318912) / 173.33
 = 0.031

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	284.0	158.0	45.4	717951.9	0.778E+04
Web	603.6	16.0	8.0	96.6	77263.3	0.225E+04
Totals	...	...	...	142.0	795215.2	0.100E+05

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 795215/142.019
 = 55.994 mm.

Distance to Centroid [C1]:
 = max(ytot, abs(Saddle Width - ytot))
 = max(55.994, abs(300.0 - 55.994))
 = 244.006 mm.

Length of Inner Rib [L]:
 = Saddle Height - Outside Radius - Bpthk
 = 1900.0 - cos(1394.0/2)(25.0 + 0.0 + 0.0) - 0.0
 = 481.000 mm.

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{10027.5/142.019}$
 = 84.028 mm.

Slenderness ratio $[KL/r]$:

$$= KL/r$$

$$= 1 \cdot 481.0/84.028$$

$$= 5.724$$

Unit Force $[Force, u]$:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 557.273 / (2 \cdot 2490.0)$$

$$= 0.112 \text{ Kg/mm.}$$

Moment at base of inner Rib $[M_{base,c}]$:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.112 \cdot 603.62 \cdot 481.0$$

$$= 32.490 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load $[\sigma_B, \text{base}, c]$:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 32.49 / 410953.75$$

$$= 0.775 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($5.7243 < 122.9326$) per AISC E2-1 $[S_{ca}]$:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (5.72)^2 / (2 \cdot 122.93^2)) 260 /$$

$$(5/3 + 3 \cdot (5.72) / (8 \cdot 122.93) - (5.72^3) / (8 \cdot 122.93^3))$$

$$= 154.2 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 4.97 / 154.21 + (32.49 \cdot 244.006 / 10027.528) / 173.33$$

$$= 0.037$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	330.0 mm.
Nominal Bolt Diameter	Bnd	30.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.9088 cm ²
Saddle Load Q0 (Weight)	Q0	29486.5 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	341.6 Kg
Maximum Transverse Force	Ft	295.2 Kg
Maximum Longitudinal Force	Fl	557.3 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction $[\tau_b, l]$:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 557.3 / (3.909 \cdot 4)$$

$$= 3.5 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction $[\tau_b, t]$:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 295.2 / (3.909 \cdot 4)$$

$$= 1.9 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load $[Bltarear_l]$:

$$= 0.0 \text{ (} Q_0 > Q_L \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load $[Bltarear_s]$:

$$= FI / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 557.27 / (103.2 \cdot 4.0)$$

$$= 0.1324 \text{ cm}^2$$

Bolt Area due to Transverse Load:



Moment on Baseplate Due to Transverse Load [Rmom]:

$$\begin{aligned} &= B \cdot Ft + \text{Sum of X Moments} \\ &= 1900.0 \cdot 295.23 + 0.0 \\ &= 560.94 \text{ Kg-m.} \end{aligned}$$

Eccentricity (e):

$$\begin{aligned} &= Rmom / QO \\ &= 560.94 / 29486.47 \\ &= 19.02 \text{ mm.} < Bplen/6 \text{ --> No Uplift in Transverse direction} \end{aligned}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$\begin{aligned} &= \max[Bltarearl, Bltarears, Bltareart] \\ &= \max[0.0, 0.1324, 0.0] \\ &= 0.1324 \text{ cm}^2 \end{aligned}$$

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

There are more than two saddles present in this model. The load on the saddle supports has been computed as the total weight divided by the number of saddles. Ensure that this is an appropriate assumption for this vessel model. It may be necessary to manually enter the bending moments at the saddle through the Saddle Input dialog. Also note that the dimension Ls is taken to be the distance between the adjoining saddles (first two or last two).

Note:

Wear Pad Width (240.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

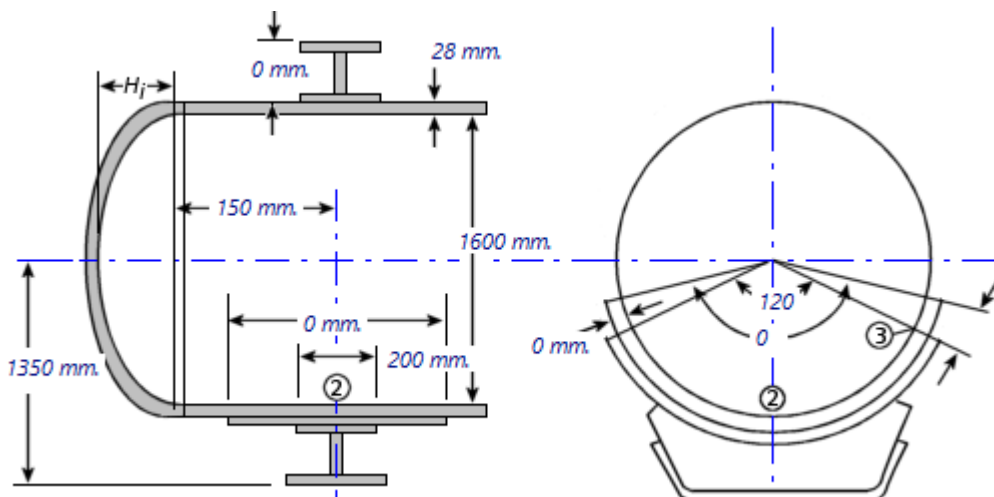
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(200.0 + 1.56 \cdot \sqrt{814.7501 \cdot 26.5}, 2 \cdot 150.0)$$

$$= 300.0000 \text{ mm.}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	814.75	mm.
Shell Thickness used in this Case	t	28.000	mm.
Stiffened Vessel Length per 4.15.6	L	6555.80	mm.
Distance from Saddle to Vessel tangent	a1 or a	150.00	mm.
Saddle Width	b1 or b	200.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1350.00	mm.
Coefficient of Friction	mu	0.60	



Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case 33787.07 Kgf
 Pressure used in Analysis P -0.000 bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-9.23	-235.80
Long. Stress at Top of Midspan	9.23	138.00
Long. Stress at Bottom of Midspan	9.23	138.00

Long. Stress at Top of Saddles	-0.28	-235.80
Long. Stress at Top of Saddles	0.28	138.00
Long. Stress at Bottom of Saddles	0.28	138.00
Tangential Shear in Shell	13.50	110.40
Tangential Shear in Head	2.33	110.40
Circ. Stress at Horn of Saddle	16.63	172.50
Circ. Compressive Stress in Shell	2.21	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(2683.9/3 + 0) \cdot 1350.0/1461.851$
 $= 2478.5\ Kg$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(649.49, 0) \cdot 1350.0/2555.0$
 $= 343.2\ Kg$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl,eq; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(7541.84, 0) \cdot 1350.0/2555.0$
 $= 3984.9\ Kg$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(7542/3 + 0) \cdot 1350.0/1461.851$
 $= 6964.8\ Kg$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 26822 + \max(343, 2479, 3985, 6965)$
 $= 33787.1\ Kg$

Longitudinal Wind Force [Fl]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 1587.169(0.0 + \pi/4(1.796 \cdot 1.2)^2)$
 $= 6369.188\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	34168.93	Kgf
Transverse Shear Load	2513.95	Kgf
Longitudinal Shear Load	7541.84	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(ShellID/2 + t + WearPadThickness)\sin(\theta/2), 2 \cdot Rm)$
 $= \min(2(1600.0/2 + 28.0 + 16.0)\sin(120.0/2), 2 \cdot 814.75)$
 $= 1461.851\ mm$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0132	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is less than $Rm/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/(3L))]$
 $= -33787 \cdot 150.0 [1 - (1 - 150.0/6555.8 + (814.75^2 - 0.0^2) / (2 \cdot 150.0 \cdot 6555.8)) / (1 + (4 \cdot 0.0)/(3 \cdot 6555.8))]$
 $= 1594.7\ Kg \cdot m$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 \cdot ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$
 $= 33787 \cdot 6556/4 \cdot ((1 + 2(815^2 - 0^2)/(6556^2)) / (1 + (4 \cdot 0)/$



$$(3 \cdot 6556) - 4 \cdot 150 / 6556 \\ = 52018.9 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t) \\ = -0.0 \cdot 814.75 / (2 \cdot 26.5) - 52018.9 / (\pi \cdot 814.8^2 \cdot 26.5) \\ = -9.23 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t) \\ = -0.0 \cdot 814.75 / (2 \cdot 26.5) + 52018.9 / (\pi \cdot 814.8^2 \cdot 26.5) \\ = 9.23 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$= P \cdot R_m / (2t) - M_1 / (\pi \cdot R_m^2 \cdot t) \\ = -0.0 \cdot 814.75 / (2 \cdot 26.5) - 1594.7 / (\pi \cdot 814.8^2 \cdot 26.5) \\ = -0.28 \text{ N./mm}^2$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot R_m / (2t) + M_1 / (\pi \cdot R_m^2 \cdot t) \\ = -0.0 \cdot 814.75 / (2 \cdot 26.5) + 1594.7 / (\pi \cdot 814.8^2 \cdot 26.5) \\ = 0.28 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3)) \\ = 33787(6555.8 - 2 \cdot 150.0) / (6555.8 + (4 \cdot 0.0 / 3)) \\ = 32240.9 \text{ Kg}$$

Shear Stress in the shell no rings, stiffened (4.15.13) [τ_3]:

$$= K_3 \cdot Q / (R_m \cdot t) \\ = 0.8799 \cdot 33787 / (814.7501 \cdot 26.5) \\ = 13.50 \text{ N./mm}^2$$

Shear Stress in the head, shell stiffened (4.15.14) [τ_3^*]:

$$= K_3 \cdot Q / (R_m \cdot t_h) \\ = 0.8799 \cdot 33787 / (814.7501 \cdot 153.5) \\ = 2.33 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t} \\ = 0.78 \cdot \sqrt{814.75 \cdot 26.5} \\ = 114.612 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2)) \\ = -0.7603 \cdot 33787 \cdot 0.1 / (26.5(200.0 + 114.61 + 114.61)) \\ = -2.21 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2) \\ = -33787 / (4 \cdot 26.5(200.0 + 114.612 + 114.612)) - \\ 3 \cdot 0.0132 \cdot 33787 / (2 \cdot 26.5^2) \\ = -16.63 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a) \\ = \min(200.0 + 1.56 \cdot \sqrt{814.75 \cdot 26.5}, 2 \cdot 150.0) \\ = 300.00 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1530.0000	mm.
Baseplate Thickness	Bpthk	25.0000	mm.
Baseplate Width	Bpwid	240.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	16.0000	mm.
Web Thickness	Webtk	16.0000	mm.
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	242.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.



Friction Coefficient mu 0.600

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(801.5 \cdot 26.5)} + 240.0)^3) / 12$$

$$= 72.477 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 72.477 + 123.848 (190.187 - 13.25)^2$$

$$= 38845.180 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	467.4	26.5	13.3	123.8	164099.1	0.388E+05
Wearplate	240.0	16.0	34.5	38.4	132480.0	0.932E+04
Web	16.0	481.0	283.0	77.0	2177967.8	0.215E+05
BasePlate	240.0	25.0	536.0	60.0	3216000.0	0.718E+05
Totals	...	...	...	299.2	5690547.0	0.141E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 5690547 / 299.208$$

$$= 190.187 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 33787.07$$

$$= 6876.4019 \text{ Kg}$$

Tension Stress, St = (Fh/As) = 3.8456 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 145.1982 N./mm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1350.0 - 801.5 \cdot \sin(120.0/2) / 1.0472$$

$$= 687.165 \text{ mm.}$$

Bending Moment, M = Fh * d = 4725.3149 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 6.2323 N./mm²
 Allowed Stress, Sa = 2/3 * Yield Str = 161.3313 N./mm²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(33787 + 382)240.0 / (2 \cdot 1530.0 \cdot 161.331)}$$

$$= 22.107 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (800.0 + 28.0 + 16.0)$$

$$= 1461.851 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1461.8508 / (4 - 1)$$

$$= 487.284 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot B_{pwid} / 2$$

$$= 487.2836 \cdot 240.0 / 2$$

$$= 584.740 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 34168.934 / (1530.0 \cdot 240.0)$$

$$= 9.305 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 584.7 \cdot 9.31$$

$$= 5441.164 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 29.44 + 38.983$$

$$= 68.423 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 5441.2 / 68.4227$$

$$= 7.799 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	16.0	184.0	108.0	29.4	317951.9	0.179E+04
Web	243.6	16.0	8.0	39.0	31186.1	730.
Totals	...	...	...	68.4	349138.1	0.252E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 200.0 - 16.0$$

$$= 184.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 349138.094 / 68.423$$

$$= 51.027 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{Saddle Width} - y_{tot})$$

$$= \max(51.027, 200.0 - 51.027)$$

$$= 148.973 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(2516.2 / 68.423)}$$

$$= 60.642 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1350.0 - \cos(120.0/2) (800.0 + 28.0 + 16.0) - 25.0$$

$$= 903.000 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19904E+09 / 242.0)}$$

$$= 127.422$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 903.0/60.642$$

$$= 14.891$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 7541.8 / (2 \cdot 1530.0) \cdot 487.284 \cdot 903.0 / 2$$

$$= 542.255 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ (14.8907 < 127.4221) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (14.89)^2 / (2 \cdot 127.42^2)) 242 /$$

$$(5/3 + 3 \cdot (14.89) / (8 \cdot 127.42) - (14.89^3) / (8 \cdot 127.42^3))$$

$$= 140.5 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 7.8 / 140.53 + (542.26 \cdot 148.973 / 25162118) / 161.33$$

$$= 0.251$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	184.0	108.0	29.4	317951.9	0.238E+04
Web	487.3	16.0	8.0	78.0	62372.3	602.
Totals	...	...	...	107.4	380324.2	0.298E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 380324.25 / 107.405$$

$$= 35.410 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(35.41, \text{abs}(200.0 - 35.41))$$

$$= 164.590 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (R_o + W_{pdthk})^2 - (Pitch/2)^2 } - B_{pthk}$$

$$= 1350.0 - \sqrt{ (844.0 + 16.0)^2 - (487.284/2)^2 } - 25.0$$

$$= 516.932 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (2984.3 / 107.405) }$$

$$= 52.712 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 516.932 / 52.712$$

$$= 9.807$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 7541.839 / (2 \cdot 1530.0)$$

$$= 2.465 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 2.465 \cdot 487.284 \cdot 516.932$$

$$= 620.839 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 620.839 / 181316.016$$

$$= 33.579 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (9.8068 < 127.4221) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.81)^2 / (2 \cdot 127.42^2)) / 242 /$$

$$(5/3 + 3 \cdot (9.81) / (8 \cdot 127.42) - (9.81^3) / (8 \cdot 127.42^3))$$

$$= 142.3 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 9.72/142.31 + (620.84 \cdot 164.59/2984.275) / 161.33$$

$$= 0.276$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (400.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{ Rm \cdot t }, 2a)$$

$$= \min(300.0 + 1.56 \cdot \sqrt{ 1364.75 \cdot 26.5 }, 2 \cdot 500.0)$$

$$= 596.6703 \text{ mm.}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1364.75	mm.
Shell Thickness used in this Case	t	28.000	mm.
Stiffened Vessel Length per 4.15.6	L	6555.80	mm.
Distance from Saddle to Vessel tangent	a1 or a	500.00	mm.
Saddle Width	b1 or b	300.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	689.25	mm.
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1900.00	mm.
Coefficient of Friction	mu	0.10	

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		32757.10	Kgf
Pressure used in Analysis	P	-1.034	bars

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		-4.80	-217.80
Long. Stress at Top of Midspan		4.80	138.00
Long. Stress at Bottom of Midspan		-0.53	-217.80
Long. Stress at Bottom of Midspan		0.53	138.00
Long. Stress at Top of Saddles		-2.66	-217.80
Long. Stress at Top of Saddles		2.66	138.00
Long. Stress at Bottom of Saddles		-2.67	-217.80
Long. Stress at Bottom of Saddles		2.67	138.00
Tangential Shear in Shell		7.82	110.40
Tangential Shear in Head		8.12	110.40
Circ. Stress at Horn of Saddle		20.18	172.50
Addl. Stress in Head as Stiffener		133.69	172.50
Circ. Compressive Stress in Shell		1.54	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3.0(2683.9/3 + 0) \cdot 1900.0/2414.4792$$

$$= 2112.0 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(1688.71, 0) \cdot 1900.0/3100.0$
 $= 1035.0 \text{ Kg}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(7541.84, 0) \cdot 1900.0/3100.0$
 $= 4622.4 \text{ Kg}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr}(F_{t,Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(7542/3 + 0) \cdot 1900.0/2414.4792$
 $= 5934.8 \text{ Kg}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 26822 + \max(1035, 2112, 4622, 5935)$
 $= 32757.1 \text{ Kg}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1587.169(0.0 + \pi/4(2.896 \cdot 1.2)^2)$
 $= 16560.307 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	33619.45	Kgf
Transverse Shear Load	2513.95	Kgf
Longitudinal Shear Load	7541.84	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(2700.0/2 + 28.0 + 16.0) \sin(120.0/2), 2 \cdot 1364.75)$
 $= 2414.479 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0132	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K11* = 0.1923	

Note: Dimension a is less than $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$
 $= -32757 \cdot 500.0 [1 - (1 - 500.0/6555.8 + (1364.75^2 - 689.25^2)/(2 \cdot 500.0 \cdot 6555.8)) / (1 + (4 \cdot 689.25)/(3 \cdot 6555.8))]$
 $= -69.1 \text{ Kg-m}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 [(1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L]$
 $= 32757 \cdot 6556/4 [(1 + 2(1365^2 - 689^2)/(6556^2)) / (1 + (4 \cdot 689)/(3 \cdot 6556)) - 4 \cdot 500/6556]$
 $= 33748.9 \text{ Kg-m}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 \cdot t)$
 $= -1.034 \cdot 1364.75/(2 \cdot 26.5) - 33748.9/(\pi \cdot 1364.8^2 \cdot 26.5)$
 $= -4.80 \text{ N./mm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$
 $= -1.034 \cdot 1364.75/(2 \cdot 26.5) + 33748.9/(\pi \cdot 1364.8^2 \cdot 26.5)$
 $= -0.53 \text{ N./mm}^2$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:
 $= P \cdot R_m/(2t) - M_1/(\pi \cdot R_m^2 \cdot t)$
 $= -1.034 \cdot 1364.75/(2 \cdot 26.5) - 69.1/(\pi \cdot 1364.8^2 \cdot 26.5)$



$$= -2.66 \text{ N./mm}^2$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot R_m / (2t) + M1 / (\pi \cdot R_m^2 t)$$

$$= -1.034 \cdot 1364.75 / (2 \cdot 26.5) + -69.1 / (\pi \cdot 1364.8^2 \cdot 26.5)$$

$$= -2.67 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 32757(6555.8 - 2 \cdot 500.0) / (6555.8 + (4 \cdot 689.25 / 3))$$

$$= 24347.4 \text{ Kg}$$

Shear Stress in the shell no rings, stiffened (4.15.13) [τ_3]:

$$= K3 \cdot Q / (R_m \cdot t)$$

$$= 0.8799 \cdot 32757 / (1364.75 \cdot 26.5)$$

$$= 7.82 \text{ N./mm}^2$$

Shear Stress in the head, shell stiffened (4.15.14) [τ_3^*]:

$$= K3 \cdot Q / (R_m \cdot t_h)$$

$$= 0.8799 \cdot 32757 / (1364.75 \cdot 25.5)$$

$$= 8.12 \text{ N./mm}^2$$

Membrane stress in the head as a stiffener (4.15.16) [σ_5]:

$$= K4 \cdot Q / (R_m \cdot t_h) + ((P \cdot R_i) / (2 \cdot t_h)) \cdot (R_i / h_m)$$

$$= 0.4011 \cdot 32757 / (1364.75 \cdot 25.5) + ((25.02 \cdot 1351.5) / (2 \cdot 25.5)) \cdot (1351.5 / 689.25)$$

$$= 133.69 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1364.75 \cdot 26.5)}$$

$$= 148.335 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 32757 \cdot 0.1 / (26.5(300.0 + 148.34 + 148.34))$$

$$= -1.54 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_7^*]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -32757 / (4 \cdot 26.5(300.0 + 148.335 + 148.335)) - 12 \cdot 0.013 \cdot 32757 \cdot 1364.75 / (6555.799 \cdot 26.5^2)$$

$$= -20.18 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(300.0 + 1.56 \cdot \sqrt{(1364.75 \cdot 26.5)}, 2 \cdot 500.0)$$

$$= 596.67 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2490.0000	mm.
Baseplate Thickness	Bpthk	25.0000	mm.
Baseplate Width	Bpwid	340.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	16.0000	mm.
Web Thickness	Webtk	16.0000	mm.
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	242.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.100	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1351.5 \cdot 26.5)} + 400.0)^3) / 12$$

$$= 107.816 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:
 = ShellInertia + AShell(C1 - Y)²
 = 107.816 + 184.235(175.499 - 13.25)²
 = 48607.152 cm⁴

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	695.2	26.5	13.3	184.2	244111.5	0.486E+05
Wearplate	400.0	16.0	34.5	64.0	220800.0	0.127E+05
Web	16.0	481.0	283.0	77.0	2177968.5	0.237E+05
BasePlate	340.0	25.0	536.0	85.0	4556001.0	0.111E+06
Totals	...	...	...	410.2	7198881.0	0.196E+06

Distance to Centroid [C1]:
 = AY / A
 = 7198881/410.195
 = 175.499 mm.

Angle [β]:
 = 180 - Saddle Angle/2
 = 180 - 120.0/2
 = 120.0

Saddle Splitting Coefficient [K1]:
 = (1 + cos(β) - 0.5•sin(β)²)/(π - β + sin(β)cos(β))
 = (1 + cos(120.0) - 0.5•sin(120.0)²)/(π - 2.094 + sin(120.0)cos(120.0))
 = 0.2035

Saddle Splitting Force [Fh]:
 = K1 • Q
 = 0.204 • 32757.102
 = 6666.7808 Kgf

Tension Stress, St = (Fh/As) = 2.8934 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 145.1984 N./mm²

Saddle Splitting Dimension [d]:
 = B - R • sin(theta/2)/(theta/2 in radians)
 = 1900.0 - 1351.5 • sin(120.0/2)/1.0472
 = 782.319 mm.

Bending Moment, M = Fh * d = 5215.6484 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 4.5895 N./mm²
 Allowed Stress, Sa = 2/3 * Yield Str = 161.3315 N./mm²

Minimum Thickness of Baseplate per Moss:
 = $\sqrt{\frac{3(Q + \text{SaddleWt})\text{BasePlateWidth}}{(2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}}$
 = $\sqrt{\frac{3(32757 + 862)340.0}{(2 \cdot 2490.0 \cdot 161.332)}}$
 = 20.459 mm.

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 = 2 • cos(90 - Saddle Angle/2)(Inside Radius + Shell Thk + Wear Plate Thk)
 = 2 • cos(90 - 120.0/2)(1350.0 + 28.0 + 16.0)
 = 2414.479 mm.

Distance between Ribs [e]:
 = Web Length / (Nribs - 1)
 = 2414.479/(5 - 1)
 = 603.620 mm.

Baseplate Pressure Area [Ap]:
 = e • Bpwid / 2
 = 603.6198 • 340.0/2
 = 1026.154 cm²

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 33619.453 / (2490.0 \cdot 340.0)$$

$$= 3.971 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 1026.2 \cdot 3.97$$

$$= 4074.973 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 45.44 + 48.29$$

$$= 93.730 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 4075.0 / 93.7296$$

$$= 4.264 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	16.0	284.0	158.0	45.4	717951.9	0.577E+04
Web	301.8	16.0	8.0	48.3	38631.7	0.256E+04
Totals	...	...	...	93.7	756583.6	0.833E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 300.0 - 16.0$$

$$= 284.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 756584 / 93.73$$

$$= 80.720 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{Saddle Width} - y_{tot})$$

$$= \max(80.72, 300.0 - 80.72)$$

$$= 219.280 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 8331.9 / 93.73 }$$

$$= 94.283 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1900.0 - \cos(120.0 / 2) (1350.0 + 28.0 + 16.0) - 25.0$$

$$= 1178.000 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 0.19904\text{E}+09 / 242.0) }$$

$$= 127.422$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 1178.0 / 94.283$$

$$= 12.494$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 7541.8 / (2 \cdot 2490.0) \cdot 603.62 \cdot 1178.0 / 2$$

$$= 538.437 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($12.4943 < 127.4221$) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3)) \\
 &= (1 - (12.49)^2 / (2 \cdot 127.42^2)) 242 / \\
 &\quad (5/3 + 3 \cdot (12.49) / (8 \cdot 127.42) - (12.49^3) / (8 \cdot 127.42^3)) \\
 &= 141.4 \text{ N./mm}^2
 \end{aligned}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= Sc/Sca + (Rm \cdot C1 / I) / Sba \\
 &= 4.26/141.39 + (538.44 \cdot 219.28/83318912) / 161.33 \\
 &= 0.116
 \end{aligned}$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	284.0	158.0	45.4	717951.9	0.778E+04
Web	603.6	16.0	8.0	96.6	77263.3	0.225E+04
Totals	...	...	...	142.0	795215.2	0.100E+05

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= AY / A \\
 &= 795215/142.019 \\
 &= 55.994 \text{ mm.}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= \max(ytot, \text{abs(Saddle Width - ytot)}) \\
 &= \max(55.994, \text{abs(300.0 - 55.994)}) \\
 &= 244.006 \text{ mm.}
 \end{aligned}$$

Length of Inner Rib [L]:

$$\begin{aligned}
 &= \text{Saddle Height} - \text{Outside Radius} - Bpthk \\
 &= 1900.0 - \cos(1394.0/2) (25.0 + 0.0 + 0.0) - 0.0 \\
 &= 481.000 \text{ mm.}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{ \text{Total Inertia} / \text{Total Area} } \\
 &= \sqrt{ (10027.5/142.019) } \\
 &= 84.028 \text{ mm.}
 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned}
 &= KL/r \\
 &= 1 \cdot 481.0/84.028 \\
 &= 5.724
 \end{aligned}$$

Unit Force [Force,u]:

$$\begin{aligned}
 &= FI / (2 \cdot \text{Baseplate Length}) \\
 &= 7541.839 / (2 \cdot 2490.0) \\
 &= 1.514 \text{ Kg/mm.}
 \end{aligned}$$

Moment at base of inner Rib [Mbase,c]:

$$\begin{aligned}
 &= \text{Unit Force} \cdot e \cdot L \\
 &= 1.514 \cdot 603.62 \cdot 481.0 \\
 &= 439.709 \text{ Kg-m.}
 \end{aligned}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$\begin{aligned}
 &= \text{Bending Moment} / \text{Section Modulus} \\
 &= 439.709/410953.75 \\
 &= 10.493 \text{ N./mm}^2
 \end{aligned}$$

Compressive Allowable, $KL/r < Cc$ (5.7243 < 127.4221) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3)) \\
 &= (1 - (5.72)^2 / (2 \cdot 127.42^2)) 242 / \\
 &\quad (5/3 + 3 \cdot (5.72) / (8 \cdot 127.42) - (5.72^3) / (8 \cdot 127.42^3)) \\
 &= 143.6 \text{ N./mm}^2
 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= Sc/Sca + (Mbase,c \cdot C1/I) / Sba \\
 &= 5.54/143.6 + (439.71 \cdot 244.006/10027.528) / 161.33 \\
 &= 0.104
 \end{aligned}$$



Review notes about nozzle loadings and supports in the Warnings report.

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Conical Reinforcement Calculations, ASME VIII Div. 1, App. 1

Input Values:

SHELL-2

Design Internal Pressure, Small End	25.098	bars
Design Internal Pressure, Large End	25.098	bars
Temperature for Internal Pressure	85.0	°C
Design External Pressure	1.034	bars
Temperature for External Pressure	85.0	°C

CONE Properties:

Cone Material		SA-516 70	+Q+IT
Cone Allowable Stress at Temperature	Ss	138.00	N./mm ²
Cone Allowable Stress At Ambient		138.00	N./mm ²
Longitudinal Joint Efficiency of Cone	Ec,E2	1.00	
Circumferential Joint Efficiency of Cone		1.00	
Actual Thickness of Cone	Tc	32.0000	mm.
Corrosion Allowance for Cone	Cac	1.5000	mm.
Diameter Basis for Cone and Cylinders		ID	
Diameter of Small End of Cone	Ds	1600.000	mm.
Diameter of Large End of Cone	Dl	2700.000	mm.
Half Apex Angle for Cone		30.00	degrees
Axial Length of Cone	Lc	1905.000	mm.

SMALL-END Properties:

Attached Small End Material Specification		SA-516 70	+Q
Operating Allowable Stress	Ss	138.00	N./mm ²
Ambient Allowable Stress		138.00	N./mm ²
Longitudinal Joint Efficiency	Es,E1	1.00	
Actual Thickness	Ts	28.0000	mm.
Corrosion Allowance	Cas	1.5000	mm.
Axial Length of Attached Element	Ls	300.000	mm.

LARGE-END Properties:

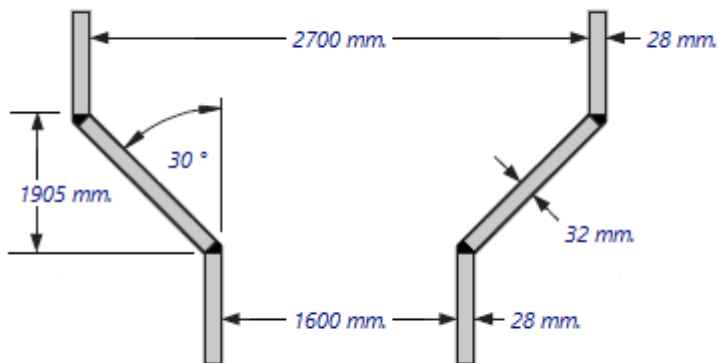
Attached Large End Material Specification		SA-516 70	+Q+IT
Operating Allowable Stress	Sl	138.00	N./mm ²
Ambient Allowable Stress		138.00	N./mm ²
Longitudinal Joint Efficiency	El,E1	1.00	
Actual Thickness	Tl	28.0000	mm.
Corrosion Allowance	Cal	1.5000	mm.
Axial Length of Attached Element	Ll	4375.800	mm.

Type of Reinforcement at Large End of Cone: None

Type of Reinforcement at Small End of Cone: None

Force Arl & Ars calculation, cone -> flange No

Force Arl & Ars calculation, cone -> cylinder No



Cone Geometry

Calculated Values:

Elastic Modulus Data from ASME Section II Part D at 85 °C

Elastic Modulus of Cone Material	0.199E+09 KPa. at 85 °C
Elastic Modulus of Small Cylinder Material	0.199E+09 KPa. at 85 °C
Elastic Modulus of Large Cylinder Material	0.199E+09 KPa. at 85 °C
Elastic Modulus of Large End Reinforcement	0.199E+09 KPa. at 85 °C
Elastic Modulus of Small End Reinforcement	0.199E+09 KPa. at 85 °C

Elastic Modulus Data from ASME Section II Part D at 85 °C

Elastic Modulus of Cone Material	0.199E+09 KPa. at 85 °C
Elastic Modulus of Small Cylinder Material	0.199E+09 KPa. at 85 °C
Elastic Modulus of Large Cylinder Material	0.199E+09 KPa. at 85 °C
Elastic Modulus of Large End Reinforcement	0.199E+09 KPa. at 85 °C
Elastic Modulus of Small End Reinforcement	0.199E+09 KPa. at 85 °C

Axial forces and moments are not computed for Horizontal geometries.
 Compute them manually and enter the loads using the Force/Moment dialog.
 Place the loads at the ends of the cone as needed.

Note: Both ends of the Cone are Lines of Support

Maximum Centroid Reinforcement Distance Large End	47.7736 mm.
Maximum Centroid Reinforcement Distance Small End	37.0321 mm.

Note: No ring was found close enough to the large end to be considered.

Note: No ring was found close enough to the small end to be considered.

Reinforcement Calculations for Cone / Large Cylinder:

Required Area of Reinforcement for Large End Under Internal Pressure

Note: Large end is in compression, 1.0 will be used for E1 and E2.

Large end ratio of pressure to allowable stress [P/Ss E1]: 0.01819

Large End Maximum half apex angle w/o reinforcement [δ]:
 $= 326.6 \cdot \sqrt{P / (Ss \cdot E1)}$ per 1-5 equation 1
 $= 326.6 \cdot \sqrt{25.098 / (137.997 \cdot 1.0)}$
 $= 44.047$ degrees

Large end actual half apex angle [alpha]: 30.000 degrees

Because delta is greater than alpha, reinforcement calculations are not required



for the large end under internal pressure.

Required Area of Reinforcement for Large End Under External Pressure

Large end ratio of pressure to allowable stress [P/Ss E1]: 0.00075

Large End Maximum half apex angle w/o reinforcement [δ]:
 $= 104 \cdot \sqrt{(P / (Ss \cdot E1))}$ per 1-8 equation 1
 $= 104 \cdot \sqrt{(1.034 / (137.997 \cdot 1.0))}$
 $= 2.847$ degrees

Large end actual half apex angle [alpha]: 30.000 degrees

Intermediate Value [k]:
 $= \max(Y / (Srl \cdot Erl), 1)$
 $= \max(0.27467E+11 / (138.0 \cdot 0.19904E+09), 1)$
 $= 1.0000$

where [Y] is:
 $= \text{Large End All. Stress} \cdot \text{Large End Elastic Modulus (Ext. temp.)}$
 $= 138.0 \cdot 0.19904E+09$
 $= 27467233280.0 \text{ (N./mm}^2\text{)}$

Allowable Stress of Large End Material (Ext. Temp)	138.0	N./mm ²
Allowable Stress of Cone Material (Ext. Temp)	138.0	N./mm ²

Required Area of Reinforcement, Large End, External [Arl]:
 $= (k \cdot QL \cdot RI \cdot \tan(\text{angle})) / (Ss \cdot E1) \cdot (1 - 1/4 \cdot ((P \cdot RI - QL) / QL) \cdot (\delta / \alpha))$
 $= (1.0 \cdot 7.2663 \cdot 1378.0 \cdot 0.577 / (138 \cdot 1.0)) \cdot$
 $(1 - 1/4 \cdot ((1.03 \cdot 1378.0 - 7.266) / 7.266) \cdot (2.847 / 30.0))$
 $= 4.011 \text{ cm}^2$

Force per Length, Cone Large End External Pressure, compressive side [QL-]:
 $= P_{ext}(RI/2) + f1$
 $= 1.034(1378.0/2) + 0.0$
 $= 7.266 \text{ Kg/mm.}$

Axial Load per Unit Circumference excluding Pressure [f1]:
 $= F_{axial} / (\pi(DI + TI)) + \text{Moment} / (\pi(RI - TI/2)^2)$
 $= +.0 / (\pi(2756.0 - 28.0)) + .0 / (\pi(1378.0 - 28.0/2)^2)$
 $= 0.0 \text{ Kg/mm.}$

Force per Length, Cone Large End External Pressure, tensile side [QL+]:
 $= P_{ext}(RI/2) + f1$
 $= 1.034(1378.0/2) + 0.0$
 $= 7.266 \text{ Kg/mm.}$

Axial Load per Unit Circumference excluding Pressure [f1]:
 $= F_{axial} / (\pi(DI + TI)) - \text{Moment} / (\pi(RI - TI/2)^2)$
 $= +.0 / (\pi(2756.0 - 28.0)) - 0.0 / (\pi(1378.0 - 28.0/2)^2)$
 $= 0.0 \text{ Kg/mm.}$

Available Area of Reinforcement, Large End, External [Ael]:
 $= 0.55[\sqrt{(DI \cdot ts)(ts - t)} + \sqrt{(DI \cdot tc / \cos(\alpha))(tc - tr)}]$ per 1-8 eqn. 3
 $= 0.55[\sqrt{(2756.0 \cdot 26.5)(26.5 - 10.722)} +$
 $\sqrt{(2756.0 \cdot 30.5/0.866)(30.5 - 8.04)}]$
 $= 61.9373 \text{ cm}^2$

Summary of Reinforcement Area, Large End, External Pressure:

Area of reinforcement required per App. 1-8(1)	4.0108	cm ²
Area of reinforcement in shell per App. 1-8(2)	61.9373	cm ²
Area of reinforcement in stiffening ring	0.0000	cm ²

Intermediate Results, Large End, External Pressure:

Area Available in Cone, Shell, and Reinforcement [AT]:
 $= LL \cdot ts/2 + Lc \cdot tc/2 + As$
 $= 4375.8 \cdot 26.5/2 + 1982.807 \cdot 30.5/2 + 0.0$
 $= 882.172 \text{ cm}^2$

Determine the Meridian Length [M]:

$$= -RL \cdot \tan(\alpha)/2 + LL/2 + (RL^2 - RS^2)/(3 \cdot RL \cdot \tan(\alpha))$$

$$= -1378.0 \cdot \tan(30.0)/2 + 4375.8/2 + (1378.0^2 - 828.0^2)/(3 \cdot 1378.0 \cdot \tan(30.0))$$

$$= 2298.450 \text{ mm.}$$

Force per Unit Length on the Cone/Shell Junction [FL]:

$$= P \cdot M + \tan(\alpha)(\text{Force}/(\pi \cdot D) + \text{Moment}/(\pi \cdot \text{Radius}^2))$$

$$= 1.034 \cdot 2298.5 + \tan(30.0)(0.0/(\pi \cdot 2756.0) + 0.0/(\pi \cdot 1378.0^2))$$

$$= 24.240 \text{ Kg/mm.}$$

Actual Buckling Stress associated with force FL, [B]:

$$= 0.75(FL \cdot DL / ATL)$$

$$= 0.75(24.24 \cdot 2756.0/882.172)$$

$$= 5.570 \text{ N./mm}^2$$

Material Strain associated with this stress [A]: 0.000056

Required Moment of Inertia, Large End, External Pressure [I's]:

$$= A \cdot DI^2 \cdot Atl / 10.9$$

$$= 0.560E-04 \cdot 2756.0002 \cdot 2756.0002 \cdot 882.17/10.9$$

$$= 3440271.00 \text{ mm.}^4$$

Available Moment of Inertia, Large End, External Pressure:

	Area	Centroid	Ar*Ce	Dist	I	Ar*Di ²
Shl	39.389	0.0000	0.000	-24.4844	23.051	23612.938
Con	52.347	-42.9076	-2246.100	18.4232	375.356	17767.518
Sec	0.000	13.2500	0.000	-37.7344	0.000	0.000
TOT	91.736		-2246.100		398.407	41380.457

Centroid of Section -24.4844 Moment of Inertia 812.21

Summary of Large End Inertia Calculations

Available Moment of Inertia (Large End)	812.211	cm ⁴
Required Moment of Inertia (Large End)	344.027	cm ⁴

Reinforcement Calculations for Cone / Small Cylinder:

Required Area of Reinforcement for Small End under Internal Pressure

Small end ratio of pressure to allowable stress [P/Ss E1]: 0.01819

Small End Maximum half apex angle w/o reinforcement [δ]:

$$= 89 \cdot \sqrt{(P / (Ss \cdot E1))} \text{ per 1-5 equation 6}$$

$$= 89 \cdot \sqrt{(25.098 / (137.997 \cdot 1.0))}$$

$$= 12.003 \text{ degrees}$$

Small end actual half apex angle [alpha]: 30.000 degrees

Intermediate Value [k]:

$$= \max(Y / (Sr \cdot Ers), 1)$$

$$= \max(0.27467E+11 / (138.0 \cdot 0.19904E+09), 1)$$

$$= 1.0000$$

where [Y] is:

$$= \text{Small End All. Stress} \cdot \text{Small End Elastic Modulus (Int. temp.)}$$

$$= 138.0 \cdot 0.19904E+09$$

$$= 27467335680.0 \text{ (N./mm}^2\text{)}^2$$

Decay Length, Cone Small End:

$$= 1.4 \cdot \sqrt{(Rs / (ts - ca))}$$

$$= 1.4 \cdot \sqrt{(801.5 / (28.0 - 1.5))}$$

$$= 204.034 \text{ mm.}$$

Required Area of Reinforcement, Small End, Internal [Ars]:

$$= k \cdot QS \cdot Rs / (Ss \cdot E1) \cdot (1 - \delta/\alpha) \cdot \tan(\alpha) \text{ per 1-5 equation 7}$$

$$= 1.0 \cdot 102.5678 \cdot 801.5 / (138 \cdot 1.0) \cdot (1 - 12.0/30.0) \cdot 0.5774$$

$$= 20.23 \text{ cm}^2$$

Force per Length, Cone Small End, tensile side [QS+]:

$$= P(Rs/2) + f2$$



$$= 25.098(801.5/2) + 0.0$$

$$= 102.568 \text{ Kg/mm.}$$

Axial Load per Unit Circumference excluding Pressure [f2]:

$$= \text{Faxial}/(\pi(Ds + Ts)) + \text{Moment}/(\pi(Rs + Ts/2)^2)$$

$$= -0.0/(\pi(1603.0+28.0)) + 0.0/(\pi(801.5+28.0/2)^2)$$

$$= 0.0 \text{ Kg/mm.}$$

Force per Length, Cone Small End, compressive side [QS-]:

$$= P(Rs/2) + f2$$

$$= 25.098(801.5/2) + -0.0$$

$$= 102.568 \text{ Kg/mm.}$$

Axial Load per Unit Circumference excluding Pressure [f2]:

$$= \text{Faxial}/(\pi(Ds + Ts)) - \text{Moment}/(\pi(Rs + Ts/2)^2)$$

$$= -0.0/(\pi(1603.0+28.0)) - 0.0/(\pi(801.5+28.0/2)^2)$$

$$= -0.0 \text{ Kg/mm.}$$

Area of Reinforcement Available in Small End Shell [Aes]:

$$= 0.78[\sqrt{(Rs \cdot ts)} \cdot (ts - t) + \sqrt{(Rs \cdot tc / \cos(\alpha))} \cdot (tc - tr)] \text{ per 1-5 eqn. 8}$$

$$= 0.78[\sqrt{(801.5 \cdot 26.5)} \cdot (26.5 - 14.739) + \sqrt{(801.5 \cdot 30.5 / 0.866)} \cdot (30.5 - 17.019)]$$

$$= 31.0362 \text{ cm}^2$$

Summary of Reinforcement Area, Small End, Internal Pressure:

Area of reinforcement required per App. 1-5(7)	20.2345	cm ²
Area of reinforcement in shell per App. 1-5(8)	31.0362	cm ²
Area of reinforcement in stiffening ring	0.0000	cm ²

Required Area of Reinforcement for Small End Under External Pressure

Note: Small end is in compression, 1.0 will be used for E1 and E2.

Allowable Stress of Small End Material (Ext. Temp)	138.0	N./mm ²
Allowable Stress of Cone Material (Ext. Temp)	138.0	N./mm ²

Intermediate Value [k]:

$$= \max(Y / (Srs \cdot Ers), 1)$$

$$= \max(0.27467E+11 / (138.0 \cdot 0.19904E+09), 1)$$

$$= 1.0000$$

where [Y] is:

$$= \text{Small End All. Stress} \cdot \text{Small End Elastic Modulus (Ext. temp.)}$$

$$= 138.0 \cdot 0.19904E+09$$

$$= 27467335680.0 \text{ (N./mm}^2\text{)}$$

Area of Reinforcement Required in Small End Shell [Ars]:

$$= k \cdot QS \cdot Rs \cdot \tan(\alpha) / (Ss \cdot E1)$$

$$= (1.0 \cdot 4.3661 \cdot 828.0 \cdot 0.5774 / (138 \cdot 1.0))$$

$$= 1.483 \text{ cm}^2$$

Force per Length, Cone Small End External Pressure, compressive side [QS-]:

$$= \text{Pext}(Rs/2) + f2$$

$$= 1.034(828.0/2) + 0.0$$

$$= 4.366 \text{ Kg/mm.}$$

Axial Load per Unit Circumference excluding Pressure [f2]:

$$= \text{Faxial}/(\pi(Ds + Ts)) + \text{Moment}/(\pi(Rs - Ts/2)^2)$$

$$= +0.0/(\pi(1656.0-28.0)) + 0.0/(\pi(828.0-28.0/2)^2)$$

$$= 0.0 \text{ Kg/mm.}$$

Force per Length, Cone Small End External Pressure, tensile side [QS+]:

$$= \text{Pext}(Rs/2) + f2$$

$$= 1.034(828.0/2) + 0.0$$

$$= 4.366 \text{ Kg/mm.}$$

Axial Load per Unit Circumference excluding Pressure [f2]:

$$= \text{Faxial}/(\pi(Ds + Ts)) - \text{Moment}/(\pi(Rs - Ts/2)^2)$$

$$= +0.0/(\pi(1656.0-28.0)) - 0.0/(\pi(828.0-28.0/2)^2)$$

$$= 0.0 \text{ Kg/mm.}$$

Area of Reinforcement Available in Small End Shell [Aes]:

$$= 0.55 \left[\sqrt{(Ds \cdot ts)(ts-t)} + \sqrt{(Ds \cdot tc / \cos(\alpha))(tc-tr)} \right]$$

$$= 0.55 \left[\sqrt{(1656.0 \cdot 26.5)(26.5 - 143.661)} + \sqrt{(1656.0 \cdot 30.5 / 0.866)(30.5 - 5.951)} \right]$$

$$= 32.6076 \text{ cm}^2$$

Summary of Reinforcement Area, Small End, External Pressure:

Area of reinforcement required per App. 1-8(6)	1.4833	cm ²
Area of reinforcement in shell per App. 1-8(7)	32.6076	cm ²
Area of reinforcement in stiffening ring	0.0000	cm ²

Intermediate Results, Small End, External Pressure:

Area Available in Cone, Shell, and Reinforcement	342.13	cm ²
Force per Unit Length on Shell / Cone Junction	8.56	Kgf/mm.
Actual Buckling Stress associated with this Force	3.05	N./mm ²
Material Strain associated with this stress	0.000031	

Required Moment of Inertia, Small End, External Pressure [I's]:

$$= A \cdot Ds^2 \cdot Ats / 10.9$$

$$= 0.306E-04 \cdot 1656.0 \cdot 1656.0 \cdot 342.13 / 10.9$$

$$= 263677.31 \text{ mm}^4$$

Available Moment of Inertia, Small End, External Pressure:

	Area	Centroid	Ar*Ce	Dist	I	Ar*Di ²
Shl	30.532	0.0000	0.000	18.9793	17.868	10998.208
Con	40.577	33.2602	1349.616	-14.2809	191.570	8275.580
Sec	0.000	13.2500	0.000	5.7293	0.000	0.000
TOT	71.110		1349.616		209.437	19273.787

Centroid of Section	18.9793	Moment of Inertia	402.18
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Summary of Small End Inertia Calculations

Available Moment of Inertia (Small End)	0.4022E+03	cm ⁴
Required Moment of Inertia (Small End)	0.2637E+02	cm ⁴



Because the 1/2 apex angle is less than 30 degrees, the following discontinuity stress evaluation results are for information only.

The following calculations are only required per 1-5(g) and do include external loads due to wind or seismic. These discontinuity stresses are computed at the shell/cone junction and do not include effects of local stiffening from a junction ring.

Results for Discontinuity Stresses per Bednar p. 236 2nd Edition:

Stress Type	Stress	Allowable	Location
Tensile Stress	181.23	483.99	Small Cyl. Long.
Compres. Stress	-104.06	-483.99	Small Cyl. Long.
Membrane Stress	146.06	-207.00	Small End Tang.
Tensile Stress	146.39	483.99	Cone Longitudinal
Compres. Stress	-68.97	-483.99	Cone Longitudinal
Compres. Stress	146.31	-207.00	Cone Tangential
Tensile Stress	373.87	483.99	Large Cyl. Long.
Compres. Stress	-244.61	-483.99	Large Cyl. Long.
Membrane Stress	-20.08	-207.00	Large End Tang.
Tensile Stress	298.29	483.99	Cone Longitudinal
Compres. Stress	-168.61	-483.99	Cone Longitudinal
Compres. Stress	-19.66	-207.00	Cone Tangential

An asterisk (*) denotes that this stress was not applicable for this combination of loads.

Maximum Allowable Pressure Calculations for Cone to Shell Junction:

Pressure Case	Pressure bars	Reason for Failure at this Pressure
---------------	---------------	-------------------------------------



MAWP	26.656	Thickness due to internal pressure, Cone Large End
MAPnc	27.988	Thickness due to internal pressure, Cone Large End

These pressures were determined by iteration.

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Input, Nozzle Desc: CH. IN NOZ A1

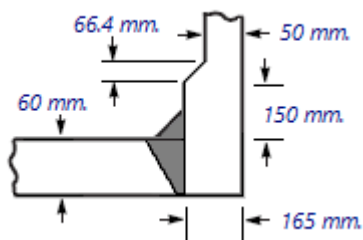
From: 10

Pressure for Reinforcement Calculations	P	191.019	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Hemispherical Head	D	1600.00	mm.
Head Finished (Minimum) Thickness	t	60.0000	mm.
Head Internal Corrosion Allowance	c	1.5000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	495.2000	mm.
User Entered Minimum Design Metal Temperature		-35.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	50.0000	mm.
Hub Height of Integral Nozzle	h	150.0000	mm.
Height of Beveled Transition	L'	66.3953	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		165.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	60.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: CH. IN NOZ A1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	1.969	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
 $= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
 $= 191.02 \cdot 801.5 / (2 \cdot 138 \cdot 1.0 - 0.2 \cdot 191.02)$
 $= 56.2547 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 191.02 \cdot 306.3 / (138 \cdot 1.0 - 0.6 \cdot 191.02)$
 $= 46.2418 \text{ mm.}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 115.0 / 0.5773$
 $= 199.1859 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

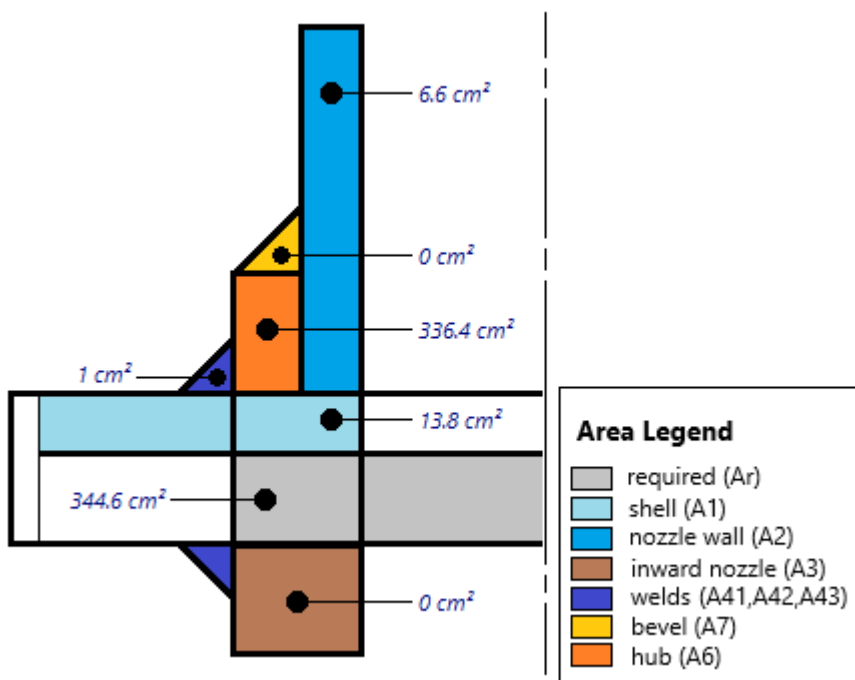
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1225.2000	mm.
Parallel to Vessel Wall, opening length	d	612.6000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	146.2500	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$





Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	344.616	NA	NA
Area in Shell	A1	13.755	NA	NA
Area in Nozzle Wall	A2	6.605	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	336.375	NA	NA
TOTAL AREA AVAILABLE	Atot	357.735	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (612.6 \cdot 56.2547 \cdot 1.0 + 2 \cdot 163.5 \cdot 56.2547 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 344.616 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 612.6(1.0 \cdot 58.5 - 1.0 \cdot 56.255) - 2 \cdot 163.5$$

$$(1.0 \cdot 58.5 - 1.0 \cdot 56.2547) \cdot (1 - 1.0)$$

$$= 13.755 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 146.25)(48.5 - 46.24)1.0$$

$$= 6.605 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(Tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2$$

$$= (2 \cdot \min(146.3, 300.0, 150.0)) \cdot (165.0 - 50.0) \cdot 1.0$$

$$= 336.375 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 47.7418 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 3.0000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 57.7547 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 57.7547 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 9.8312 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.831, \max(57.7547, 3.0)]$$

$$= 9.8312 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(47.7418, 9.8312)$$

$$= 47.7418 \text{ mm.}$$

Available Nozzle Neck Thickness = 50.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):



Sustained :	81.1,	Allowable :	138.0 N./mm ²	Passed
Expansion :	0.0,	Allowable :	263.8 N./mm ²	Passed
Occasional :	55.9,	Allowable :	183.5 N./mm ²	Passed
Shear :	17.2,	Allowable :	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -46 °C

Weld Size Calculations, Description: CH. IN NOZ A1

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (344.6164 - 13.7546 + 2 \cdot 163.5 \cdot 1.0 \cdot (1.0 \cdot 58.5 - 56.2547)) \cdot 138) \\
 &= 475904.19 \text{ Kgf}
 \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2+A4-(W_i - Can/.707)^2 \cdot f_{r2}) \cdot S_v \\
 &= (342.9802 + 1. - 0.0 \cdot 1.0) \cdot 138 \\
 &= 484032.34 \text{ Kgf}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\
 &= (342.9802 + 0.0 + 1. + (191.295)) \cdot 138 \\
 &= 753213.50 \text{ Kgf}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\
 &= (342.9802 + 0.0 + 1. + 0.0 + (191.295)) \cdot 138 \\
 &= 753213.50 \text{ Kgf}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.1416/2.0) \cdot 939.6 \cdot 10.0 \cdot 0.49 \cdot 138 \\
 &= 101765. \text{ Kgf}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\
 &= (3.1416 \cdot 388.05) \cdot (165.0 - 1.5) \cdot 0.7 \cdot 138 \\
 &= 1963342. \text{ Kgf}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\
 &= (3.1416/2.0) \cdot 939.6 \cdot (60.0 - 1.5) \cdot 0.74 \cdot 138
 \end{aligned}$$



= 899069. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (101765 + 1963342) = 2065107 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (101765 + 0 + 899069 + 0) = 1000834 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\ = (101765 + 899069 + 0) = 1000834 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 2065107 Kgf, must exceed W = 475904 Kgf or W1 = 484032 Kgf
 Path 2-2 = 1000834 Kgf, must exceed W = 475904 Kgf or W2 = 753213 Kgf
 Path 3-3 = 1000834 Kgf, must exceed W = 475904 Kgf or W3 = 753213 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 194 bars

The Drop for this Nozzle is : 152.4755 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 512.4755 mm.

Input Echo, WRC107/537 Item 1, Description: CH. IN NOZ A1 :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.5000	mm.
Vessel Diameter	Dv	1600.000	mm.
Vessel Thickness	Tv	60.000	mm.
Design Temperature	T1	85.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	1.5000	mm.
Nozzle Diameter	Dn	609.600	mm.
Nozzle Thickness	Tn	165.000	mm.
Nozzle Material		SA-350 LF2	
Nozzle UNS Number		K03011	
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm ²
Design Internal Pressure	Dp	191.019	bars
Include Pressure Thrust		No	

External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-19579.9	Kgf
Longitudinal Shear (SUS)	(V1) V1	19579.9	Kgf
Circumferential Shear (SUS)	(Vc) V2	14684.9	Kgf
Circumferential Moment (SUS)	(Mc) M1	30532.8	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	30532.8	Kg-m.
Torsional Moment (SUS)	Mt	35230.2	Kg-m.

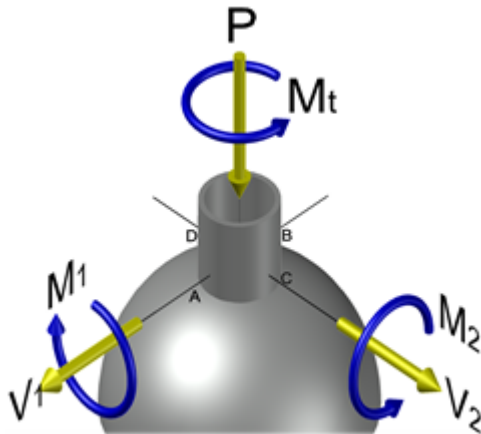
Use Interactive Control

No

WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 939.6 + 2 \cdot 1.65 \cdot \sqrt{(830.75 (60.0 - 1.5))}$
 $= 1667.090 \text{ mm.}$

WRC 107 Stress Calculation for SUSained loads:

Radial Load	P	-19579.9	Kgf
Circumferential Shear	(VC) V2	14684.9	Kgf
Longitudinal Shear	(VL) V1	19579.9	Kgf
Circumferential Moment	(MC) M1	30532.8	Kg-m.
Longitudinal Moment	(ML) M2	30532.8	Kg-m.
Torsional Moment	MT	35230.2	Kg-m.

Dimensionless Param: U = 2.13 TAU = 5.00 (2.37) RHO = 0.36

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.01627	(A,B,C,D)
$M(x) / P$	SP 1	0.01378	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.01386	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.01415	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.01386	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.01415	(A,B,C,D)
$N(y) * T / P$	SP 1	0.02290	(A,B,C,D)
$M(y) / P$	SP 1	0.00493	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.01526	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.00437	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.01526	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.00437	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of	Stress Intensity Values at
---------	----------------------------



Stress	Load	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
Rad.	Bend. P	4.6	-4.6	4.6	-4.6	4.6	-4.6	4.6	-4.6
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-5.5	-5.5	5.5	5.5
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-33.7	33.7	33.7	-33.7
Rad.	Memb. ML	-5.5	-5.5	5.5	5.5	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-33.7	33.7	33.7	-33.7	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-33.6	24.5	44.7	-31.9	-33.6	24.5	44.7	-31.9
Tang.	Memb. P	1.3	1.3	1.3	1.3	1.3	1.3	1.3	1.3
Tang.	Bend. P	1.7	-1.7	1.7	-1.7	1.7	-1.7	1.7	-1.7
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-6.1	-6.1	6.1	6.1
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-10.4	10.4	10.4	-10.4
Tang.	Memb. ML	-6.1	-6.1	6.1	6.1	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-10.4	10.4	10.4	-10.4	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-13.5	4.0	19.4	-4.7	-13.5	4.0	19.4	-4.7
Shear VC		1.7	1.7	-1.7	-1.7	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-2.2	-2.2	2.2	2.2
Shear MT		4.3	4.3	4.3	4.3	4.3	4.3	4.3	4.3
Tot. Shear		5.9	5.9	2.6	2.6	2.0	2.0	6.5	6.5
Str. Int.		35.3	26.1	45.0	32.2	33.8	24.7	46.3	33.4

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	131.3	131.3	131.3	131.3	131.3	131.3	131.3	131.3
Rad.	Pl (SUS)	-4.6	-4.6	6.4	6.4	-4.6	-4.6	6.4	6.4
Rad.	Q (SUS)	-29.0	29.0	38.3	-38.3	-29.0	29.0	38.3	-38.3
Long.	Pm (SUS)	131.3	131.3	131.3	131.3	131.3	131.3	131.3	131.3
Long.	Pl (SUS)	-4.8	-4.8	7.3	7.3	-4.8	-4.8	7.3	7.3
Long.	Q (SUS)	-8.7	8.7	12.1	-12.1	-8.7	8.7	12.1	-12.1
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	1.7	1.7	-1.7	-1.7	-2.2	-2.2	2.2	2.2
Shear	Q (SUS)	4.3	4.3	4.3	4.3	4.3	4.3	4.3	4.3
Pm (SUS)		131.3	131.3	131.3	131.3	131.3	131.3	131.3	131.3
Pm+Pl (SUS)		128.3	128.3	139.9	139.9	128.9	128.9	140.5	140.5
Pm+Pl+Q (Total)		119.4	157.4	176.3	126.8	118.0	156.0	177.6	128.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	131.31	138.00	Passed
Pm+Pl (SUS)	140.46	207.00	Passed
Pm+Pl+Q (TOTAL)	177.61	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: CH. OUT NOZ B1

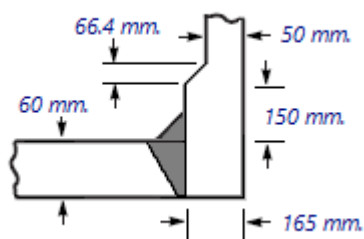
From: 10

Pressure for Reinforcement Calculations	P	191.070	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Hemispherical Head	D	1600.00	mm.
Head Finished (Minimum) Thickness	t	60.0000	mm.
Head Internal Corrosion Allowance	c	1.5000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	495.2000	mm.
User Entered Minimum Design Metal Temperature		-35.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-45.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	50.0000	mm.
Hub Height of Integral Nozzle	h	150.0000	mm.
Height of Beveled Transition	L'	66.3953	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		165.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	60.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: CH. OUT NOZ B1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	1.969	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
 $= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
 $= 191.07 \cdot 801.5 / (2 \cdot 138 \cdot 1.0 - 0.2 \cdot 191.07)$
 $= 56.2702 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 191.07 \cdot 306.3 / (138 \cdot 1.0 - 0.6 \cdot 191.07)$
 $= 46.2555 \text{ mm.}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 115.0 / 0.5773$
 $= 199.1859 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

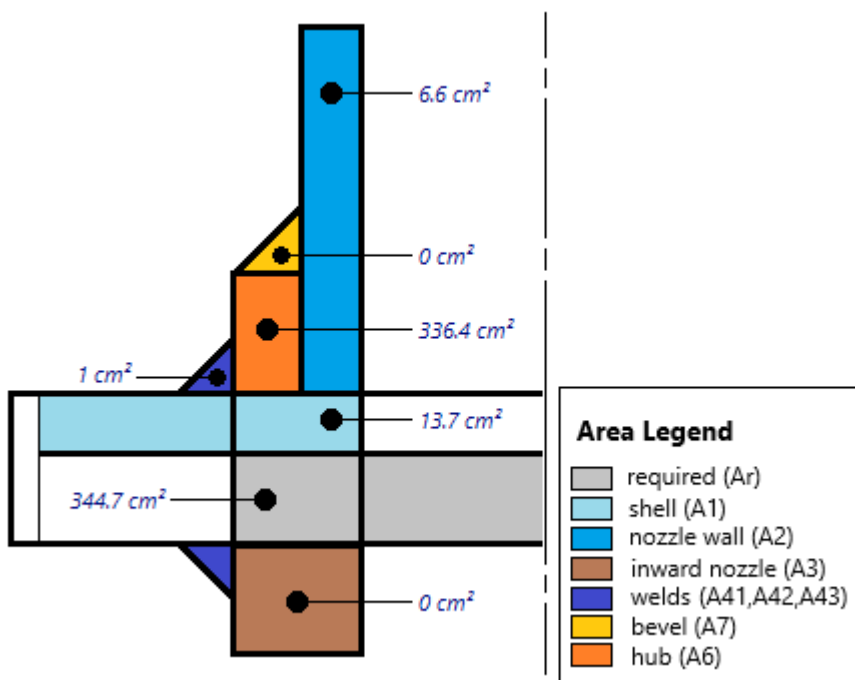
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1225.2000	mm.
Parallel to Vessel Wall, opening length	d	612.6000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	146.2500	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	344.711	NA	NA
Area in Shell	A1	13.660	NA	NA
Area in Nozzle Wall	A2	6.565	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	336.375	NA	NA
TOTAL AREA AVAILABLE	Atot	357.600	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (612.6 \cdot 56.2702 \cdot 1.0 + 2 \cdot 163.5 \cdot 56.2702 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 344.711 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 612.6(1.0 \cdot 58.5 - 1.0 \cdot 56.27) - 2 \cdot 163.5$$

$$(1.0 \cdot 58.5 - 1.0 \cdot 56.2702) \cdot (1 - 1.0)$$

$$= 13.660 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 146.25)(48.5 - 46.26)1.0$$

$$= 6.565 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(Tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2$$

$$= (2 \cdot \min(146.3, 300.0, 150.0)) \cdot (165.0 - 50.0) \cdot 1.0$$

$$= 336.375 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 47.7555 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 3.0000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 57.7702 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 57.7702 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 9.8312 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.831, \max(57.7702, 3.0)]$$

$$= 9.8312 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(47.7555, 9.8312)$$

$$= 47.7555 \text{ mm.}$$

Available Nozzle Neck Thickness = 50.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):



Sustained :	81.2,	Allowable :	138.0 N./mm ²	Passed
Expansion :	0.0,	Allowable :	263.8 N./mm ²	Passed
Occasional :	55.9,	Allowable :	183.5 N./mm ²	Passed
Shear :	17.2,	Allowable :	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -46 °C

Weld Size Calculations, Description: CH. OUT NOZ B1

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (344.7111 - 13.6599 + 2 \cdot 163.5 \cdot 1.0 \cdot (1.0 \cdot 58.5 - 56.2702)) \cdot 138) \\
 &= 476099.56 \text{ Kgf}
 \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2+A4-(W_i - Can/.707)^2 \cdot f_{r2}) \cdot S_v \\
 &= (342.9402 + 1. - 0.0 \cdot 1.0) \cdot 138 \\
 &= 483976.09 \text{ Kgf}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\
 &= (342.9402 + 0.0 + 1. + (191.295)) \cdot 138 \\
 &= 753157.19 \text{ Kgf}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\
 &= (342.9402 + 0.0 + 1. + 0.0 + (191.295)) \cdot 138 \\
 &= 753157.19 \text{ Kgf}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.1416/2.0) \cdot 939.6 \cdot 10.0 \cdot 0.49 \cdot 138 \\
 &= 101765. \text{ Kgf}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\
 &= (3.1416 \cdot 388.05) \cdot (165.0 - 1.5) \cdot 0.7 \cdot 138 \\
 &= 1963342. \text{ Kgf}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\
 &= (3.1416/2.0) \cdot 939.6 \cdot (60.0 - 1.5) \cdot 0.74 \cdot 138
 \end{aligned}$$



= 899069. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (101765 + 1963342) = 2065107 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (101765 + 0 + 899069 + 0) = 1000834 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\ = (101765 + 899069 + 0) = 1000834 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 2065107 Kgf, must exceed W = 476099 Kgf or W1 = 483976 Kgf
 Path 2-2 = 1000834 Kgf, must exceed W = 476099 Kgf or W2 = 753157 Kgf
 Path 3-3 = 1000834 Kgf, must exceed W = 476099 Kgf or W3 = 753157 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 194 bars

The Drop for this Nozzle is : 152.4755 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 512.4755 mm.

Input Echo, WRC107/537 Item 1, Description: CH. OUT NOZ B1:

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.5000	mm.
Vessel Diameter	Dv	1600.000	mm.
Vessel Thickness	Tv	60.000	mm.
Design Temperature	T1	85.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	1.5000	mm.
Nozzle Diameter	Dn	609.600	mm.
Nozzle Thickness	Tn	165.000	mm.
Nozzle Material		SA-350 LF2	
Nozzle UNS Number		K03011	
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm ²
Design Internal Pressure	Dp	191.070	bars
Include Pressure Thrust		No	

External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-19579.9	Kgf
Longitudinal Shear (SUS)	(V1) V1	19579.9	Kgf
Circumferential Shear (SUS)	(Vc) V2	14684.9	Kgf
Circumferential Moment (SUS)	(Mc) M1	30532.8	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	30532.8	Kg-m.
Torsional Moment (SUS)	Mt	35230.2	Kg-m.

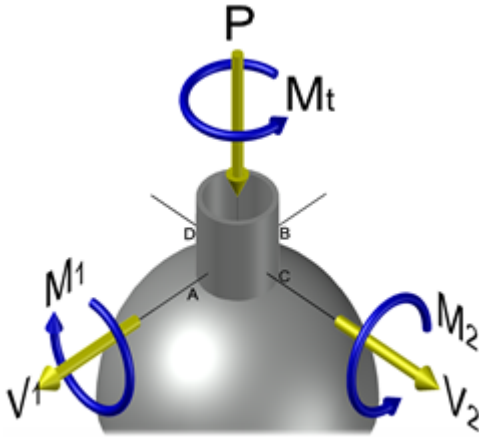
Use Interactive Control

No

WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 939.6 + 2 \cdot 1.65 \cdot \sqrt{(830.75 (60.0 - 1.5))}$
 $= 1667.090 \text{ mm.}$

WRC 107 Stress Calculation for SUSained loads:

Radial Load	P	-19579.9	Kgf
Circumferential Shear	(VC) V2	14684.9	Kgf
Longitudinal Shear	(VL) V1	19579.9	Kgf
Circumferential Moment	(MC) M1	30532.8	Kg-m.
Longitudinal Moment	(ML) M2	30532.8	Kg-m.
Torsional Moment	MT	35230.2	Kg-m.

Dimensionless Param: U = 2.13 TAU = 5.00 (2.37) RHO = 0.36

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.01627	(A,B,C,D)
$M(x) / P$	SP 1	0.01378	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.01386	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.01415	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.01386	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.01415	(A,B,C,D)
$N(y) * T / P$	SP 1	0.02290	(A,B,C,D)
$M(y) / P$	SP 1	0.00493	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.01526	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.00437	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.01526	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.00437	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of	Stress Intensity Values at
---------	----------------------------



Stress	Load	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
Rad.	Bend. P	4.6	-4.6	4.6	-4.6	4.6	-4.6	4.6	-4.6
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-5.5	-5.5	5.5	5.5
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-33.7	33.7	33.7	-33.7
Rad.	Memb. ML	-5.5	-5.5	5.5	5.5	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-33.7	33.7	33.7	-33.7	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-33.6	24.5	44.7	-31.9	-33.6	24.5	44.7	-31.9
Tang.	Memb. P	1.3	1.3	1.3	1.3	1.3	1.3	1.3	1.3
Tang.	Bend. P	1.7	-1.7	1.7	-1.7	1.7	-1.7	1.7	-1.7
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-6.1	-6.1	6.1	6.1
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-10.4	10.4	10.4	-10.4
Tang.	Memb. ML	-6.1	-6.1	6.1	6.1	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-10.4	10.4	10.4	-10.4	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-13.5	4.0	19.4	-4.7	-13.5	4.0	19.4	-4.7
Shear VC		1.7	1.7	-1.7	-1.7	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-2.2	-2.2	2.2	2.2
Shear MT		4.3	4.3	4.3	4.3	4.3	4.3	4.3	4.3
Tot. Shear		5.9	5.9	2.6	2.6	2.0	2.0	6.5	6.5
Str. Int.		35.3	26.1	45.0	32.2	33.8	24.7	46.3	33.4

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	131.3	131.3	131.3	131.3	131.3	131.3	131.3	131.3
Rad.	Pl (SUS)	-4.6	-4.6	6.4	6.4	-4.6	-4.6	6.4	6.4
Rad.	Q (SUS)	-29.0	29.0	38.3	-38.3	-29.0	29.0	38.3	-38.3
Long.	Pm (SUS)	131.3	131.3	131.3	131.3	131.3	131.3	131.3	131.3
Long.	Pl (SUS)	-4.8	-4.8	7.3	7.3	-4.8	-4.8	7.3	7.3
Long.	Q (SUS)	-8.7	8.7	12.1	-12.1	-8.7	8.7	12.1	-12.1
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	1.7	1.7	-1.7	-1.7	-2.2	-2.2	2.2	2.2
Shear	Q (SUS)	4.3	4.3	4.3	4.3	4.3	4.3	4.3	4.3
Pm (SUS)		131.3	131.3	131.3	131.3	131.3	131.3	131.3	131.3
Pm+Pl (SUS)		128.3	128.3	140.0	140.0	128.9	128.9	140.5	140.5
Pm+Pl+Q (Total)		119.5	157.4	176.3	126.9	118.0	156.0	177.6	128.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	131.35	138.00	Passed
Pm+Pl (SUS)	140.50	207.00	Passed
Pm+Pl+Q (TOTAL)	177.65	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

Input, Nozzle Desc: MANHOLE M1

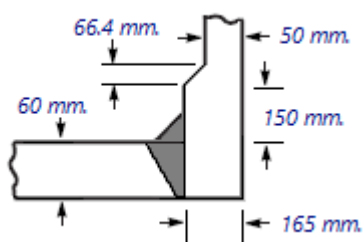
From: 10

Pressure for Reinforcement Calculations	P	191.060	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Hemispherical Head	D	1600.00	mm.
Head Finished (Minimum) Thickness	t	60.0000	mm.
Head Internal Corrosion Allowance	c	1.5000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	546.0000	mm.
User Entered Minimum Design Metal Temperature		-35.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-160.02	deg
Diameter		20.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	50.0000	mm.
Flange Material		SA-350 LF2	+IT
Hub Height of Integral Nozzle	h	150.0000	mm.
Height of Beveled Transition	L'	66.3953	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		165.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		60.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: MANHOLE M1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 20.000 in.
Actual Thickness Used in Calculation 1.969 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $191.06 \cdot 801.5 / (2 \cdot 138 \cdot 1.0 - 0.2 \cdot 191.06)$
= 56.2672 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $191.06 \cdot 255.5 / (138 \cdot 1.0 - 0.6 \cdot 191.06)$
= 38.5818 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 115.0/0.5773
= 199.1859 mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

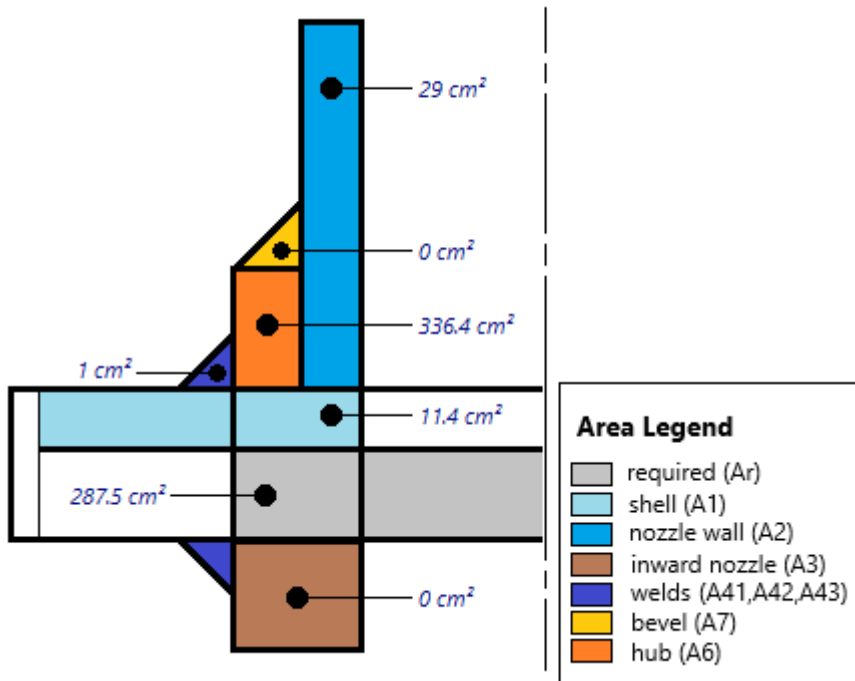
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1022.0000	mm.
Parallel to Vessel Wall, opening length	d	511.0000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	146.2500	mm.

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, 138.0/138.0)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, 138.0/138.0)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1.0, 1.0)
= 1.000



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	287.525	NA	NA
Area in Shell	A1	11.410	NA	NA
Area in Nozzle Wall	A2	29.011	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	336.375	NA	NA
TOTAL AREA AVAILABLE	Atot	377.796	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (511.0 \cdot 56.2672 \cdot 1.0 + 2 \cdot 163.5 \cdot 56.2672 \cdot 1.0 \cdot (1 - 1.0))$
 $= 287.525 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 511.0(1.0 \cdot 58.5 - 1.0 \cdot 56.267) - 2 \cdot 163.5$
 $(1.0 \cdot 58.5 - 1.0 \cdot 56.2672) \cdot (1 - 1.0)$
 $= 11.410 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 146.25)(48.5 - 38.58)1.0$
 $= 29.011 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 1.000 \text{ cm}^2$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(146.3, 300.0, 150.0)) \cdot (165.0 - 50.0) \cdot 1.0$$

$$= 336.375 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 40.0818 \text{ mm.}$
Wall Thickness per UG16(b),	$tr_{16b} = 3.0000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure	$trb_1 = 57.7672 \text{ mm.}$
Wall Thickness	$tb_1 = \max(trb_1, tr_{16b}) = 57.7672 \text{ mm.}$
Wall Thickness	$tb_2 = \max(trb_2, tr_{16b}) = 3.0000 \text{ mm.}$
Wall Thickness per table UG-45	$tb_3 = 9.8312 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[9.831, \max(57.7672, 3.0)]$$

$$= 9.8312 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(40.0818, 9.8312)$$

$$= 40.0818 \text{ mm.}$$

Available Nozzle Neck Thickness = 50.0000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
--	--------

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
--	--------

Gov. MDMT of the nozzle to shell joint welded assembly :	-46 °C
--	--------

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-60 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

$$\text{Design Pressure/Ambient Rating} = 191.06/255.10 = 0.749$$

Weld Size Calculations, Description: MANHOLE M1

Intermediate Calc. for nozzle/shell Welds	T_{min}	19.0000 mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_{r1} \cdot (E_1 \cdot t - tr)) S_v)$$

$$= \max(0, (287.5253 - 11.4097 + 2 \cdot 163.5 \cdot 1.0 \cdot (1.0 \cdot 58.5 - 56.2672)) 138)$$

$$= 398810.69 \text{ Kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.



Weld Load [W1]:

$$\begin{aligned} &= (A2+A4-(W1-Can/.707)^2 \cdot fr2) \cdot Sv \\ &= (365.3858 + 1. - 0.0 \cdot 1.0) \cdot 138 \\ &= 515560.44 \text{ Kgf} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv \\ &= (365.3858 + 0.0 + 1. + (191.295)) \cdot 138 \\ &= 784741.56 \text{ Kgf} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S \\ &= (365.3858 + 0.0 + 1. + 0.0 + (191.295)) \cdot 138 \\ &= 784741.56 \text{ Kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.1416/2.0) \cdot 838.0 \cdot 10.0 \cdot 0.49 \cdot 138 \\ &= 90761. \text{ Kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 337.25) \cdot (165.0 - 1.5) \cdot 0.7 \cdot 138 \\ &= 1706319. \text{ Kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 838.0 \cdot (60.0 - 1.5) \cdot 0.74 \cdot 138 \\ &= 801852. \text{ Kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (90761 + 1706319) = 1797080 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (90761 + 0 + 801852 + 0) = 892613 \text{ Kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (90761 + 801852 + 0) = 892613 \text{ Kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 1797080 Kgf, must exceed W = 398810 Kgf or W1 = 515560 Kgf
Path 2-2 = 892613 Kgf, must exceed W = 398810 Kgf or W2 = 784741 Kgf
Path 3-3 = 892613 Kgf, must exceed W = 398810 Kgf or W3 = 784741 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 198 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 118.5024 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 478.5024 mm.

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Input, Nozzle Desc: LIQ OUT NOZ B3

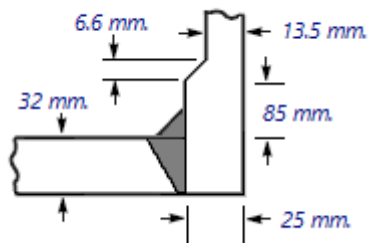
From: 50

Pressure for Reinforcement Calculations	P	25.046	bars
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	85	°C
Parent Material		SA-516 70	+Q+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cone at Nozzle Location	D	1744.49	mm.
Equivalent Length of Conical Section	Le	1368.9332	mm.
Cone Half Apex Angle	Alpha	30.00	Degrees
Shell Finished (Minimum) Thickness	t	32.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1203.18	mm.
User Entered Minimum Design Metal Temperature		-35.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-90.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-350 LF2	+IT
Hub Height of Integral Nozzle	h	85.0000	mm.
Height of Beveled Transition	L'	6.6468	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		25.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	32.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: LIQ OUT NOZ B3

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 3.438 in.
 Actual Thickness Used in Calculation 0.531 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Conical Transition, t_r [Int. Press]
 $= P \cdot D / (2 \cdot \cos(a) \cdot (S_v \cdot E - 0.6 \cdot P))$ Appendix 1-4(e)
 $= 25.046 \cdot 1747.949 / (2 \cdot \cos(30.0) \cdot (137.997 \cdot 1.0 - 0.6 \cdot 25.046))$
 $= 18.5189$ mm.

Reqd Cone thickness at Nozzle Location under External Pres. : 9.1683 mm.

Reqd thk per UG-37(a) of Nozzle Wall, t_{rn} [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 25.05 \cdot 45.16 / (138 \cdot 1.0 - 0.6 \cdot 25.05)$
 $= 0.8288$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.5068 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
 $= 2.5 \cdot \text{Corroded Hub Thickness}$
 $= 2.5 \cdot 23.5$, less than the hub height, use (e-2)
 $= 58.7500$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

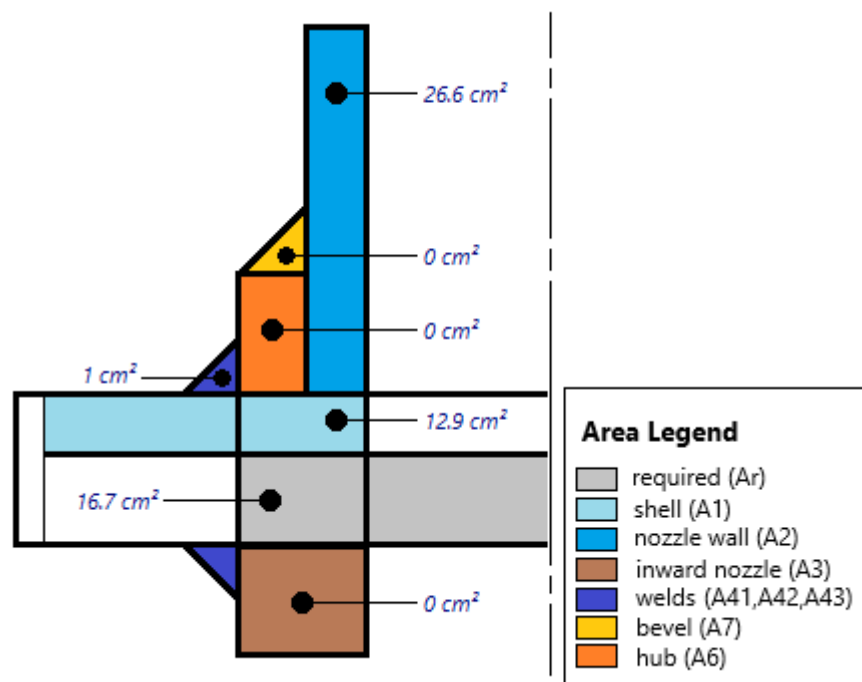
Parallel to Vessel Wall (Diameter Limit)	D1	198.3252	mm.
Parallel to Vessel Wall	$R_n + t_n + t$	99.1626	mm.
Normal to Vessel Wall (Thickness Limit), no pad	T_{lnp}	58.7500	mm.

Hub Height was ≥ 2.5 times Hub Thickness, using sketch UG-40 (e-2).

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	16.727	3.463	NA
Area in Shell	A1	12.940	24.658	NA
Area in Nozzle Wall	A2	26.639	27.017	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	40.578	52.675	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (90.3252 \cdot 18.5189 \cdot 1.0 + 2 \cdot 23.5 \cdot 18.5189 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 16.727 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 108.0(1.0 \cdot 30.5 - 1.0 \cdot 18.519) - 2 \cdot 23.5$$

$$(1.0 \cdot 30.5 - 1.0 \cdot 18.5189) \cdot (1 - 1.0)$$

$$= 12.940 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 58.75)(23.5 - 0.83)1.0$$

$$= 26.639 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

There are no hub area calculations because Figure UG-40 (e-2) was used.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 2.3288$ mm.
 Wall Thickness per UG16(b), $t_{r16b} = 3.0000$ mm.
 Wall Thickness, shell/head, internal pressure $t_{rb1} = 20.0189$ mm.
 Wall Thickness $t_{b1} = \max(t_{rb1}, t_{r16b}) = 20.0189$ mm.
 Wall Thickness, shell/head, external pressure $t_{rb2} = 2.2567$ mm.
 Wall Thickness $t_{b2} = \max(t_{rb2}, t_{r16b}) = 3.0000$ mm.
 Wall Thickness per table UG-45 $t_{b3} = 6.7578$ mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[t_{b3}, \max(t_{b1}, t_{b2})]$
 $= \min[6.7578, \max(20.0189, 3.0)]$
 $= 6.7578$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(t_a, t_b)$
 $= \max(2.3288, 6.7578)$
 $= 6.7578$ mm.

Available Nozzle Neck Thickness = $0.875 \times 13.487 = 11.801$ mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C
 Calculated Minimum Design Metal Temperature -104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C
 Calculated Minimum Design Metal Temperature -104 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -46 °C
 Flange MDMT with Temp reduction per UCS-66(i)(2) -79 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $25.05/51.00 = 0.491$

Weld Size Calculations, Description: LIQ OUT NOZ B3

Intermediate Calc. for nozzle/shell Welds $T_{min} = 19.0000$ mm.

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld $6.0000 = \text{Min per Code}$ $7.0700 = 0.7 \times W_o$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr))S_v)$
 $= \max(0, (16.7272 - 12.9396 + 2 \cdot 23.5 \cdot 1.0 \cdot (1.0 \cdot 30.5 - 18.5189))138)$
 $= 13253.65$ Kgf

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A4-(W_i-Can/.707)^2 \cdot fr2) \cdot Sv$$

$$= (26.6387 + 1. - 0.0 \cdot 1.0) \cdot 138$$

$$= 38891.87 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2(Hub \text{ Thickness}) \cdot t \cdot fr1)) \cdot Sv$$

$$= (26.6387 + 0.0 + 1. + (14.335)) \cdot 138$$

$$= 59063.39 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+(2 \cdot (Hub \text{ Thickness}) \cdot t \cdot fr1)) \cdot Sv$$

$$= (26.6387 + 0.0 + 1. + 0.0) \cdot 138$$

$$= 59063.39 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 137.3252 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 14873. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 56.9126) \cdot (25.0 - 1.5) \cdot 0.7 \cdot 138$$

$$= 41387. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 137.3252 \cdot (32.0 - 1.5) \cdot 0.74 \cdot 138$$

$$= 68508. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (14873 + 41387) = 56261 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (14873 + 0 + 68508 + 0) = 83382 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (14873 + 68508 + 0) = 83382 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 56260	Kgf, must exceed W = 13253 Kgf	or W1 = 38891 Kgf
Path 2-2 = 83381	Kgf, must exceed W = 13253 Kgf	or W2 = 59063 Kgf
Path 3-3 = 83381	Kgf, must exceed W = 13253 Kgf	or W3 = 59063 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 27 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 2.7068 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 284.7068 mm.

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Input, Nozzle Desc: SS IN NOZ. A2

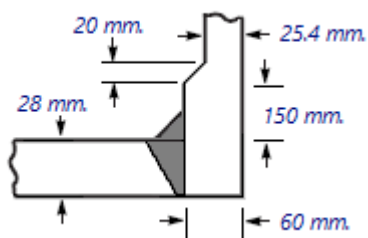
From: 60

Pressure for Reinforcement Calculations	P	25.000	bars
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	85	°C
Parent Material		SA-516 70	+Q+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2700.00	mm.
Design Length of Section	L	4375.7998	mm.
Shell Finished (Minimum) Thickness	t	28.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		6608.17	mm.
User Entered Minimum Design Metal Temperature		-35.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		12.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		120	
Flange Material		SA-350 LF2	+IT
Hub Height of Integral Nozzle	h	150.0000	mm.
Height of Beveled Transition	L'	19.9763	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		60.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	27.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.





Reinforcement CALCULATION, Description: SS IN NOZ. A2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 10.750 in.
Actual Thickness Used in Calculation 1.000 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $25.0 \cdot 1351.5 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$
= 24.7547 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $25.0 \cdot 138.03 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$
= 2.5281 mm.

Required Nozzle thickness under External Pressure per UG-28 : 1.1344 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
= 2.5 • Corroded Hub Thickness
= 2.5 • 58.5, less than the hub height, use (e-2)
= 146.2500 mm.

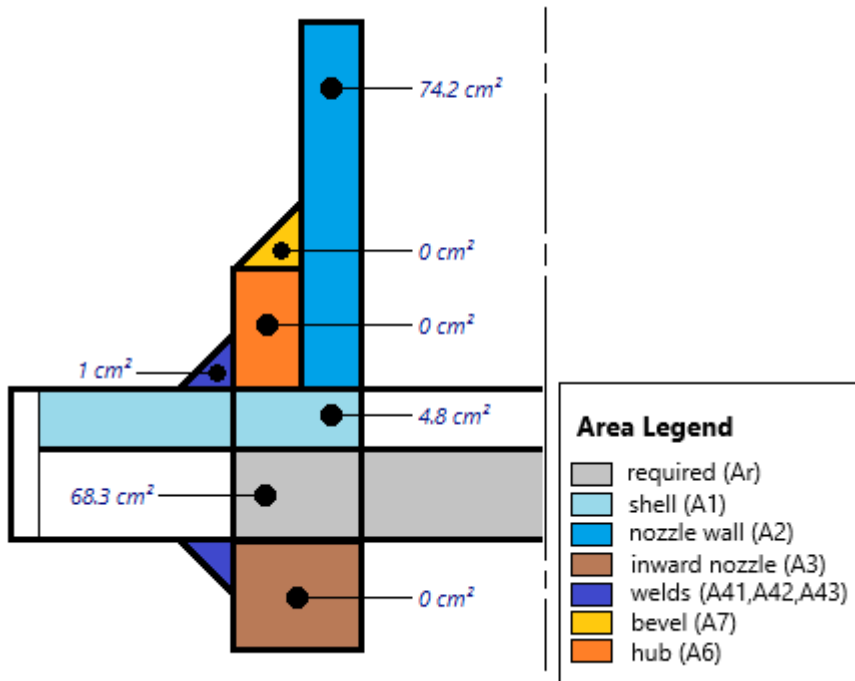
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	552.1000	mm.
Parallel to Vessel Wall, opening length	d	276.0500	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	66.2500	mm.

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, $138.0 / 138.0$)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, $138.0 / 138.0$)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1.0, 1.0)
= 1.000



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	68.335	14.799	NA
Area in Shell	A1	4.818	43.556	NA
Area in Nozzle Wall	A2	74.163	76.009	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	79.981	120.565	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (276.05 \cdot 24.7547 \cdot 1.0 + 2 \cdot 58.5 \cdot 24.7547 \cdot 1.0 \cdot (1 - 1.0))$
 $= 68.335 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 276.05(1.0 \cdot 26.5 - 1.0 \cdot 24.755) - 2 \cdot 58.5$
 $(1.0 \cdot 26.5 - 1.0 \cdot 24.7547) \cdot (1 - 1.0)$
 $= 4.818 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 66.25)(58.5 - 2.53)1.0$
 $= 74.163 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 1.000 \text{ cm}^2$

There are no hub area calculations because Figure UG-40 (e-2) was used.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 4.0281$ mm.
 Wall Thickness per UG16(b), $tr_{16b} = 3.0000$ mm.
 Wall Thickness, shell/head, internal pressure $trb_1 = 26.2547$ mm.
 Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 26.2547$ mm.
 Wall Thickness, shell/head, external pressure $trb_2 = 2.5134$ mm.
 Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 3.0000$ mm.
 Wall Thickness per table UG-45 $tb_3 = 9.8312$ mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb_3, \max(tb_1, tb_2)]$
 $= \min[9.831, \max(26.2547, 3.0)]$
 $= 9.8312$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(t_a, t_b)$
 $= \max(4.0281, 9.8312)$
 $= 9.8312$ mm.

Available Nozzle Neck Thickness = $0.875 * 25.400 = 22.225$ mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	18.5,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	326.5 N./mm ²	Passed
Occasional	:	6.6,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	7.8,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C
 Calculated Minimum Design Metal Temperature -104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -46 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -46 °C
 Flange MDMT with Temp reduction per UCS-66(i)(2) -79 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $25.00/51.00 = 0.490$

Weld Size Calculations, Description: SS IN NOZ. A2

Intermediate Calc. for nozzle/shell Welds T_{min} 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o$ mm.



Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (68.3355 - 4.8178 + 2 \cdot 58.5 \cdot 1.0 \cdot (1.0 \cdot 26.5 - 24.7547)) \cdot 138)$
 $= 92252.38 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i - \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (74.1627 + 1. - 0.0 \cdot 1.0) \cdot 138$
 $= 105765.38 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2(\text{Hub Thickness}) \cdot t \cdot f_{r1})) \cdot S_v$
 $= (74.1627 + 0.0 + 1. + (31.005)) \cdot 138$
 $= 149394.14 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+(2 \cdot (\text{Hub Thickness}) \cdot t \cdot f_{r1})) \cdot S_v$
 $= (74.1627 + 0.0 + 1. + 0.0) \cdot 138$
 $= 149394.14 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 393.05 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 42570. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 167.275) \cdot (60.0 - 1.5) \cdot 0.7 \cdot 138$
 $= 302815. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 393.05 \cdot (28.0 - 1.5) \cdot 0.74 \cdot 138$
 $= 170368. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (42570 + 302815) = 345385 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (42570 + 0 + 170368 + 0) = 212938 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (42570 + 170368 + 0) = 212938 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 345385 Kgf, must exceed W = 92252 Kgf	or W1 = 105765 Kgf
Path 2-2 = 212938 Kgf, must exceed W = 92252 Kgf	or W2 = 149394 Kgf
Path 3-3 = 212938 Kgf, must exceed W = 92252 Kgf	or W3 = 149394 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 27 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars



The Drop for this Nozzle is : 14.3811 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 442.3811 mm.

Input Echo, WRC297 Item 1, Description: SS IN NOZ. A2 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	2700.000	mm.
Shell Thickness	28.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm.
Nozzle Diameter	273.050	mm.
Nozzle Thickness	60.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUStained loads.*

Design Internal Pressure	Dp	25.00	bars
Radial Load	P	-1835.61	Kgf
Circumferential Shear	Vc	1376.71	Kgf
Longitudinal Shear	Vl	1835.61	Kgf
Circumferential Moment	Mc	1431.23	Kg-m.
Longitudinal Moment	Ml	1431.23	Kg-m.
Torsional Moment	Mt	1651.42	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (6.551) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 393.05 + 2 \cdot 1.65 \cdot \sqrt{1364.75(28.0 - 1.5)}$
 $= 1020.622 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2729.500	mm.
Nozzle Outside Diameter	(d) =	393.050	mm.
Vessel Thickness used	(T) =	26.500	mm.
Nozzle Thickness used	(t) =	58.500	mm.
T / t	=	0.453	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.461$$

Nr/P	=	0.083
Mr/P	=	0.176

M0/P = 0.051
 N0/P = 0.110
 MrD/Mc = 0.298
 NrDL/Mc = 0.095
 M0d/Mc = 0.092
 N0DL/Mc = 0.058
 MrD/Ml = 0.148
 NrDL/Ml = 0.081
 M0D/Ml = 0.047
 N0DL/Ml = 0.128

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-6	-6	6	6
Outplane Bending (ML)	-14	14	14	-14
Normal Pressure Stress	126	128	126	128
Outplane Stress Summary	116	131	157	115

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	27	-27	27	-27
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-45	45	45	-45
Inplane Pressure Stress	63	63	63	63
Inplane Stress Summary	43	79	141	-2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	116	131	158	118

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	-2	-2	2	2
Outplane Bending (Mc)	-28	28	28	-28
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0



Normal Pressure Stress	63	63	63	63
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Outplane Stress Summary	42	83	104	32
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	27	-27	27	-27
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-91	91	91	-91
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	126	128	126	128

Inplane Stress Summary	59	189	251	17
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	3	3
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	59	189	251	33
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-6	-6	6	6
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	8	5	8

Hoop Stress Summary	2	5	15	18
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Axial Membrane (P)	0	0	0	0
Axial Bending (P)	4	-4	4	-4
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-3	-3	3	3
Axial Bending (ML)	-6	6	6	-6
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-2	2	17	-4
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	1	1	0	0
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	5	5	17	22
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-2	-2	2	2
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	8	5	8

Hoop Stress Summary	5	8	11	14
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	4	-4	4	-4
Axial Membrane (Mc)	-3	-3	3	3
Axial Bending (Mc)	-15	15	15	-15
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-11	11	26	-13
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	0	0	1	1
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	17	11	26	28
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	126	128	126	128	126	128	126	128
Circ. Pl (SUS)	-4	-4	8	8	-2	-2	6	6
Circ. Q (SUS)	-7	7	21	-21	-64	64	118	-118



Long. Pm (SUS)	63	63	63	63	63	63	63	63
Long. Pl (SUS)	-2	-2	6	6	0	0	4	4
Long. Q (SUS)	-18	18	72	-72	-21	21	35	-35
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	126.0	128.0	126.0	128.0	126.0	128.0	126.0	128.0
Pm+Pl (SUS)	122.0	124.0	134.0	136.0	124.0	126.0	132.0	134.0
Pm+Pl+Q (Total)	115.1	131.1	155.3	118.1	60.1	190.0	250.1	32.5

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	128.00	138.00	Passed
Pm+Pl (SUS)	136.00	207.00	Passed
Pm+Pl+Q (TOTAL)	250.06	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	5	8	5	8	5	8	5	8
Circ. Pl (SUS)	-4	-4	8	8	0	0	4	4
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-3	-3	3	3	-3	-3	3	3
Long. Q (SUS)	-2	2	10	-10	-11	11	19	-19
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	5.0	8.0	5.0	8.0	5.0	8.0	5.0	8.0
Pm+Pl (SUS)	2.0	5.0	13.0	16.0	6.0	9.0	9.0	12.0
Pm+Pl+Q (Total)	4.5	4.3	15.4	21.1	17.1	10.4	24.1	26.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	8.00	138.00	Passed
Pm+Pl (SUS)	16.00	207.00	Passed
Pm+Pl+Q (TOTAL)	26.08	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

Input, Nozzle Desc: SS OUT NOZ. B2

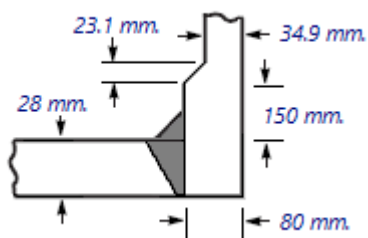
From: 60

Pressure for Reinforcement Calculations	P	25.000	bars
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	85	°C
Parent Material		SA-516 70	+Q+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2700.00	mm.
Design Length of Section	L	4375.7998	mm.
Shell Finished (Minimum) Thickness	t	28.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		4558.17	mm.
User Entered Minimum Design Metal Temperature		-35.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		18.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		120	
Flange Material		SA-350 LF2	+IT
Hub Height of Integral Nozzle	h	150.0000	mm.
Height of Beveled Transition	L'	23.1373	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	27.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.





Reinforcement CALCULATION, Description: SS OUT NOZ. B2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 15.250 in.
Actual Thickness Used in Calculation 1.375 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $25.0 \cdot 1351.5 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$
= 24.7547 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $25.0 \cdot 195.18 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$
= 3.5749 mm.

Required Nozzle thickness under External Pressure per UG-28 : 1.3831 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / $\tan(30)$
= $45.075 / 0.5773$
= 78.0722 mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

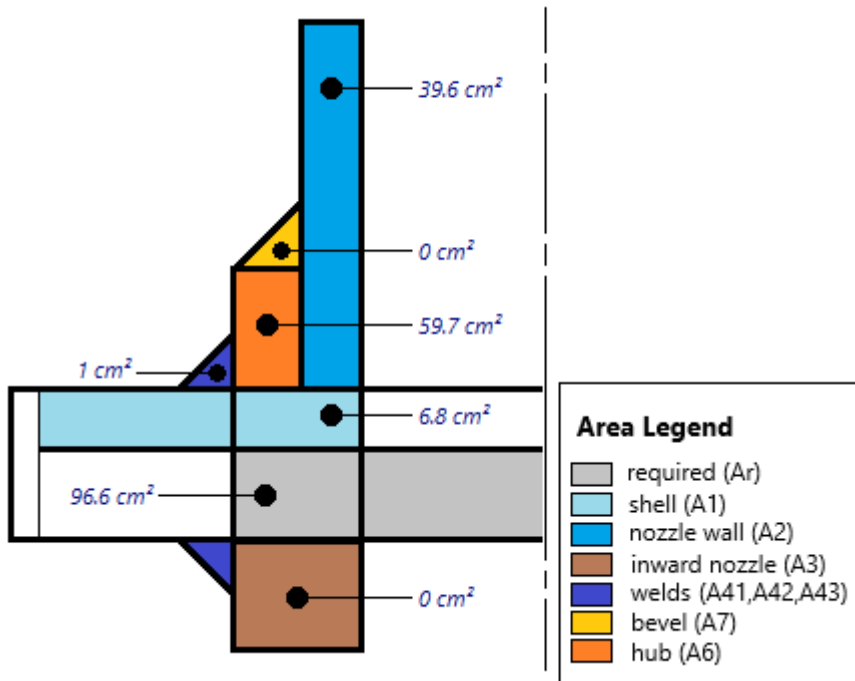
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	780.7000	mm.
Parallel to Vessel Wall, opening length	d	390.3500	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	66.2500	mm.

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 138.0 / 138.0)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 138.0 / 138.0)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1.0, 1.0)$
= 1.000



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	96.630	20.926	NA
Area in Shell	A1	6.813	61.590	NA
Area in Nozzle Wall	A2	39.551	42.456	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	59.724	59.724	NA
TOTAL AREA AVAILABLE	Atot	107.088	164.770	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (390.35 \cdot 24.7547 \cdot 1.0 + 2 \cdot 78.5 \cdot 24.7547 \cdot 1.0 \cdot (1 - 1.0))$
 $= 96.630 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 390.35 (1.0 \cdot 26.5 - 1.0 \cdot 24.755) - 2 \cdot 78.5$
 $(1.0 \cdot 26.5 - 1.0 \cdot 24.7547) \cdot (1 - 1.0)$
 $= 6.813 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2$
 $= (2 \cdot 66.25) (33.42 - 3.57) 1.0$
 $= 39.551 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 1.000 \text{ cm}^2$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2$$

$$= (2 \cdot \min(66.3, 400.0, 150.0)) \cdot (80.0 - 34.9) \cdot 1.0$$

$$= 59.724 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 5.0749 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 3.0000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 26.2547 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 26.2547 \text{ mm.}$
 Wall Thickness, shell/head, external pressure $trb2 = 2.5134 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 9.8312 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[9.831, \max(26.2547, 3.0)]$
 $= 9.8312 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(5.0749, 9.8312)$
 $= 9.8312 \text{ mm.}$

Available Nozzle Neck Thickness = $0.875 \cdot 34.925 = 30.559 \text{ mm.} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	16.3,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	328.7 N./mm ²	Passed
Occasional	:	6.7,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	6.2,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-46 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-79 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $25.00/51.00 = 0.490$

Weld Size Calculations, Description: SS OUT NOZ. B2

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000 mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (96.6302 - 6.8126 + 2 \cdot 78.5 \cdot 1.0 \cdot (1.0 \cdot 26.5 - 24.7547)) \cdot 138)$
 $= 130242.60 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (99.2758 + 1. - 0.0 \cdot 1.0) \cdot 138$
 $= 141103.20 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (99.2758 + 0.0 + 1. + (41.605)) \cdot 138$
 $= 199647.77 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (99.2758 + 0.0 + 1. + 0.0 + (41.605)) \cdot 138$
 $= 199647.77 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 547.35 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 59282. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 234.425) \cdot (80.0 - 1.5) \cdot 0.7 \cdot 138$
 $= 569461. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 547.35 \cdot (28.0 - 1.5) \cdot 0.74 \cdot 138$
 $= 237249. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (59282 + 569461) = 628743 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (59282 + 0 + 237249 + 0) = 296531 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (59282 + 237249 + 0) = 296531 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 628743 Kgf, must exceed W = 130242 Kgf	or W1 = 141103 Kgf
Path 2-2 = 296531 Kgf, must exceed W = 130242 Kgf	or W2 = 199647 Kgf
Path 3-3 = 296531 Kgf, must exceed W = 130242 Kgf	or W3 = 199647 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 26 bars



Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 28.0310 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 456.0311 mm.

Input Echo, WRC297 Item 1, Description: SS OUT NOZ. B2:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	2700.000	mm.
Shell Thickness	28.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm.
Nozzle Diameter	387.350	mm.
Nozzle Thickness	80.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	25.00	bars
Radial Load	P	-2753.42	Kgf
Circumferential Shear	Vc	2065.06	Kgf
Longitudinal Shear	Vl	2753.42	Kgf
Circumferential Moment	Mc	3220.26	Kg-m.
Longitudinal Moment	Ml	3220.26	Kg-m.
Torsional Moment	Mt	3715.68	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (6.842) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 547.35 + 2 \cdot 1.65 \cdot \sqrt{(1364.75(28.0 - 1.5))}$$

$$= 1174.922 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2729.500	mm.
Nozzle Outside Diameter	(d) =	547.350	mm.
Vessel Thickness used	(T) =	26.500	mm.
Nozzle Thickness used	(t) =	78.500	mm.
T / t	=	0.338	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.035$$

$$Nr/P = 0.069$$



Mr/P = 0.142
 M0/P = 0.041
 N0/P = 0.087
 MrD/Mc = 0.286
 NrDL/Mc = 0.094
 M0d/Mc = 0.088
 N0DL/Mc = 0.075
 MrD/Ml = 0.110
 NrDL/Ml = 0.076
 M0D/Ml = 0.035
 N0DL/Ml = 0.111

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-9	-9	9	9
Outplane Bending (ML)	-17	17	17	-17
Normal Pressure Stress	126	128	126	128
Outplane Stress Summary	112	130	165	114

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	32	-32	32	-32
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-54	54	54	-54
Inplane Pressure Stress	63	63	63	63
Inplane Stress Summary	38	80	158	-14

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	113	130	166	129

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	-6	-6	6	6
Outplane Bending (Mc)	-43	43	43	-43
Outplane Membrane (ML)	0	0	0	0



Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	63	63	63	63

Outplane Stress Summary	26	94	125	19
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	32	-32	32	-32
Inplane Membrane (Mc)	-7	-7	7	7
Inplane Bending (Mc)	-140	140	140	-140
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	126	128	126	128

Inplane Stress Summary	13	231	310	-34
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	4	4
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	26	231	310	54
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-9	-9	9	9
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	9	6	9

Hoop Stress Summary	0	3	18	21
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-4	4	4	-4
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	0	1	12	0
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	1	1	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	2	4	18	22

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-6	-6	6	6
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	9	6	9
Hoop Stress Summary	3	6	15	18

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-13	13	13	-13
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-9	11	22	-10

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	1	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	13	11	22	29

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	126	128	126	128	126	128	126	128
Circ. Pl (SUS)	-6	-6	12	12	-5	-5	9	9

Circ. Q (SUS)	-8	8	26	-26	-108	108	172	-172
Long. Pm (SUS)	63	63	63	63	63	63	63	63
Long. P1 (SUS)	-4	-4	8	8	-3	-3	9	9
Long. Q (SUS)	-22	22	86	-86	-34	34	52	-52
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	0	0	0	0	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	126.0	128.0	126.0	128.0	126.0	128.0	126.0	128.0
Pm+P1 (SUS)	120.0	122.0	138.0	140.0	121.0	123.0	135.0	137.0
Pm+P1+Q (Total)	112.1	130.1	164.5	129.1	26.1	231.0	307.0	55.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	128.00	138.00	Passed
Pm+P1 (SUS)	140.00	207.00	Passed
Pm+P1+Q (TOTAL)	307.05	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	6	9	6	9	6	9	6	9
Circ. P1 (SUS)	-6	-6	12	12	-3	-3	9	9
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. P1 (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	-2	2	6	-6	-11	11	15	-15
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	6.0	9.0	6.0	9.0	6.0	9.0	6.0	9.0
Pm+P1 (SUS)	1.0	3.0	18.0	21.0	3.0	6.0	15.0	18.0
Pm+P1+Q (Total)	1.0	3.0	18.0	22.0	13.0	12.0	20.0	28.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	9.00	138.00	Passed
Pm+P1 (SUS)	21.00	207.00	Passed
Pm+P1+Q (TOTAL)	28.00	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



Input, Nozzle Desc: VENT NOZ V1

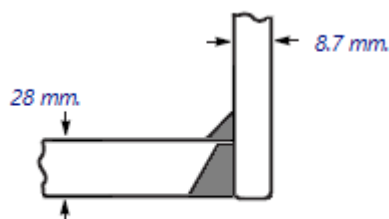
From: 60

Pressure for Reinforcement Calculations	P	25.000	bars
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	85	°C
Parent Material		SA-516 70	+Q+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2700.00	mm.
Design Length of Section	L	4375.7998	mm.
Shell Finished (Minimum) Thickness	t	28.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		3258.17	mm.
User Entered Minimum Design Metal Temperature		-35.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		27.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: VENT NOZ V1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.687 in.
 Actual Thickness Used in Calculation 0.344 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 25.0 \cdot 1351.5 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$
 $= 24.7547$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 25.0 \cdot 22.92 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$
 $= 0.4199$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3495 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.3250	mm.
Parallel to Vessel Wall	Rn+tn+t	56.6625	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	18.0940	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: VENT NOZ V1.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.9199	mm.
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 26.2547	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 26.2547	mm.
Wall Thickness, shell/head, external pressure	trb2	= 2.5134	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm.
Wall Thickness per table UG-45	tb3	= 4.9200	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.92, \max(26.2547, 3.0)]$
 $= 4.9200$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.9199, 4.92)$
 $= 4.9200$ mm.

Available Nozzle Neck Thickness = $0.875 \cdot 8.738 = 7.645$ mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46	°C
Calculated Minimum Design Metal Temperature	-104	°C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104	°C

Weld Size Calculations, Description: VENT NOZ V1

Intermediate Calc. for nozzle/shell Welds	Tmin	7.2376	mm.
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Results Per UW-16.1:

Required Thickness	Actual Thickness
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Nozzle Weld $5.0663 = 0.7 * t_{min}$. $7.0700 = 0.7 * W_o$ mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.3370 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 278.3370 mm.

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Input, Nozzle Desc: PR. IN P1

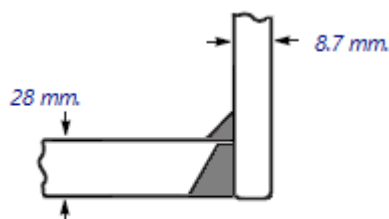
From: 60

Pressure for Reinforcement Calculations	P	25.000	bars
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	85	°C
Parent Material		SA-516 70	+Q+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2700.00	mm.
Design Length of Section	L	4375.7998	mm.
Shell Finished (Minimum) Thickness	t	28.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		3008.17	mm.
User Entered Minimum Design Metal Temperature		-35.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	27.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: PR. IN P1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 1.687 in.
 Actual Thickness Used in Calculation 0.344 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 25.0 \cdot 1351.5 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$
 $= 24.7547$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 25.0 \cdot 22.92 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$
 $= 0.4199$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3495 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.3250	mm.
Parallel to Vessel Wall	Rn+tn+t	56.6625	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	18.0940	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: PR. IN P1.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.9199	mm.
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 26.2547	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 26.2547	mm.
Wall Thickness, shell/head, external pressure	trb2	= 2.5134	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm.
Wall Thickness per table UG-45	tb3	= 4.9200	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.92, \max(26.2547, 3.0)]$
 $= 4.9200$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.9199, 4.92)$
 $= 4.9200$ mm.

Available Nozzle Neck Thickness = $0.875 \cdot 8.738 = 7.645$ mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46	°C
Calculated Minimum Design Metal Temperature	-104	°C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104	°C

Weld Size Calculations, Description: PR. IN P1

Intermediate Calc. for nozzle/shell Welds	Tmin	7.2376	mm.
---	------	--------	-----

Results Per UW-16.1:

Required Thickness	Actual Thickness
--------------------	------------------



Nozzle Weld $5.0663 = 0.7 * t_{min}$. $7.0700 = 0.7 * W_o$ mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.3370 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 278.3370 mm.

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Input, Nozzle Desc: PR. IN NOZ. P2

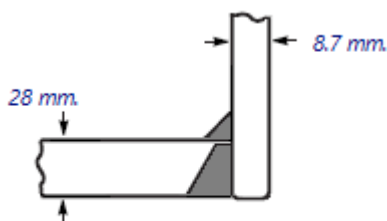
From: 60

Pressure for Reinforcement Calculations	P	25.000	bars
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	85	°C
Parent Material		SA-516 70	+Q+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2700.00	mm.
Design Length of Section	L	4375.7998	mm.
Shell Finished (Minimum) Thickness	t	28.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		6108.17	mm.
User Entered Minimum Design Metal Temperature		-35.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		27.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: PR. IN NOZ. P2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.687 in.
 Actual Thickness Used in Calculation 0.344 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 25.0 \cdot 1351.5 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$
 $= 24.7547$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 25.0 \cdot 22.92 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$
 $= 0.4199$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3495 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.3250	mm.
Parallel to Vessel Wall	Rn+tn+t	56.6625	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	18.0940	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: PR. IN NOZ. P2.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.9199	mm.
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 26.2547	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 26.2547	mm.
Wall Thickness, shell/head, external pressure	trb2	= 2.5134	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm.
Wall Thickness per table UG-45	tb3	= 4.9200	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.92, \max(26.2547, 3.0)]$
 $= 4.9200$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.9199, 4.92)$
 $= 4.9200$ mm.

Available Nozzle Neck Thickness = $0.875 \cdot 8.738 = 7.645$ mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46	°C
Calculated Minimum Design Metal Temperature	-104	°C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104	°C

Weld Size Calculations, Description: PR. IN NOZ. P2

Intermediate Calc. for nozzle/shell Welds	Tmin	7.2376	mm.
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Results Per UW-16.1:

Required Thickness	Actual Thickness
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Nozzle Weld $5.0663 = 0.7 * t_{min}$. $7.0700 = 0.7 * W_o$ mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.3370 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 278.3370 mm.

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Input, Nozzle Desc: SAFETY VALVE S1

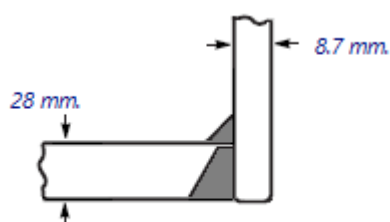
From: 60

Pressure for Reinforcement Calculations	P	25.000	bars
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	85	°C
Parent Material		SA-516 70	+Q+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2700.00	mm.
Design Length of Section	L	4375.7998	mm.
Shell Finished (Minimum) Thickness	t	28.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		5858.17	mm.
User Entered Minimum Design Metal Temperature		-35.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	27.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: SAFETY VALVE S1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 1.687 in.
Actual Thickness Used in Calculation 0.344 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $25.0 \cdot 1351.5 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$
= 24.7547 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $25.0 \cdot 22.92 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$
= 0.4199 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3495 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.3250 mm.
Parallel to Vessel Wall	Rn+tn+t	56.6625 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	18.0940 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: SAFETY VALVE S1.
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.9199 mm.
Wall Thickness per UG16(b),	tr16b	= 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 26.2547 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 26.2547 mm.
Wall Thickness, shell/head, external pressure	trb2	= 2.5134 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000 mm.
Wall Thickness per table UG-45	tb3	= 4.9200 mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[4.92, max(26.2547, 3.0)]
= 4.9200 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.9199, 4.92)
= 4.9200 mm.

Available Nozzle Neck Thickness = $0.875 \cdot 8.738 = 7.645$ mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104 °C

Weld Size Calculations, Description: SAFETY VALVE S1

Intermediate Calc. for nozzle/shell Welds	Tmin	7.2376 mm.
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Results Per UW-16.1:

Required Thickness	Actual Thickness
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Nozzle Weld $5.0663 = 0.7 * t_{min}$. $7.0700 = 0.7 * W_o$ mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.3370 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 278.3370 mm.

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Input, Nozzle Desc: LEVEL GUAGE L1

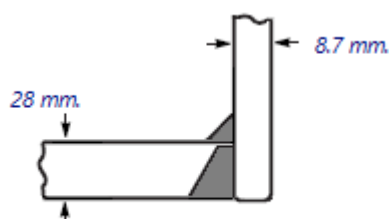
From: 60

Pressure for Reinforcement Calculations	P	25.000	bars
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	85	°C
Parent Material		SA-516 70	+Q+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2700.00	mm.
Design Length of Section	L	4375.7998	mm.
Shell Finished (Minimum) Thickness	t	28.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Cylinder/Cone Centerline	L1	700.0000	mm.
Distance from Bottom/Left Tangent		3008.17	mm.
User Entered Minimum Design Metal Temperature		-35.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		30.48	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: LEVEL GUAGE L1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.687 in.
Actual Thickness Used in Calculation	0.344 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 25.0 \cdot 1351.5 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$
 $= 24.7547$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 25.0 \cdot 22.92 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$
 $= 0.4199$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3799 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	120.9052 mm.
Parallel to Vessel Wall	Rn+tn+t	60.4526 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	18.0940 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: LEVEL GUAGE L1.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104 °C

Weld Size Calculations, Description: LEVEL GUAGE L1

Intermediate Calc. for nozzle/shell Welds	Tmin	7.2376 mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.0663 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm.

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: LEVEL GUAGE L1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.687 in.
Actual Thickness Used in Calculation	0.344 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]



$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 25.0 \cdot 1351.5 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$$

$$= 24.7547 \text{ mm.}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 25.0 \cdot 22.92 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$$

$$= 0.4199 \text{ mm.}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.3799 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.3250	mm.
Parallel to Vessel Wall	Rn+tn+t	56.6625	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	18.0940	mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: LEVEL GUAGE L1.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.9199	mm.
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 26.2547	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 26.2547	mm.
Wall Thickness, shell/head, external pressure	trb2	= 2.5134	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm.
Wall Thickness per table UG-45	tb3	= 4.9200	mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[4.92, \max(26.2547, 3.0)]$$

$$= 4.9200 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(1.9199, 4.92)$$

$$= 4.9200 \text{ mm.}$$

Available Nozzle Neck Thickness = $0.875 \cdot 8.738 = 7.645 \text{ mm.} \rightarrow \text{OK}$

Weld Size Calculations, Description: LEVEL GUAGE L1

Intermediate Calc. for nozzle/shell Welds Tmin 7.2376 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.0663 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 18.8386 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 351.4631 mm.

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Input, Nozzle Desc: LEVEL GUAGE L2

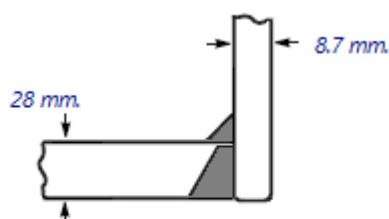
From: 60

Pressure for Reinforcement Calculations	P	25.057	bars
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	85	°C
Parent Material		SA-516 70	+Q+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2700.00	mm.
Design Length of Section	L	4375.7998	mm.
Shell Finished (Minimum) Thickness	t	28.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Cylinder/Cone Centerline	L1	700.0000	mm.
Distance from Bottom/Left Tangent		3008.17	mm.
User Entered Minimum Design Metal Temperature		-35.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-30.48	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	27.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle 90 degrees to the Longitudinal axis.



Reinforcement CALCULATION, Description: LEVEL GUAGE L2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.687 in.
Actual Thickness Used in Calculation	0.344 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 25.06 \cdot 1351.5 / (138 \cdot 1.0 - 0.6 \cdot 25.06)$
 $= 24.8121$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 25.06 \cdot 22.92 / (138 \cdot 1.0 - 0.6 \cdot 25.06)$
 $= 0.4209$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3799 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	120.9047 mm.
Parallel to Vessel Wall	Rn+tn+t	60.4523 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	18.0940 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: LEVEL GUAGE L2.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104 °C

Weld Size Calculations, Description: LEVEL GUAGE L2

Intermediate Calc. for nozzle/shell Welds	Tmin	7.2376 mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.0663 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm.

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: LEVEL GUAGE L2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.687 in.
Actual Thickness Used in Calculation	0.344 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]



$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 25.06 \cdot 1351.5 / (138 \cdot 1.0 - 0.6 \cdot 25.06)$$

$$= 24.8121 \text{ mm.}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 25.06 \cdot 22.92 / (138 \cdot 1.0 - 0.6 \cdot 25.06)$$

$$= 0.4209 \text{ mm.}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.3799 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.3250	mm.
Parallel to Vessel Wall	Rn+tn+t	56.6625	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	18.0940	mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: LEVEL GUAGE L2.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.9209	mm.
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 26.3121	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 26.3121	mm.
Wall Thickness, shell/head, external pressure	trb2	= 2.5134	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm.
Wall Thickness per table UG-45	tb3	= 4.9200	mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[4.92, \max(26.3121, 3.0)]$$

$$= 4.9200 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(1.9209, 4.92)$$

$$= 4.9200 \text{ mm.}$$

Available Nozzle Neck Thickness = $0.875 \cdot 8.738 = 7.645 \text{ mm.} \rightarrow \text{OK}$

Weld Size Calculations, Description: LEVEL GUAGE L2

Intermediate Calc. for nozzle/shell Welds	Tmin	7.2376	mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.0663 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 18.8386 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 351.4631 mm.

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Input, Nozzle Desc: LEVEL GUAGE L3

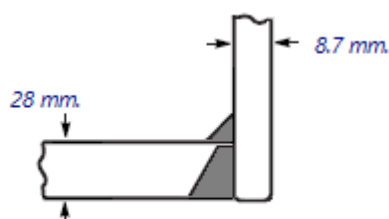
From: 60

Pressure for Reinforcement Calculations	P	25.000	bars
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	85	°C
Parent Material		SA-516 70	+Q+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2700.00	mm.
Design Length of Section	L	4375.7998	mm.
Shell Finished (Minimum) Thickness	t	28.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Cylinder/Cone Centerline	L1	700.0000	mm.
Distance from Bottom/Left Tangent		3658.17	mm.
User Entered Minimum Design Metal Temperature		-35.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		30.48	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle 90 degrees to the Longitudinal axis.



Reinforcement CALCULATION, Description: LEVEL GUAGE L3

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.687 in.
Actual Thickness Used in Calculation	0.344 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $25.0 \cdot 1351.5 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$
= 24.7547 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $25.0 \cdot 22.92 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$
= 0.4199 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3799 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	120.9052 mm.
Parallel to Vessel Wall	Rn+tn+t	60.4526 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	18.0940 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: LEVEL GUAGE L3.
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104 °C

Weld Size Calculations, Description: LEVEL GUAGE L3

Intermediate Calc. for nozzle/shell Welds	Tmin	7.2376 mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.0663 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm.

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: LEVEL GUAGE L3

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.687 in.
Actual Thickness Used in Calculation	0.344 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]



$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 25.0 \cdot 1351.5 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$$

$$= 24.7547 \text{ mm.}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 25.0 \cdot 22.92 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$$

$$= 0.4199 \text{ mm.}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.3799 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.3250	mm.
Parallel to Vessel Wall	Rn+tn+t	56.6625	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	18.0940	mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: LEVEL GUAGE L3.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.9199	mm.
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 26.2547	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 26.2547	mm.
Wall Thickness, shell/head, external pressure	trb2	= 2.5134	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm.
Wall Thickness per table UG-45	tb3	= 4.9200	mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[4.92, \max(26.2547, 3.0)]$$

$$= 4.9200 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(1.9199, 4.92)$$

$$= 4.9200 \text{ mm.}$$

Available Nozzle Neck Thickness = $0.875 \cdot 8.738 = 7.645 \text{ mm.} \rightarrow \text{OK}$

Weld Size Calculations, Description: LEVEL GUAGE L3

Intermediate Calc. for nozzle/shell Welds Tmin 7.2376 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.0663 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 18.8386 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 351.4631 mm.

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Input, Nozzle Desc: LEVEL GUAGE L4

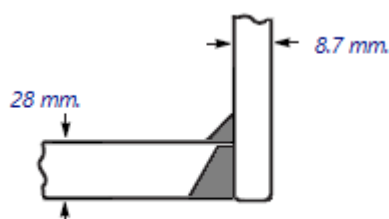
From: 60

Pressure for Reinforcement Calculations	P	25.057	bars
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	85	°C
Parent Material		SA-516 70	+Q+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2700.00	mm.
Design Length of Section	L	4375.7998	mm.
Shell Finished (Minimum) Thickness	t	28.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Cylinder/Cone Centerline	L1	700.0000	mm.
Distance from Bottom/Left Tangent		3658.17	mm.
User Entered Minimum Design Metal Temperature		-35.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-30.48	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	27.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle 90 degrees to the Longitudinal axis.



Reinforcement CALCULATION, Description: LEVEL GUAGE L4

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.687 in.
Actual Thickness Used in Calculation	0.344 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 25.06 \cdot 1351.5 / (138 \cdot 1.0 - 0.6 \cdot 25.06)$
 $= 24.8121$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 25.06 \cdot 22.92 / (138 \cdot 1.0 - 0.6 \cdot 25.06)$
 $= 0.4209$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3799 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	120.9047 mm.
Parallel to Vessel Wall	Rn+tn+t	60.4523 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	18.0940 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: LEVEL GUAGE L4.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104 °C

Weld Size Calculations, Description: LEVEL GUAGE L4

Intermediate Calc. for nozzle/shell Welds	Tmin	7.2376 mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.0663 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: LEVEL GUAGE L4

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.687 in.
Actual Thickness Used in Calculation	0.344 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]



$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 25.06 \cdot 1351.5 / (138 \cdot 1.0 - 0.6 \cdot 25.06)$$

$$= 24.8121 \text{ mm.}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 25.06 \cdot 22.92 / (138 \cdot 1.0 - 0.6 \cdot 25.06)$$

$$= 0.4209 \text{ mm.}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.3799 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.3250	mm.
Parallel to Vessel Wall	Rn+tn+t	56.6625	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	18.0940	mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: LEVEL GUAGE L4.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.9209	mm.
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 26.3121	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 26.3121	mm.
Wall Thickness, shell/head, external pressure	trb2	= 2.5134	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm.
Wall Thickness per table UG-45	tb3	= 4.9200	mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[4.92, \max(26.3121, 3.0)]$$

$$= 4.9200 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(1.9209, 4.92)$$

$$= 4.9200 \text{ mm.}$$

Available Nozzle Neck Thickness = $0.875 \cdot 8.738 = 7.645 \text{ mm.} \rightarrow \text{OK}$

Weld Size Calculations, Description: LEVEL GUAGE L4

Intermediate Calc. for nozzle/shell Welds	Tmin	7.2376	mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.0663 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 18.8386 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 351.4631 mm.

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Input, Nozzle Desc: DRAIN NOZ. D2

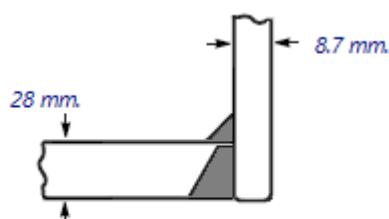
From: 60

Pressure for Reinforcement Calculations	P	25.098	bars
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	85	°C
Parent Material		SA-516 70	+Q+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2700.00	mm.
Design Length of Section	L	4375.7998	mm.
Shell Finished (Minimum) Thickness	t	28.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Cylinder/Cone Centerline	L1	700.0000	mm.
Distance from Bottom/Left Tangent		4508.17	mm.
User Entered Minimum Design Metal Temperature		-35.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle 90 degrees to the Longitudinal axis.



Reinforcement CALCULATION, Description: DRAIN NOZ. D2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.687 in.
Actual Thickness Used in Calculation	0.344 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $25.1 \cdot 1351.5 / (138 \cdot 1.0 - 0.6 \cdot 25.1)$
= 24.8529 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $25.1 \cdot 22.92 / (138 \cdot 1.0 - 0.6 \cdot 25.1)$
= 0.4216 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3799 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	120.9045 mm.
Parallel to Vessel Wall	Rn+tn+t	60.4522 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	18.0940 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: DRAIN NOZ. D2.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104 °C

Weld Size Calculations, Description: DRAIN NOZ. D2

Intermediate Calc. for nozzle/shell Welds	Tmin	7.2376 mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.0663 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: DRAIN NOZ. D2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.687 in.
Actual Thickness Used in Calculation	0.344 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]



$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 25.1 \cdot 1351.5 / (138 \cdot 1.0 - 0.6 \cdot 25.1)$$

$$= 24.8529 \text{ mm.}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 25.1 \cdot 22.92 / (138 \cdot 1.0 - 0.6 \cdot 25.1)$$

$$= 0.4216 \text{ mm.}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.3799 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.3250	mm.
Parallel to Vessel Wall	Rn+tn+t	56.6625	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	18.0940	mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: DRAIN NOZ. D2.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.9216	mm.
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 26.3529	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 26.3529	mm.
Wall Thickness, shell/head, external pressure	trb2	= 2.5134	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm.
Wall Thickness per table UG-45	tb3	= 4.9200	mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[4.92, \max(26.3529, 3.0)]$$

$$= 4.9200 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(1.9216, 4.92)$$

$$= 4.9200 \text{ mm.}$$

Available Nozzle Neck Thickness = $0.875 \cdot 8.738 = 7.645 \text{ mm.} \rightarrow \text{OK}$

Weld Size Calculations, Description: DRAIN NOZ. D2

Intermediate Calc. for nozzle/shell Welds	Tmin	7.2376	mm.
---	------	--------	-----

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.0663 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 18.8386 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 351.4631 mm.

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Input, Nozzle Desc: TEMP IN. T1

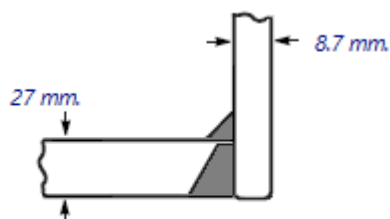
From: 70

Pressure for Reinforcement Calculations	P	25.058	bars
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	85	°C
Parent Material		SA-516 70	+Q+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	2700.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	27.0000	mm.
Head Internal Corrosion Allowance	c	1.5000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	700.0000	mm.
User Entered Minimum Design Metal Temperature		-35.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: TEMP IN. T1



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.687 in.
 Actual Thickness Used in Calculation 0.344 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 25.06 \cdot 2703.0 \cdot 0.899 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 25.06)$
 $= 22.1051$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 25.06 \cdot 22.92 / (138 \cdot 1.0 - 0.6 \cdot 25.06)$
 $= 0.4209$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3799 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	113.3825	mm.
Parallel to Vessel Wall	Rn+tn+t	56.6913	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	18.0940	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: TEMP IN. T1.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $25.06 / 51.00 = 0.491$

Weld Size Calculations, Description: TEMP IN. T1

Intermediate Calc. for nozzle/shell Welds	Tmin	7.2376	mm.
---	------	--------	-----

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.0663 = 0.7 \cdot tmin.$	$7.0700 = 0.7 \cdot Wo$ mm.

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: TEMP IN. T1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 1.687 in.
 Actual Thickness Used in Calculation 0.344 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 25.06 \cdot 2703.0 \cdot 0.899 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 25.06)$
 $= 22.1051 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 25.06 \cdot 22.92 / (138 \cdot 1.0 - 0.6 \cdot 25.06)$
 $= 0.4209 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3799 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	111.3250 mm.
Parallel to Vessel Wall	Rn+tn+t	55.6625 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	18.0940 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: TEMP IN. T1.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.9209 mm.
Wall Thickness per UG16(b),	tr16b	= 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 26.0510 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 26.0510 mm.
Wall Thickness, shell/head, external pressure	trb2	= 2.5115 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000 mm.
Wall Thickness per table UG-45	tb3	= 4.9200 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.92, \max(26.051, 3.0)]$
 $= 4.9200 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.9209, 4.92)$
 $= 4.9200 \text{ mm.}$

Available Nozzle Neck Thickness = $0.875 \cdot 8.738 = 7.645 \text{ mm.} \rightarrow \text{OK}$

Weld Size Calculations, Description: TEMP IN. T1

Intermediate Calc. for nozzle/shell Welds Tmin 7.2376 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.0663 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 9.1979 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 337.3570 mm.

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Input Echo, Tubesheet Number 1, Description: U-TS

Shell Data:

Main Shell Description: SHELL-1

Shell Maximum Design Pressure	Psd,max	25.00 bars
Shell Maximum Operating Pressure	Psox,max	25.00 bars
Shell Minimum Operating Pressure	Psox,min	0.00 bars
Shell Thickness	ts	28.0000 mm.
Shell Internal Corrosion Allowance	cas	1.5000 mm.
Shell External Corrosion Allowance	casext	0.0000 mm.
Inside Diameter of Shell	Ds	1600.000 mm.
Shell Temperature for Internal Pressure	Ts	85.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: CH. COVER

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	191.00 bars
Channel Maximum Operating Pressure	Ptox,max	191.00 bars
Channel Minimum Operating Pressure	Ptox,min	0.00 bars
Channel Thickness	tc	140.0000 mm.
Channel Internal Corrosion Allowance	cac	1.5000 mm.
Channel External Corrosion Allowance	cacext	0.0000 mm.
Inside Diameter of Channel	Dc	1600.000 mm.
Channel Design Temperature	TEMPC	85.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	1906
Tube Wall Thickness	et	2.7700 mm.
Tube Outside Diameter	D	25.4000 mm.
Straight Tube Length (bet. inner tubsht faces) L		5200.00 mm.
Design Temperature of the Tubes		85.00 °C
Tube Material		SA-334 6
Tube Material UNS Number		K03006
Is this a Welded Tube		No
Tube Material Specification used	Smls. & wld. tube	
Tube Allowable Stress at Temperature		118.00 N./mm ²
Tube Allowable Stress At Ambient		118.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	223.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm.
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.7700 mm.
Groove Weld Leg	ag	2.7700 mm.
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	770.800 mm.
Largest Center-to-Center Tube Distance	Ul	76.2000 mm.
Length of Expanded Portion of Tube	ltx	0.0000 mm.
Tube-side pass partition groove depth	hg	0.0000 mm.

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral with Channel, Conf. e



Tubesheet Design Metal Temperature T 85.00 °C
 Tubesheet Material SA-350 LF2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K03011
 Tubesheet Allowable Stress at Temperature S 138.00 N./mm²
 Tubesheet Allowable Stress at Ambient Tt 138.00 N./mm²
 Thickness of Tubesheet h 400.0000 mm.
 Tubesheet Corr. Allowance (Shell side) Cats 1.5000 mm.
 Tubesheet Corr. Allowance (Channel side) Catc 1.5000 mm.
 Tubesheet Outside Diameter A 1880.000 mm.

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped? Yes
 Is the Tubesheet Flat on Tubeside? Yes
 Is the Tubesheet Flat on Shellside? No
 Step 1 Diameter on the Shellside ds1 1632.00 mm.
 Step 1 Depth on the Shellside hs1 6.00 mm.
 Step 2 Diameter on the Shellside ds2 1597.00 mm.
 Step 2 Depth on the Shellside hs2 6.00 mm.
 Calculated Tubesheet Diameter per Div 2, 4.18.4 1880.00 mm.

Note: Tubesheet diameter is now: 1880.000 mm. Div 2, 4.18.4.

Area of the Untubed Lanes AL 1181.6 cm²
 Junction Stress Reduction option Perform Plastic Calculation
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side Shell
 Flange Outside Diameter A 1880.000 mm.
 Flange Inside Diameter B 1600.000 mm.
 Flange Face Outside Diameter Fod 1635.000 mm.
 Flange Face Inside Diameter Fid 1600.000 mm.
 Gasket Outside Diameter Go 1629.000 mm.
 Gasket Inside Diameter Gi 1603.000 mm.
 Small end Hub thk. g0 28.0000 mm.
 Large end Hub thk. g1 32.0000 mm.
 Gasket Factor, m 3.75
 Gasket Design Seating Stress y 62.06 N./mm²
 Flange Facing Sketch Code Sketch 1a
 Gasket Thickness tg 3.1750 mm.
 Full face Gasket Flange Option Program Selects

Bolting Information:

Diameter of Bolt Circle C 1776.000 mm.
 Nominal Bolt Diameter dB 36.0000 mm.
 Thread Series TEMA Metric Thread
 Number of Bolts n 48
 Bolt Material SA-320 L7
 Bolt Allowable Stress At Temperature Sb 172.00 N./mm²
 Bolt Allowable Stress At Ambient Sa 172.00 N./mm²
 Weld between Flange and Shell/Channel 0.0000 mm.

Note: The full bolt load Sa*Ab, was used for all load cases.

Tubesheet Integral with Channel
 Tubesheet Extended as Flange Yes
 Thickness of Extended Portion of Tubesheet Tf 388.0000 mm.
 Is Bolt Load Transferred to the Tubesheet Yes

TCA	ODCA	SLN	D/T	L/D	Factor A	B
2.7700	25.40	5970.80	9.17	50.0000	0.0130823	122.73

EMAWP = $(2.167/(D/T) - 0.0833) * B = 187.7951$ bars

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.6538	25.40	5970.80	38.85	50.0000	0.0007287	72.86

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 72.855) / (3 \cdot 38.8524) = 25.0009$ bars

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.9308 mm.
Allowable Internal Pressure at Corroded thickness	281.95 bars
Required Internal Design Pressure	191.00 bars
Allowable External Pressure at Corroded thickness	187.80 bars
Required External Design Pressure	25.00 bars
Required Thickness due to Shell Side pressure	0.6538 mm.

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (G_{oc} - G_{ic}) / 2$	13.000 mm.
Basic Gasket Width,	$b_0 = N / 2$	6.500 mm.
Effective Gasket Width,	$b = \sqrt{b_0} \cdot 2.5$	6.422 mm.
Gasket Reaction Diameter,	$G = G_o - 2.0 \cdot b$	1616.156 mm.

Results for ASME U-tube Tubesheet Calculations for Configuration e:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 770.8 + 25.4$$

$$= 1567.000 \text{ mm.}$$

Basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.75 - 25.4) / 31.75$$

$$= 0.200$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$= G_s / D_o \text{ (Configurations d, e, f)}$$

$$= 1616.1555 / 1567.0$$

$$= 1.0314$$

Ratio [pc]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 1600.0 / 1567.0$$

$$= 1.0211$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$

$$= 0 / 400.0$$

$$= 0.000 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(dt - 2t_t (E_t/T/E) \cdot (S_t/T/S) \cdot (p); dt - 2t_t)$$

$$= \max(25.4 - 2 \cdot 2.77 (0.19904E+09 / 1.19904E+09) \cdot (117/138) \cdot (0); 25.4 - 2 \cdot 2.77)$$

$$= 25.4000 \text{ mm.}$$

Effective tube pitch [p*]:

$$= p / \sqrt{1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2) }$$



$$= 31.75 / (1 - 4 \cdot \min(1181.56; 4 \cdot 1567.0 \cdot 31.75) / (\pi \cdot 1567.0^2))$$

$$= 32.7697 \text{ mm.}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (32.7697 - 25.4) / 32.7697$$

$$= 0.2249$$

$$h/p = 12.598425; \mu^* = 0.224894; E^* = 36884020 \text{ KPa.}$$

$$E^*/E = 0.185307; \nu^* = 0.424937$$

Note: As h/p (12.598) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:
 β_s , k_s , λ_s , δ_s and ω_s are all = 0

Channel coefficient [β_c]:

$$= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{((D_c + t_c)t_c)}$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1600.0 + 140.0)140.0)$$

$$= 0.0037 \text{ 1/mm.}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_s^2))$$

$$= 0.004 \cdot 1990422 \cdot 140.0^3 / (6(1 - 0.3^2))$$

$$= 3684289280.0000 \text{ bars} \cdot \text{mm.}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(1600.0 + 140.0 / \cos(0)) \cdot 0.0037 / 400.0^3 \cdot (1.0 + 400.0 \cdot 0.004 \cdot (1.0 + 1.0) / 2 + (400.0 \cdot 0.004)^2 / 2 \cdot 1.0)$$

$$= 2138649.8 \text{ bars}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 1600.0^2 / (4 \cdot 0.19904E+09 \cdot 140.0) \cdot (1 - 1600.0 \cdot 0.3 / (2(1600.0 + 140.0)))$$

$$= 0.19798E-01 \text{ mm./N./mm}^2$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

Parameter [ω_c]:

$$= p_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + Cq\Delta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.021 \cdot 0.36843E+1 [0.019798(1.0 + 0.004 \cdot 1.0 \cdot 400.0) - 0(1.0/400.0 + 1.0 \cdot 0.004/2)]$$

$$= 67848.2578 \text{ mm.}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1880.0 / 1567.0$$

$$= 1.1997$$

Coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.42) / 36884020 \cdot (2138649 + 0.19904E+09 \cdot \ln(1.2))$$

$$= 3.8995$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (P_s , P_t) [Mts]:

$$= D_o^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_t - 1) \cdot (p_t^2 + 1) \cdot P_t]$$

$$= 1567.0^2 / 16 \cdot [(1.031 - 1) \cdot (1.031^2 + 1) \cdot 25.0 -$$



$$\begin{aligned} & (1.021 - 1) \cdot (1.021^2 + 1) \cdot 191.0] \\ & = -1012498.6250 \text{ bars} \cdot \text{mm}^2 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M*]
$$\begin{aligned} & = Mts + (\omega_c \cdot Pt) + W* \cdot (C - Gs) / (2 \cdot \pi \cdot Do) \\ & = -1012498 + (12959017) + 621295 \cdot (1776.0 - 1616.156) / (2 \cdot \pi \cdot 1567.0) \\ & = 12935642.0000 \text{ bars} \cdot \text{mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:
$$\begin{aligned} & = ((M*) - Do^2/32 \cdot F(Ps - Pt)) / (1 + F) \\ & = ((12935642) - 1567.0^2/32 \cdot 3.9(25.0 - 191.0)) / (1 + 3.9) \\ & = 12778221.0000 \text{ bars} \cdot \text{mm}^2 \end{aligned}$$

Maximum bending moment acting on center of tubesheet [Mo]:
$$\begin{aligned} & = Mp + Do^2/64 \cdot (3 + \nu) \cdot (Ps - Pt) \\ & = 12778221 + 1567.0^2/64 \cdot (3 + 0.425) \cdot (25.0 - 191.0) \\ & = -9034949.0000 \text{ bars} \cdot \text{mm}^2 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:
$$\begin{aligned} & = \max(|Mp|; |Mo|) \\ & = \max(|12778221|; |-9034949.0|) \\ & = 12778221.0000 \text{ bars} \cdot \text{mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:
$$\begin{aligned} & = 6 \cdot M / (\mu \cdot (h - hg')^2) \\ & = 6 \cdot 12778221 / (0.2249(400.0 - 0)^2) \\ & = 213.0829 \text{ N./mm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:
$$\begin{aligned} & = 2 \cdot S \\ & = 2 \cdot 138.0 \\ & = 276.00 \text{ N./mm}^2 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:
$$\begin{aligned} & = 6 \cdot M / (\mu \cdot (h - hg')^2) \\ & = 6 \cdot 12606255 / (0.2249(349.1758 - 0)^2) \\ & = 275.8646 \text{ N./mm}^2 \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:
$$\begin{aligned} & = h + Cats + Catc \\ & = 349.1758 + 0 + 0 \\ & = 349.1758 \text{ mm.} \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:
$$\begin{aligned} & = \max(HreqB; HreqS) \\ & = \max(349.1758; 294.5393) \\ & = 349.1758 \text{ mm.} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [|Ps - Pt|]:
$$\begin{aligned} & = |Ps - Pt| \\ & = |25.0 - 191.0| \\ & = 166.000 \text{ bars} \end{aligned}$$

Shear Stress check [τ_{limit}]:
$$\begin{aligned} & = 4 \cdot \mu \cdot h / Do \cdot \min(0.8 \cdot S; 0.533 \cdot Sy) \\ & = 4 \cdot 0.2 \cdot 400.0 / 1567.0 \cdot \min(0.8 \cdot 138.0; 0.533 \cdot 229.6) \\ & = 22.545 \text{ N./mm}^2 \end{aligned}$$

|Ps - Pt| is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:
$$\begin{aligned} & = 1 / (4 \cdot \mu) \cdot (Do / \min(0.8 \cdot S; 0.533 \cdot Sy)) \cdot |Ps - Pt| + Cats + Catc \\ & = 1 / (4 \cdot 0.2) \cdot (1567.0 / 110.4) \cdot |25.0 - 191.0| + 0 \end{aligned}$$

= 294.5393 mm.

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Channel thickness of: 140.000 mm.
 Min. Channel len. adjacent to the tubesheet is: 851.915mm.

Axial Channel Membrane Stress [σ_c, m]:
 $= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$
 $= 1600.0^2 / (4 \cdot 140.0 (1600.0 + 140.0) \cdot \cos(0)) \cdot 191.0$
 $= 50.184 \text{ N./mm}^2$

Axial Channel Bending Stress [σ_c, b]:
 $= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$
 $= 6 \cdot 82828016 / 140.0^2 \cdot [(0.019798 \cdot 1.0 \cdot 0.004 - 0 \cdot 1.0) \cdot 191.0 - 6((1 - 0.425) / 36884020) \cdot 1567.0 / 400.0^3 \cdot (1.0 + 0.004 \cdot 400.0 / 2 \cdot 1.0) \cdot (12778221 + 1567.0^2 / 32 \cdot (25.0 - 191.0))]$
 $= 155.288 \text{ N./mm}^2$

Channel Membrane + Bending Stress [σ_c]:
 $= |\sigma_{c,m}| + |\sigma_{c,b}| \text{ (Should be } \leq 1.5 \cdot S_c)$
 $= 205.471 \text{ N./mm}^2 \text{ (Should be } \leq 207.00)$

Tube Metal Area [A_t]:
 $= \pi (d_o - t) t$
 $= \pi (25.4 - 2.77) 2.77$
 $= 1.969 \text{ cm}^2$

Tube Weld Size Results per UW-20.3:

Tube Strength [F_t]:
 $= \pi \cdot t (d_o - t) S$
 $= \pi \cdot 2.77 (25.4 - 2.77) 117.998$
 $= 2369.519 \text{ Kgf}$

Groove Weld Strength [F_g]:
 $= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$
 $= 0.85 \cdot \pi \cdot 2.77 (25.4 + 0.67 \cdot 2.77) 118.0$
 $= 2369.5188 \text{ Kgf}$

Weld Strength Factor [f_w]:
 $= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$
 $= 117.998 / 117.998$
 $= 1.000$

Design Strength Ratio [f_d]:
 $= 1 \text{ for full strength welds}$

Fillet Weld Strength [F_f]:
 $= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$
 $= 0.55 \cdot \pi \cdot 2.77 (25.4 + 0.67 \cdot 2.77) 118.0$
 $= 1569.6356 \text{ Kgf}$

Minimum Length of Weld [a_r]:
 $= 2(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o)$
 $= 2(\sqrt{(0.75 \cdot 25.4)^2 + 1.07 \cdot 2.77 \cdot (25.4 - 2.77) 1.0 \cdot 1.0} - 0.75 \cdot 25.4)$
 $= 3.372 \text{ mm.}$

Minimum Required Fillet Weld Leg [a_{fr}]: 1.6859 mm.
 Minimum Required Groove Weld Leg [a_{gr}]: 1.6859 mm.

Tube-Tubesheet Jt allowable, 2369.52 is $\geq$ tube strength 2369.52 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. ($P_{sd,max} + P_{td,max}$):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	213.1 <=	276.0 N./mm ²	Ok
Tubesheet shear stress	81.3 <=	110.4 N./mm ²	Ok
Stress in Channel at Tubesheet	205.5 <=	207.0 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm.)	Required	Actual	P/F
Tubesheet Thickness :	349.176	400.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.686	2.770	Ok
Tube-Tubesheet Groove Weld Leg :	1.686	2.770	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	368.571 64.409	234	276	94 110	Ps+Pt D1	Ok
D2uc	133.514 ...	34	276	12 110	Ps+Pt D2	Ok
D3uc	349.176 ...	213	276	81 110	Ps+Pt D3	Ok
D1c	370.952 64.409	237	276	94 110	Ps+Pt-c D1	Ok
D2c	136.454 ...	34	276	12 110	Ps+Pt-c D2	Ok
D3c	351.749 ...	216	276	82 110	Ps+Pt-c D3	Ok
Max:	370.9520 64.409 mm.	0.857	0.854	(Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Req'd. Thk. Shell Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc	...	...	...	234	484	Ok
D2uc	...	...	...	29	207	Ok
D3uc	...	...	...	205	207	Ok
D1c	...	...	...	240	484	Ok
D2c	...	...	...	29	207	Ok
D3c	...	...	...	210	484	Ok
Max	...	...	...	0.993		

Note:
 Because there was no net pressure differential across the tubesheet due to
 the specified pressures psd,min and ptd,min; load case D4 was not processed as
 the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm.)	Required	Actual	P/F
-----------------	----------	--------	-----

Tubesheet thickness :	370.952	400.000	Ok
Tubesheet thickness flanged extension :	64.409	388.000	Ok
Tube thickness :	1.931	2.770	Ok
Tube-tubesheet fillet weld leg :	1.686	2.770	Ok
Tube-tubesheet groove weld leg :	1.686	2.770	Ok

Given Shell Thickness: 28.0000 mm.
 Given Channel Thickness: 140.0000 mm.

Min Channel length of thk, (140.000) adj. to tubesheet: 851.915 mm.

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	222.346	1.000	200.656	0.998
Tubesheet Shear Stress	223.745	1.000	223.745	1.000
Tubesheet-Channel Junction	223.745	1.006	223.745	1.126
Tube Pressure Stress	281.948	1.000	187.794	1.000
Tubesheet Extension Stress	No Calc	No Calc	26.535	...
Minimum MAWP	222.346		26.535	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	224.962	1.000	204.436	1.000
Tubesheet Shear Stress	224.962	0.998	225.436	1.000
Tubesheet-Channel Junction	224.962	1.000	225.436	1.115
Tube Pressure Stress	281.948	1.000	187.794	1.000
Tubesheet Extension Stress	No Calc	No Calc	26.535	...
Minimum MAPnc	224.962		26.535	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Channel Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):
 = max(tubesheet thk/4, min(tubesheet thk, channel thickness))
 = max(400.0/4, min(400.0, 140.0))
 = 140.000 mm.

Warning:
 The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm..
 This component must be impact tested.

Shell Side MDMT calculation:

Governing thickness for the shell side:
 = tubesheet thickness/4
 = 400.0/4
 = 100.000 mm.

Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

where the MDMT reduction ratio per UCS 66 (b)(1)(b) is:

= max(pt/Tubeside MAPnc, ps/Shellside MAPnc), must be ≤ 1
= max(191.0/224.96, 25.0/26.53)
= 0.942

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: One or more nozzles do not meet proper Spacing Requirements!**

Please review all reports carefully!

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